



TEMENOS
Learning Community

Transact

Technical Essentials

Guide (R22) (English)
2022 Q3

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Document History

Author	Version	Release Date	Comments
TLC CaTS	1.0	1-Jan-22	Initial document.

Introduction

Lesson Overview

I will describe about the course Transact Technical Essentials .

| Lesson Overview



Transact Technical Essentials » Objective and Learning Outcomes

This course will allow you to, Define the Technical Essentials of Application Structure, User, Services and COB You will be able to, Identify how the records are stored in Transact Create USER and configure security Describe Services Demonstrate COB

Transact Technical Essentials » Objective and Learning Outcomes

This course will allow you to:

- Define the Technical Essentials of Application Structure, User, Services and COB

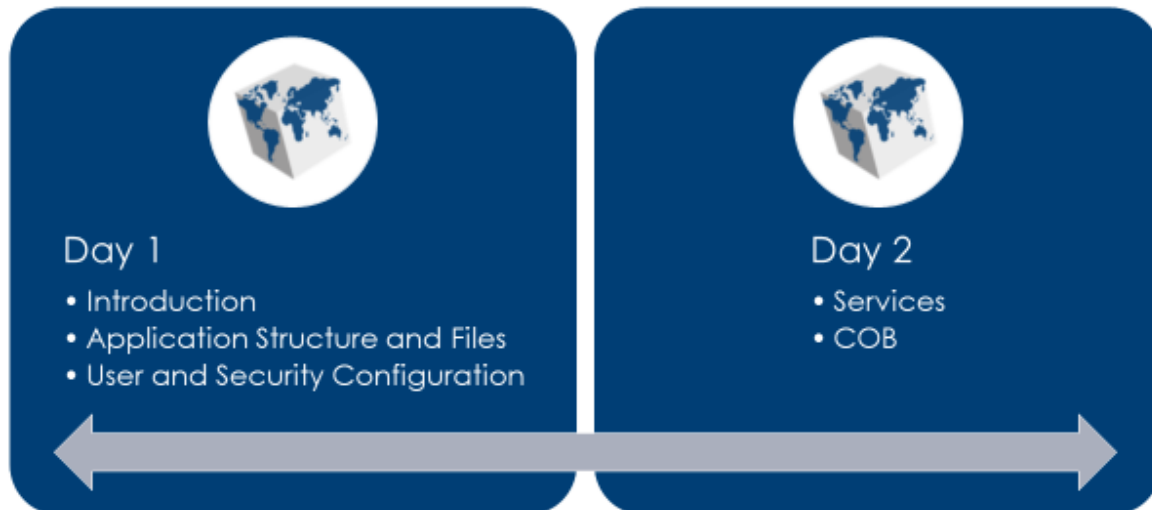
In particular you will be able to:

- Identify how the records are stored in Transact
- Create USER and configure security
- Describe Services
- Demonstrate COB

Transact Technical Essentials » Timetable

Here's the timetable. We've just started on the Introduction lesson. Following this we'll cover Application Structure and Files, User and Security Configurations. On the 2nd day we'll look at Services and COB.

| Transact Technical Essentials » Timetable



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Lesson Summary

I described about your course Transact Technical Essentials

| Lesson Summary



Application Structure and Files

Lesson Overview

In this lesson, we will see how applications stores data in Transact

| Lesson Overview

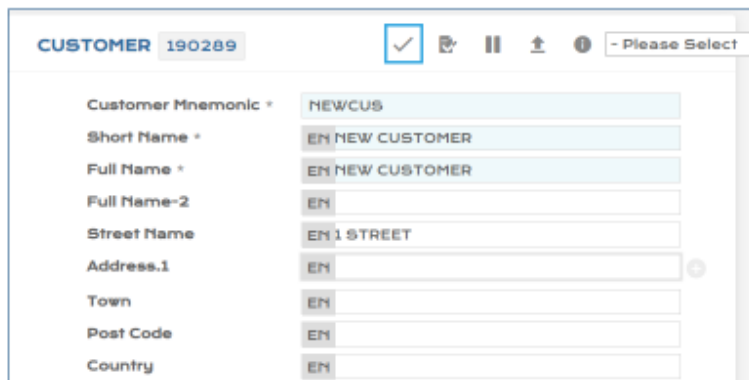


Slide 5

We have seen in the previous session that a Transact product is made up of Application, services, enquiries and versions. An application records the data entered and stores it in an associated file, field by field

| Application

- An application records the data entered and stores it in an associated file, field by field.
- What happens after we enter the data?



The screenshot shows a Transact application form for a customer. The form is titled "CUSTOMER 190289" and includes a toolbar with icons for save, print, and other actions. The form fields are as follows:

Field	Value
Customer Mnemonic *	NEWCUS
Short Name *	EN NEW CUSTOMER
Full Name *	EN NEW CUSTOMER
Full Name-2	EN
Street Name	EN 1 STREET
Address.1	EN
Town	EN
Post Code	EN
Country	EN

Slide 6

An application is associated with not just one file/table – Up to five files may be associated with an application in Transact

| Application

- How is the data stored?
 - An application is not just one file.
 - Up to five files may be associated with an application.

Slide 7

A transaction in Transact can involve multiple stake holders. A transaction is input by a user and is authorized by his manager. Depending on values in the fields of the transaction - banks might want more than 1 level of authorization. To create a record in Transact, you need to input all the mandatory fields and then get the record authorized. Inputter is the person who inputs data into the fields in a record. Authorizer is the person who checks the record and authorizes it. The error message "EB.RTN.SAME.NAME.AUTHORISER/INPUTTER" will be displayed if the same user tries to input and also authorize the record.

| Application » Workflow » Input

- Banks usually use a four-eye view for a financial transaction.
- To facilitate this, Core Banking has two stages in a transaction:
 - Input
 - Authorization

The screenshot shows the Transact application interface for a customer input. The top bar displays 'CUSTOMER 1 F3'. The main form is titled 'CUSTOMER 190289' and contains the following fields:

Customer Mnemonic *	STEVE
Short Name *	EN STEVE
Full Name *	EN STEVE
Full Name-2	EN
Street Name	EN 1 STREET
Address.1	EN
Town	EN

On the right side of the form, there is a red box labeled '190289 Unauthorised'. A callout box points to this red box with the text: 'On clicking commit, the record is written to the unauthorised file'.

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Slide 8

When the USER authorizes the record, the record is MOVED from the unauthorized to the authorized file.

| Application » Workflow » Authorize

- Another USER has to authorize the record



CUSTOMER A 190289

CUSTOMER 190289

Please Select

Customer Pseudonym	STEVE
Short Name	EN STEVE
Full Name	EN STEVE
Street Name	EN L STREET
Country	EN UK
Sector	1001 Individual
Account Officer	1 Implementation
Industry	1000 Private Person (Name)
Target	999 Others
Nationality	US United States of America
Customer Status	1 Private Client - Standard
Residence	US United States of America
Language	1 English
Branch Name	GB0000001 Model Bank
Clas Party	No
AML Check	Null
AML Result	Null
Internet Banking?	Null
Mobile Banking?	Null
Record Status	INAU INPUT Unauthorised



Authorised file is also known as LIVE file

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Slide 9

'A' is a function which allows the user to authorize a record. 'D' is a function which allows the user to delete a record which is not yet authorized. An authorized record cannot be deleted. 'E' function allows the user to list the unauthorized records. 'I' function allows the user to input a record in an application. 'L' function is used to list live records 'S' allows user to only view the records. 'R' authorized records cannot be deleted. To remove an authorized record, the function is R – Reverse 'H' – a reversed record can be brought back using the History restore function 'Q' – audit . Not available by default 'V' – is a special function. Applicable only for some applications.

| Application » Functions

A – Authorize

D – Delete

E – List exceptions (List unauthorized records)

I – Input

L – List live records

P – Print

S – See

R – Reverse

H – History Restore

Q – Audit

V - Verify

Slide 10

Financial transactions, need version control. We need to store the changes made in every record. In Transact, the current authorized version of a transaction is kept in the Authorized file, and older versions are maintained in a history file. Unauthorized transactions are maintained in the unauthorized file. The reason for having separate history files is that over a period - this information can be archived. Deleted History file captures records deleted from unauthorized files. This enables preservation of deleted transactions for audit purposes. Input a record in USER application with ID STEVE and commit. The record will be stored in the unauthorized file.

| Multiple Files

- Create a USER record with ID STEVE for the first time
- The record will be stored in the unauthorized file

Input USER with ID STEVE
USER | STEVE



Slide 11

When the record is authorised using the A function, the record is moved from the unauthorised to the Authorised file

| Multiple Files

- Authorise the USER record with ID STEVE for the first time
- The record will be moved from the unauthorized file to the authorized/LIVE file

Authorise USER with ID STEVE

USER A STEVE



Slide 12

Information about the status of the record – is it unauthorized, is it authorized, is stored in a field called RECORD.STATUS RECORD.STATUS for unauthorized record is INAU RECORD.STATUS for an authorized record is BLANK. Authorized records are also called LIVE records

| Multiple Files » Audit field – RECORD.STATUS

- Information about the status of the record – is it unauthorized, is it authorized, is stored in a field called RECORD.STATUS
- RECORD.STATUS for unauthorized record is INAU
- RECORD.STATUS for an authorized record is BLANK. Authorized records are also called LIVE records

Slide 13

When an authorised record is modified and committed, the authorized record will continue to be available in the LIVE file. The modified/unauthorized record is available in the unauthorized file Note - this ID is now available in two file the LIVE file as well as the unauthorised

| Multiple Files

- Edit the authorized record and commit
- The authorized record will continue to be available in the Authorised file. The modified/unauthorized record is available in the unauthorized file

Modify the authorised USER with ID STEVE

USER | STEVE



- Note - this ID is now available in two files – Authorised as well as unauthorised

Slide 14

When the modification is authorized, the previous LIVE record is moved to the history file. The unauthorized record with the changes will be moved to the LIVE file. A record can be modified and authorized many times. Hence the ID of the record in the History file is ID;CURR.NO STEVE;1 CURR.NO is an audit field that indicates how many times the record has been authorized

| Multiple Files

- Authorize the changes
- The previously authorized record will be moved to history file
- The unauthorized record with the changes will be moved to LIVE

Authorize the changes

USER A STEVE



- A record can be modified and authorized several times. Hence the ID of the record in the History file is ID;CURR.NO **STEVE;1**
- **CURR.NO** is an audit field that indicates how many times the record has been authorized

Slide 15

Authorized records can only be reversed. When an authorized record is reversed, the record is available in the unauthorized file as the reversal is not yet authorized. The RECORD.STATUS is now RNAU.

| Multiple Files » REVERSE a record – R function

- Authorised records cannot be deleted. They can be moved from the authorised file to History file using R function. This is called REVERSAL
- When authorized USER record with ID STEVE is reversed, the record will be available in the unauthorized file as the reversal is not yet authorized
- The RECORD.STATUS is RNAU (Reversal Not Authorised)

Reverse the record STEVE

USER R STEVE



Slide 16

When the reversal is authorized, the record is removed from LIVE and available in the History file. The status of the record is REVE. As there can be more than one change for the same ID, the ID of a record in History file is RECORDID;CURR.NO

| Multiple Files » Authorise the reversal

- When the reversal is authorized, the record is moved to the History file.
The RECORD.STATUS is **REVE - REVERSED**

Authorize the reversal

USER A STEVE



Slide 17

To restore a record from the History file, use the H function. When H function is applied on a record that's reversed, the record will be available in the unauth file and the status will be HNAU, as the history restore of the record is not authorized. When the History restore is authorized, the record will be available in the authorised file. It will continue to exist in the History file.

| Multiple Files » History restore a reversed record

- To restore a record from the History file, use the H function
- When H function is applied on a record that's reversed, the record will be available in the unauthorised file and the status will be HNAU, as the history restore of the record is not authorized
- When the History restore is authorized, the record will be available in the authorised file. It will continue to exist in the History file.

Slide 18

Deleted history is not available by default and must be configured if required for an application as they are stored as an image ref in the back-end table. The ID of the record in deleted history is recid;date.and.time
E.g., 150502;19042581549367
Deleted records can be viewed using enquiry VIEW.DELETE.HISTORY. In the above figure we could see the Deleted history of records for the application CUSTOMER. You can view the deleted history based on application.

Multiple Files

Input USER with ID STEPHAN

USER I STEPHAN

Delete USER with ID STEPHAN

USER D STEPHAN



TEMENOS ENQ VIEW.DELETE.HISTORY						
Results 1 - 3 of 3						
Deleted Record History						
Transaction Ref	Record Status	Curr No	Date Time	Inputter	Company Name	
150501;22031167311166	INAU	1	03-11-2022 18:41:51:366	33566_OFFICER_OFS_SEAT	GB0010001	See Record Details
150502;22031167311287	INAU	1	03-11-2022 18:41:51:287	10074_OFFICER_OFS_SEAT	GB0010001	See Record Details
150500;22031167311612	INAU	1	03-11-2022 18:41:51:612	22870_OFFICER_OFS_SEAT	GB0010001	See Record Details
VIEW.DELETE.HISTORY						
Application Name *			equals	CUSTOMER		

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Slide 19

'V' is a special function available only in some applications. Performs some action depending on the application. For e.g., Verify function is used in RECORD.LOCK application to release a lock

| Multiple Files » VERIFY

- 'V' is a special function available only in some applications
- Performs some action depending on the application
- For e.g. Verify function is used in RECORD.LOCK application to release a lock

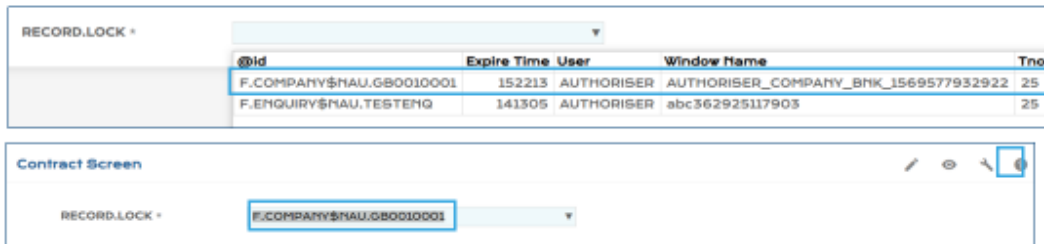
RECORD.LOCK »				
	@id	Expire Time	User	Window Name
	F.COMPANY\$NAU.GB0010001	152213	AUTHORISER	AUTHORISER_COMPANY_BNK_1569577932922
	F.ENQUIRY\$NAU.TESTENQ	141305	AUTHORISER	abc362925117903

Slide 20

Launch RECORD.LOCK application. Select the required record using the dropdown and click on Action button. This will bring up the VERIFY button. Choose VERIFY to unlock the record.

Multiple Files » VERIFY

- Select the required record to unlock and click on the action button



@Id	Expire Time	User	Window Name	Tno
F.COMPANY\$NAU.GB0010001	152213	AUTHORISER	AUTHORISER_COMPANY_BNK_1569577932922	25
F.ENGQUIRY\$NAU.TESTENQ	141305	AUTHORISER	abc362925117903	25

Contract Screen

RECORD.LOCK + F.COMPANY\$NAU.GB0010001

- Now, click on the VERIFY



RECORD.LOCK F.COMPANY\$NAU.GB0010001

Expire Time	152213
User	AUTHORISER
Window Name	AUTHORISER_COMPANY_BNK_1569577932922
Tno	25
File Name	F.COMPANY\$NAU
Record Id	GB0010001
Application	COMPANY
Date Time Sec	190927152213

Slide 21

Financial transactions need version control. We need to store the changes made in every record. In Transact, the current authorized version of a transaction is kept in the live file, and older versions are maintained in a history file. The reason for having separate history files is that over a period - this information can be archived. Deleted History file captures records deleted from unauthorized files. This enables preservation of deleted transactions for audit purposes.

| Record Status and Files » History File

Time Line	Unauthorised File	Live File	History File
Record is input by the User, but has not been authorised	- John O'Brien 1234 Green St - Single		
Record is authorised		- John O'Brien 1234 Green St - Single	
Address and status are changed	- John O'Brien 456 Lake Road - Married	- John O'Brien 1234 Green St - Single	
Change is authorised		- John O'Brien 456 Lake Road - Married	- John O'Brien 1234 Green St - Single

Slide 22

We know that a record passes through many stages – Unauthorized , Live , authorized , history and many more The audit field RECORD.STATUS holds the status of a record For a live record, RECORD.STATUS is blank/Null Live records cannot be deleted. They can be reversed. RECORD.STATUS may be one of the following

INAU – Input Not Authorized	INAO – Input Not Authorized due to Overrides	INA2 – Input requires Second Authorizer
RNAU –Reversal Not Authorized	RNA2 – Reversal requires second Authorizer	RNAO -Reversal Not Authorized awaiting Overrides
HNAU –History Not Authorized	IHLD - Input on HOLD.	REVE -Reversed

Record Status and Files

- We know that a record passes through many stages – unauthorized, live, authorized, history and more.
- The audit field RECORD.STATUS holds the status of a record.
- For a live record, RECORD.STATUS is blank/null.
- Live records cannot be deleted, they can be reversed.
- The RECORD.STATUS may have one of the following values:

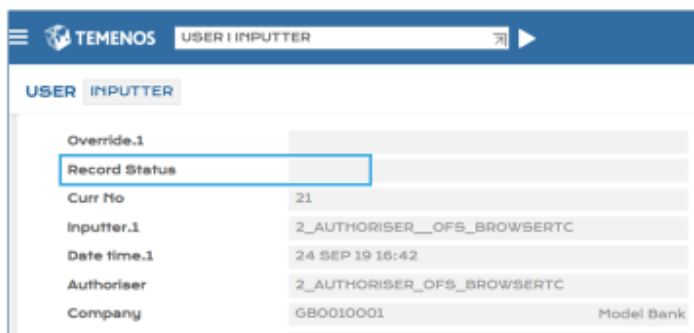
INAU	–	Input Not Authorized
INAO	–	Input Not Authorized due to Overrides
INA2	–	Input requires Second Authorizer
RNAU	–	Reversal Not Authorized
RNA2	–	Reversal requires second Authorizer
RNAO	–	Reversal Not Authorized awaiting Overrides
HNAU	–	History Not Authorized
IHLD	–	Input on HOLD.
REVE	–	Reversed

Slide 23

When a record is committed or authorized, Transact updates the following audit fields. They are no input fields attached at the end of every record. **RECORD STATUS**: Holds the status of the record. Possible values are INAU, IHLD, INAO, etc.,. If the record is in live file, **RECORD.STATUS** is Null **CURR NO.** : Holds the number of times the record was edited. **INPUTTER** : Holds the ID of the user who has inputted the record. **DATE TIME** : Based on the **COMPANY** table, the corresponding audit fields of the transaction record displays the local zone date and time when **USE.LOCAL.TIME** is set to YES in **SPF AUTHORISER** : Holds the ID of the user who has authorized the record. **CO CODE** : Defaults based on current company logged into **DEPT CODE** : Defaults to the user's department code The two fields below are populated only when a record is audited (Q function) **AUDITOR CODE** : Holds the code of the auditor who has reviewed the record. **AUDIT DATE TIME** : Holds the audit date and time.

Record Status and Files » Audit Fields

- When a record is committed or authorised, Transact updates the "Audit" fields.
- They are auto-stamped fields attached at the end of every record across applications.
- They are present in most of the Transact applications.



USER INPUTTER	
Override.1	
Record Status	
Curr No	21
Inputter.1	2_AUTHORISER__OFS_BROWSERTC
Date time.1	24 SEP 19 16:42
Authoriser	2_AUTHORISER__OFS_BROWSERTC
Company	GB0010001 Model Bank

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Slide 24

Data which is entered into Transact applications are stored in Files at database level. At database level these files are broadly classified into Live files, Unauthorized files History files Deleted History and SIM files. Live files store only authorized records. The field RECORD.STATUS will be BLANK for records in Live files. Unauthorized files (\$NAU) store unauthorized records. There are various Record Statuses that can be associated with records in Unauthorized Files. The Record Statuses are: INAU, INAO, INA2, RNAU, RNA2, RNAO, HNAU, IHLD History files (\$HIS) store old copies of authorized records. When an authorized record is edited and the changes authorized, the latest version of the record is stored in the Live file and the old version is moved to the History file. If an authorized record is reversed and the reversal authorized, that record also moves from the Live file to the History file. The Record Status of a reversed record is REVE. Format of the record ID is : <Record ID>;<Sequence Number> For Example : 100297;4 100724;3 The History file can store any number of amendments of a particular record. Deleted History Files store unauthorized records which are deleted. The ID format of these records take the form: <Transaction reference>;<Date and Time in milliseconds>. Audit fields are updated with User details (who performed the deletion operation) with the corresponding Date and Time. These records can be viewed only by using the enquiry VIEW.DELETE.HISTORY. Simulation - \$SIM files store records that are generated after simulation. These records are generated by Transact and hence there will be no audit fields for records in this file Number of files for an application depends on the TYPE of the APPLICATION in PGM.FILE

File Structure

File Type	Purpose	Naming Convention	Record Status possible
Live File	Holds only authorized records	<ApplicationName>	BLANK
Unauthorised File	Holds all the unauthorized records	<ApplicationName>\$NAU	INAU, INAO, INA2, RNAU, RNA2, RNAO, HNAU, IHLD
History File	Holds old copies of modified authorized records and reversed records.	<ApplicationName>\$HIS	BLANK REVE
Deleted History File	Holds deleted unauthorized records	<ApplicationName>\$DEL	Same status as of unauthorized transaction.
Simulation	Holds records created for simulation purpose in AA	<ApplicationName>\$SIM	No audit fields for \$SIM files

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Slide 25

TYPE field in PGM.FILE can contain H, U, L. (there are other types also) H means APPLICATION will have LIVE, \$NAU and \$HIS files. Can have \$DEL if required U means APPLICATION will have LIVE, \$NAU files. Can have \$DEL if required L means APPLICATION will have LIVE file only D means codeless Dynamic Application

| File Structure » TYPE field in PGM.FILE



PGM.FILE & CUSTOMER	
PGM.FILE CUSTOMER	
Type	H
Screen Title	ER CUSTOMER
Narrative	.BOA
Product	ST

- **TYPE** field in PGM.FILE can contain H, U, L. (there are other types also)
- **H** means APPLICATION will have LIVE, \$NAU and \$HIS files. Can have \$DEL if required
- **U** means APPLICATION will have LIVE, \$NAU files. Can have \$DEL if required
- **L** means APPLICATION will have LIVE file only
- **D** means codeless Dynamic Application

Slide 26

Note: DELETE.HISTORY in FILE.CONTROL must be set to 'YES' if deleted history needs to be maintained by Transact. can be set to Y for TYPE in PGM.FILE equal to H or U only Note: SIM.REQD in FILE.CONTROL must be set to 'YES' if \$SIM file is required

File Structure » FILE.CONTROL

- Holds the files associated with an application.
- FILE.CONTROL holds information about the number of files an application has at database level
- CUSTOMER application has four files at the database level – the LIVE file, \$HIS,\$NAU and \$DEL

PGM.FILE \$ CUSTOMER	
Type	H
Screen Title	ER CUSTOMER
Narrative	.BDA
Product	ST

FILE.CONTROL CUSTOMER	
Description.1	Customer details
Product	ST
Suffixes.1	☐ \$HIS ☒ \$NAU ☐ \$SIM ☐ \$DEL
Suffixes.2	☒ \$HIS ☐ \$NAU ☐ \$SIM ☐ \$DEL
Suffixes.3	☐ \$HIS ☐ \$NAU ☐ \$SIM ☐ \$DEL
File Type	2
File Module	ST
Classification	Cus
Sub Product	CUSTOMER
Sys Clear Files	☐ Y ☒ N ☐ B
Cus Clear Files	☒ Y ☐ N ☐ B
Branch File	☐ Ref ☒ Ref ☐ Bin
Inc File	☐
Related File.1	CUSTOMER
Local Ref Master	
Sim Reqd	☐ Y ☒ N
Delete History	☒ Y

Slide 27

PGM.FILE for ABBREVIATION has type U ABBREVIATION has LIVE and \$NAU files. Users can input and authorize records. There is no History details stored for ABBREVIATION as there is no \$HIS file

| File Structure » FILE.CONTROL

PGM.FILE ABBREVIATION	
Type	U
Screen Title	EN Abbreviation
Product	EB

FILE.CONTROL ABBREVIATION	
Description.1	Command line abbreviations
Product	EB
Suffixes.1	\$Nau
File Type	2
File Modulo	5
Classification	Int
Sub Product	SYSTEM

- PGM.FILE for ABBREVIATION has type U
- ABBREVIATION has LIVE and \$NAU files.
- Users can input and authorize records. There is no History details stored for ABBREVIATION as there is no \$HIS file

Slide 28

All applications do not have all files at the database level. From screenshots above, we can see that STMT.ENTRY has LIVE file only as PGM.FILE type is T. SIM.REQD is set as Yes and hence \$SIM file is also available for STMT.ENTRY

| File Structure » FILE.CONTROL

PGM.FILE STMT.ENTRY	
Type	L
Screen Title	EN STMT.ENTRY
Product	AC

FILE.CONTROL STMT.ENTRY	
Description.1	Details of all entries over Customer and
Description.2	Details of all entries over Customer and Internal Accounts
Product	AC Accounts
Suffixes.1	\$Sim
File Type	2
File Module	131
Classification	Fin
Sub Product	ENTRY
Bye Clear Files	Y
Cus Clear Files	N
Branch File	Bin
Sim Reqd	Y

- STMT.ENTRY type is **L** in PGM.FILE
- It will have LIVE file only. We will be able to only **View** records
- L type files will not have audit fields
- \$SIM file is also available as SIM.REQD is set to Y

Slide 29

All applications do not have all files at the database level. From screenshots above, we can see that AA.ACCOUNT.CLOSURE.DETAILS is a type “D” – DYNAMIC CODELESS application having LIVE, \$NAU and \$HIS files at the Database level

| File Structure » FILE.CONTROL

PGM.FILE AA.ACCOUNT.CLOSURE.DETAILS	
Type	D
Screen Title	EN AA.ACCOUNT.CLOSURE.DETAILS
Product	AA

FILE.CONTROL AA.ACCOUNT.CLOSURE.DETAILS	
Description.1	Account Closure Details
Product	AA
Suffixes.1	\$Nau
Suffixes.2	\$His
File Type	18
File Modulo	97
Classification	Cus

- Type is **D** in PGM.FILE meaning it is a Dynamic codeless application

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Slide 30

How are the fields displayed in any application? The system reads the STANDARD.SELECTION file for the corresponding application and populates all the fields in the application. This file holds the field names, field position, validations, etc. This is a no input file and is generated automatically by the system. For an application to have a record in this file, there must be a corresponding FILE.CONTROL record.

| File Structure » STANDARD.SELECTION

- The ID of a record in STANDARD.SELECTION is the application name, e.g., CUSTOMER, USER, ACCOUNT.
- The fields associated with an application are seen in STANDARD.SELECTION.
- D in the field 'SYS.TYPE' denotes a core field that holds data.

The ID of a record in STANDARD.SELECTION is the application name, eg. CUSTOMER, USER, ACCOUNT. The fields associated with an application are seen in STANDARD.SELECTION. D in the field 'SYS.TYPE' denotes a core field that holds data. In the above screenshot, we can see that the fields displayed in the CUSTOMER record can be seen in the STANDARD.SELECTION record with ID CUSTOMER

STANDARD.SELECTION		CUSTOMER
Sys Field Name..4	PENETRORC	
Sys Type..4	D	
Sys Field No..4 .1	3	
Sys Val Prog..4 .1	[H27F]#####S[IMATE][ACCE SS(DIRECT)]	
Sys Display Fmt..4	3DL	
Sys Alt Index..4	N	
Sys Single Mult..4	S	
Sys Leng Field..4	N	
Sys Generated..4	Y	
Sys Field Name..5	SHOOT_NAME	
Sys Type..5	D	
Sys Field No..5 .1	2	
Sys Val Prog..5 .1	[H2B]#####S[EGHTPHATE][ACCE SS_PORTABLE(DIRECT)]	
Sys Display Fmt..5	3SL	
Sys Alt Index..5	N	
Sys Single Mult..5	M	
Sys Leng Field..5	Y	
Sys Generated..5	Y	
Sys Field Name..6	NAME_1	
Sys Type..6	D	
Sys Field No..6 .1	3	
Sys Val Prog..6 .1	[H2D]#####S[EGHTPHATE][ACCE SS_PORTABLE(DIRECT)]	
Sys Display Fmt..6	3SL	
Sys Alt Index..6	N	
Sys Single Mult..6	M	
Sys Leng Field..6	Y	

CUSTOMER 190254	
Customer Mnemonic *	
Short Name *	EN
Full Name *	EN
Full Name-2	EN
Street Name	EN
Address.1	EN
Town	EN
Post Code	EN
Country	EN
+	
Relation Code.1	
Relation Customer.1	
Reverse Relation.1	
-	
Rel Deliv Opt.1.1	- Please Select
Role.1.1	
More Role Info.1.1	
Role Notes.1.1	



Lesson Summary

In this lesson, we saw how applications store data in Transact

| Lesson Summary



User and Security Configuration

Lesson Overview

In this lesson, we will see how to set up USER and permission for users

| Lesson Overview



Slide 5

Why does SMS(Security Management System) need to be a part of Transact? Irrespective of where you work today, you have a role to play in your organization. You can do certain things, and you are restricted from doing others. In a bank, there are different job profiles that an employee can have. For example, one can be a Teller, another can be a Loan Disbursal Manager, or just the housekeeper. Now should a Teller be able to disburse a loan? Will the housekeeper even be allowed access to Transact? The software that the bank uses must be able to control what an employee can and cannot do once logged on. You are going to learn how this is done in Transact.

| USER

- Different roles exist in an Organization: CFO, Senior Managers, Managers, etc.
- Access to information must be controlled based on different roles.

Slide 6

Login to core banking by giving a username and password. In Model bank, use INPUTT/123456 to login

USER

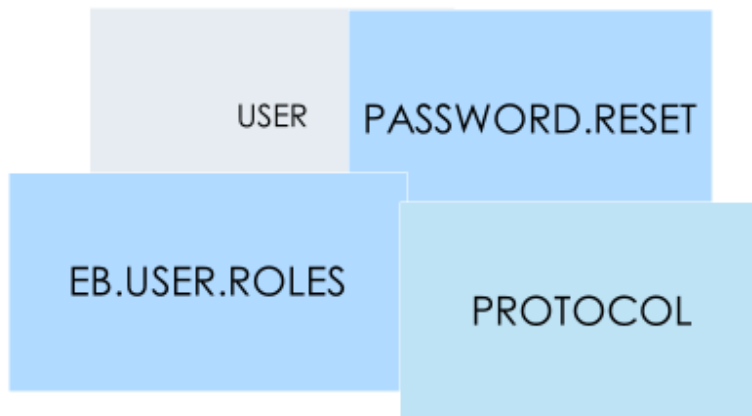
- To log into Core Banking, a user needs to supply a username and a password.
- The password is masked.



Slide 7

The above tables are used to implement security management in Transact. We will learn about these tables as we proceed with the course.

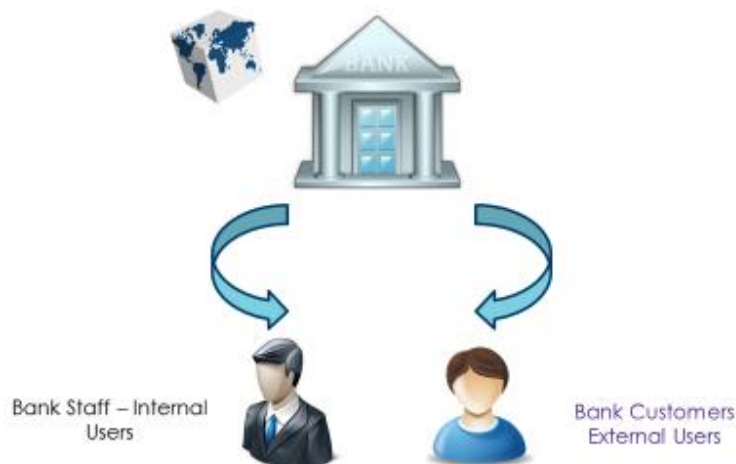
| USER » SMS » Applications



Slide 8

When we use the term “customers”, we refer to the customers of the bank . For e.g. customers who have a deposit in the bank, who have taken a loan. These customers are linked to Transact, when they do online banking. Hence we have a need to have 2 types of users. The bank employees are the internal user. We create a profile for them using the USER application. The customers of the bank are called external users and EB.EXTERNAL.USER holds the user profile of such customers.

| USER » Types



Slide 9

SignOn Name - USER ID and SIGN.ON.NAME must be different Classification - Bank employees are classified as Internal users Language – The Language of the User. Enrichments, error messages, overrides will be displayed in the Language of the USER Start Date profile – is validated against system date and not Transact date End Date Profile – user will not have access to Transact from this date

| USER » User Profile

USER

INPUTTER

✓

||

User Name *	INPUTTER	
Sign On Name *	INPUTT	
Classification *	☐ Ext ☒ Int	
Language *	1	English▼
Company.1 *	ALL	ALL COMPANIES
Department Code *	1	Implementation▼
Password Validity *	31 DEC 2099 M0601 31 DEC 2099 Every 6 months on day 1	
Start Date *	11 MAR 2022	11 MAR 2022
End Date *	31 DEC 2099	31 DEC 2099

Slide 10

Start Time and End Time – User will have access to Transact during this time. Can be combined with allowed days, Day St Time and Day End time to specify time of access for each day of the week
COMPANY.RESTR – Can be multivalued to specify access to multiple companies. Value ‘ALL’ indicates access to all companies
User Group / APPLICATION – can be multivalued to give access to different applications. ‘ALL.PG’ indicates access to all applications. This field is shown as User Group in above screenshot.
Function Allowed / FUNCTION – List of functions to which the user will have access for the application given. When ‘ALL’ is entered and validated, it will be expanded for all Transact functions

| USER

USER

INPUTTER

Start Time.1 *	0
End Time.1 *	2400
Time Out Minutes *	999
Attempts *	9
Initial Application	
Customer.1	
Account.1.1	
Company Restr.1	ALL
User Group.1	ALL.PG
Version.1	
Function Allowed.1	A 2 B C D E F H I L P R S V
Field No.1	
Data Comparison.1	- Please Select
Data From.1	
Data To.1	

Slide 11

Use the radio buttons to specify if PROTOCOL application in Transact will be updated for actions performed by this User. The preferred date format can be set in the USER profile. If Attributes field has SUPER.USER or COMMAND.LINE, the USER will have access to the command line in the browser.

USER

Sign On Off Log *

☐ Y ☒ No

Security Mgmt L *

☐ Y ☒ No

Application Log *

☐ Y ☒ No

Function Id Log *

☐ Y ☒ No

Input Day Month *

☒ Ddmm ☐ Mmdd

Clear Screen *

☒ Y ☐ No

Max Sessions

Attributes.1

Super User

Dispo Officer

Dispo Rights.1

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Slide 12

No more than two repeated characters (This is hardcoded). Last three passwords cannot be used. At first sign on, the system will ask for the Password to be input twice. Minimum of six characters and a maximum of 16 characters are allowed.

| USER » First Login » Password

- No more than two repeated characters (This is hardcoded).
- Last three passwords cannot be used.
- At first sign on, the system will ask for the Password to be input twice.
- Minimum of six characters and a maximum of 16 characters are allowed.



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Slide 13

What if a particular privilege or a set of privileges must be assigned for a group of users because they belong to the same department? Tedious to set the privileges in every single user profile Easier to create something like a set of privileges and apply it to the relevant users . For example common privileges for Managers, Teller , Head Teller , accountants , clerks .

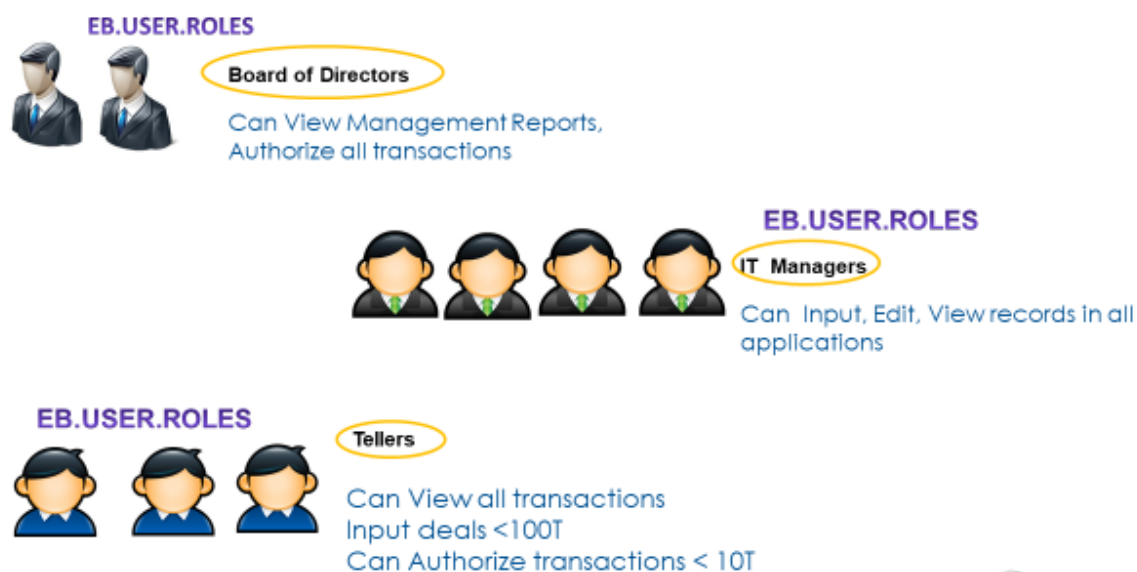
| Permissions » ROLES

- What if a particular privilege or a set of privileges must be assigned to a group of users because they belong to the same department?
- It's tedious to set the privileges in every single user profile.
- It's easier to create a set of privileges and apply it to the relevant users.
- For example, you can set common privileges for managers, tellers, head teller, accountants, clerks, etc.

Slide 14

USERS can be grouped to assign similar privileges to a set of users. Security Management System uses the EB.USER.ROLES application, to control Transact access and information and facilities by grouping together various Transact Applications. Enables the creation of logical group and specifies the group directly in a user profile, to avoid repetition of related applications, in different user profiles. Every user record existing in Transact must have a role defined for them. Specific Role needs to be defined for each company. A user is allowed to have multiple roles in a same company

| Permissions » ROLES



Slide 15

We can create different roles in EB.USER.ROLES

Permissions » EB.USER.ROLES

- Specify the APPLICATIONS and Functions for the Role

EB.USER.ROLES CRMROLE

Description *

EN CRM ROLE

Application.1 *

Version.1

Function Allowed.1

Field No.1

Data Comparison.1

Data From.1

Data To.1

Use local SMS.1

Reserved09.1

Reserved08.1

CUSTOMER

ALL

- Please Select

☐ Yes

Application.2 *

Version.2

Function Allowed.2

Field No.2

Data Comparison.2

Data From.2

Data To.2

Use local SMS.2

Reserved09.2

ACCOUNT

5

- Please Select

☐ Yes

EB.USER.ROLES ADMINROLE

Description *

EN ADMIN ROLE

Application.1 *

Version.1

Function Allowed.1

Field No.1

Data Comparison.1

Data From.1

Data To.1

Use local SMS.1

Reserved09.1

Reserved08.1

ALL.PC

ALL

- Please Select

☐ Yes

Application.2 *

Version.2

Function Allowed.2

Field No.2

Data Comparison.2

Data From.2

Data To.2

Use local SMS.2

Reserved09.2

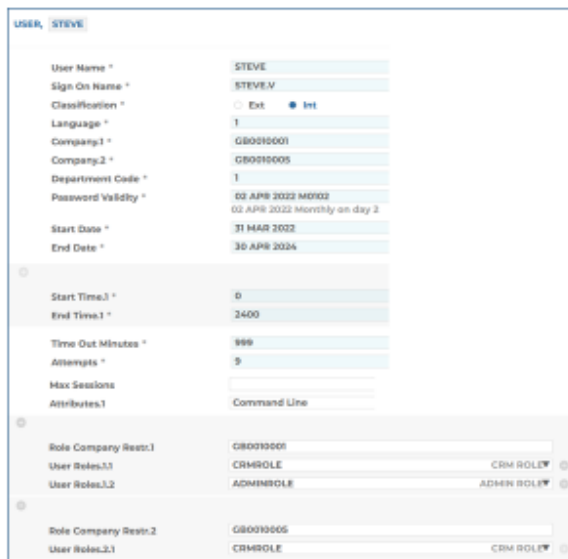
- Please Select

☐ Yes

Slide 16

In the above screenshot, we have allocated two roles to the user STEVE. The ID of the COMPANY record is attached to the field ROLE.COMPANY.RESTR and the ID of the record from EB.USER.ROLES is added to the USER.ROLES field. User can have multiple roles and different roles in different companies.

Permissions » Adding the role to the USER



User Name *	STEVE
Sign On Name *	STEVE.V
Classification *	☐ Ext ☒ Int
Language *	1
Company.1 *	GB0010001
Company.2 *	GB0010005
Department Code *	1
Password Validity *	02 APR 2022 M0102 02 APR 2022 Monthly on day 2
Start Date *	31 MAR 2022
End Date *	30 APR 2024
Start Time.1 *	0
End Time.1 *	2400
Time Out Minutes *	999
Attempts *	5
Max Sessions	
Attributes.1	Command Line
Role Company Restr.1	GB0010001
User Roles.1.1	CRMROLE CRM ROLE
User Roles.1.2	ADMINROLE ADMIN ROLE
Role Company Restr.2	GB0010005
User Roles.2.1	CRMROLE CRM ROLE

- Use the fields ROLE.COMPANY.RESTR and USER.ROLE to add roles to a USER
- A user can be associated with multiple roles

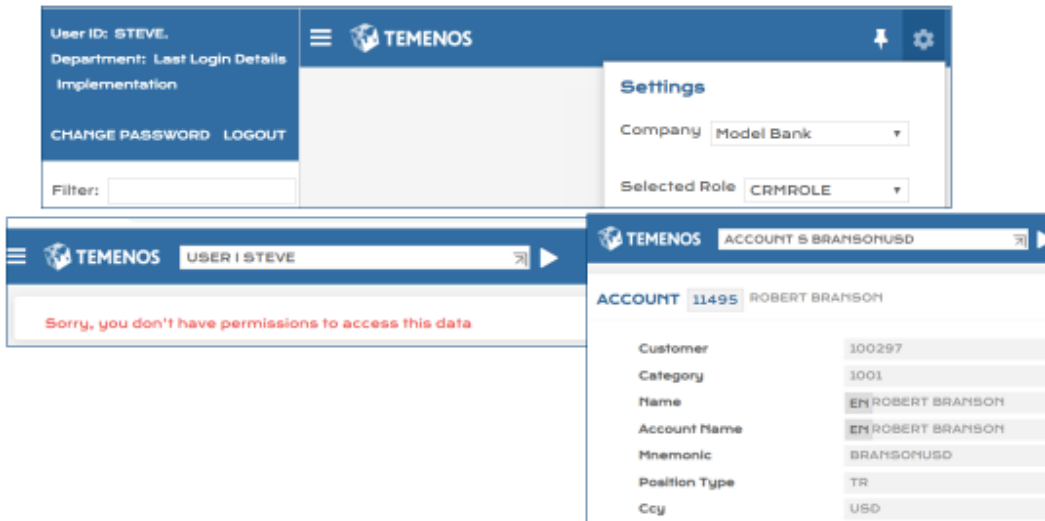
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Slide 17

When STEVE logs in, the default role is CRMROLE. Hence, STEVE does not have access to the USER application but he can view the ACCOUNT record.

Permissions » USER Role

- STEVE has logged in with CRM.ROLE



The screenshot displays the TEMENOS application interface. On the left, a sidebar shows user information: 'User ID: STEVE', 'Department: Last Login Details Implementation', and buttons for 'CHANGE PASSWORD' and 'LOGOUT'. Below this is a 'Filter:' input field. The main content area is divided into two sections. The top section, titled 'Settings', contains a 'Company' dropdown set to 'Model Bank' and a 'Selected Role' dropdown set to 'CRMROLE'. The bottom section, titled 'ACCOUNT 11495 ROBERT BRANSON', displays a table of account details:

Customer	100297
Category	1001
Name	EN ROBERT BRANSON
Account Name	EN ROBERT BRANSON
Mnemonic	BRANSONUSD
Position Type	TR
Ccy	USD

On the left side of the main content area, there is a red error message: 'Sorry, you don't have permissions to access this data'.

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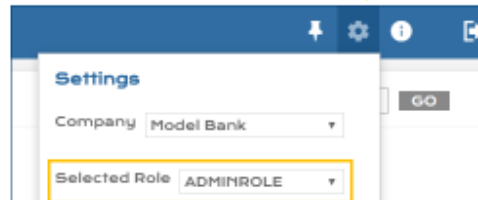
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Slide 18

When STEVE switches to ADMINROLE, USER application is accessible

| Permissions » USER role

- Switch role to ADMINROLE

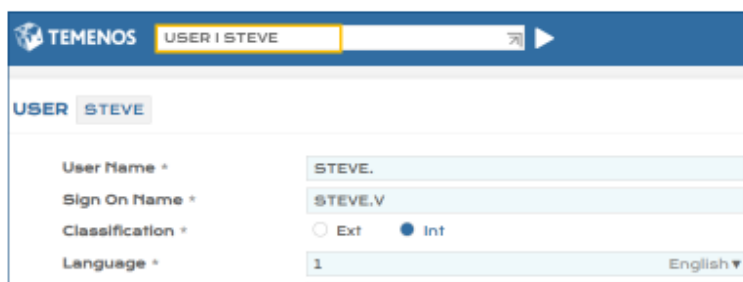


Settings

Company Model Bank

Selected Role ADMINROLE

GO



TEMENOS USER | STEVE

USER STEVE

User Name * STEVE.

Sign On Name * STEVE.V

Classification * ☐ Ext ☒ Int

Language * 1 English

Slide 19

What if you forget your password? What if your account is locked since you unsuccessfully tried different passwords and exceeded the maximum number of attempts? The application PASSWORD.RESET will allow an administrator to reset your account. You may not have access to this application. The ID of a record in PASSWORD.RESET can be any alphanumeric text. USER PW ATTEMPT: This field specifies the ID of the user whose record has been locked. When this record is authorised Transact resets the password and enables the profile at the same time. You must set a new password the next time you log in. If a user crosses the number of password attempts or if the user forgets the password, these can be set in the following fields: User Attempt : Every user is given a specific number of password attempts after which the user account gets locked. Maximum number of password attempts is specified in the application USER. Such locked user accounts can be unlocked by giving the user name in the field User Attempt. What if an administrator wants to activate a profile of an user even before the end of the deactivation period? You can achieve this by setting the following field. User Deact Perd : Specifies the ID of the user for whom the security administrator wants to reactivate the profile before the end of the deactivation period. User Reset : This field has the ID of a user whose password is to be reset. User Password : A new password must be set in this associated field. This password will expire once you login and thus the user will be forced to change it on sign-on.

PASSWORD.RESET

- ID of PASSWORD.RESET can be any meaningful text

The screenshot shows the 'Password Reset' application interface. It has a title bar with 'Password Reset' and 'RESET.PASSWORD'. Below the title bar are two tabs: 'Reset Users' (active) and 'Audit'. The main form area contains several fields:

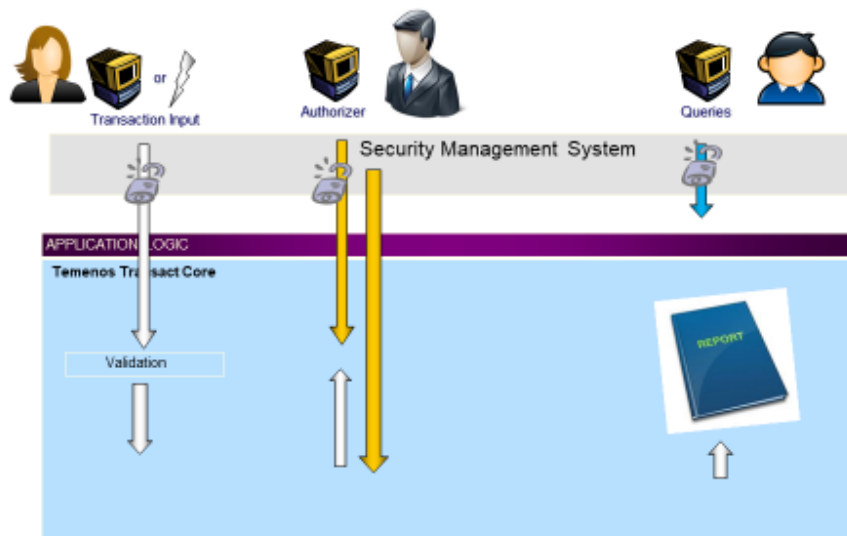
- User Pwd Attempt.1**: A field with a plus icon, linked to a callout: 'RESET password and number of attempts for a USER'.
- USER ID**: A field with a plus icon, linked to a callout: 'RESET password. Give the USER ID and new password'.
- User Attempt**: A field with a plus icon, linked to a callout: 'RESET number of attempts for a USER If max invalid attempts reached'.
- No of Attempts**: A field with a plus icon.
- User Deact Period**: A field with a plus icon.
- Deactiv Period**: A field with a plus icon.
- User Reset**: A field with a plus icon.
- User Password**: A field with a plus icon.
- User Type ***: A dropdown menu with options 'Int' (selected) and 'Ext'.

At the bottom left, there is a copyright notice: '©2022 Temenos Headquarters SA - all rights reserved.' At the bottom right, there is the Temenos Learning Community logo.

Slide 20

Transact doesn't stop with just validating the user and his/her password, the validation process continues even after user logs in successfully. Anything that a user tries to do is tracked and can proceed only if the user has necessary permissions to. Before the bank allows all users to log on to Transact and start using it, it must decide the user privileges in the system. This must be done because when a user tries to access or amend a record in any application, Transact checks to see if the user has the privilege. The permissions that are checked here include whether the user has access to the respective functions and applications. Here the user will encounter Transact SMS for the second time. To validate the record created, the user will click on the Validate button. The Transact application may reference various static tables of data to complete the record. The user does not need to have implicit permission to do so. This is not part of Transact SMS. On Commit the record is stored in the unauthorised file. Every record in Transact must be authorised. When a user tries to authorise a record, Transact must check to see if the user has the authorise permission for the application. User will not be allowed to authorise the record with insufficient permissions. Once the record is authorised, it moves to the authorised file. Static information could be updated at this stage as well, for example accounting entries etc. Close Of Business is the process which does not need any user intervention and hence no SMS check is required. The user administering COB must have the relevant SMS setting for COB related applications. SMS comes into the picture even if you want to execute an enquiry in Transact. In other words, even though you only want to view data, you must have necessary permissions to do so.

SMS workflow



Slide 21

PROTOCOL file in Transact stores all the details of actions performed by user. This file is updated by Transact and therefore records cannot be edited but only viewed. The following fields in the USER application are related to PROTOCOL SECURITY.MGMT.L Specifies whether a record should be written to the PROTOCOL file every time this User accesses any of the Security Management Applications. If Y is entered, a record is written to the PROTOCOL file every time this User accesses any of the following Applications: PASSWORD PASSWORD.ENABLE PASSWORD.RESET USER PASSWORD.RESET
Note: Security Violations are always logged, regardless of the values in Fields 25-28. APPLICATION.LOG Specifies whether or not records should be written to the PROTOCOL file recording every Application accessed by this User. If Y is entered, a record is written to the PROTOCOL file every time this User accesses a different Application. Security Violations are always logged, regardless of this field. It must be NO if SECURITY MGMT LOG (Field 26) is NO. FUNCTION.ID.LOG Specifies whether or not full details of every Application, Function and record ID accessed by this User should be recorded in the PROTOCOL file. It must be NO if APPLICATION LOG (Field 27) is NO. Security Violations are always logged, regardless of this field. Records from PROTOCOL application can be accessed in two ways By typing PROTOCOL L in the command line to list all the records in the file. By running an enquiry ENQ %PROTOCOL, which takes you to the selection box.

| PROTOCOL

- An application named PROTOCOL stores:
 - User login and logoff details.
 - Security violations by the user.
 - User access of the various applications.
 - Details of every application, function and record ID used by the user.

Slide 22

@ID : format of ID YYYYMMDD is the date on which the recorded event occurred. NNNNNNNNNN is the sequence number allocated to the event when it was recorded. The first event recorded each day is 0000000001 and so on. XX is the sequence number to identify the number of activities per one second. Process Date : Date on which the activity took place. Displayed in the format which was selected in the DATE FORMAT field in USER application. TIME : Identifies the time at which the recorded event took place. TIME.MSECS : Time when the activity took place denoted in the format hh:mm:ss:msc TERMINAL ID : The input is in the format NN XX. The first part of the terminal id denotes the database user number and the second part denotes operating system port number. COMPANY ID : Denotes the company in which the activity took place. USER : Holds the USER.ID of the user who has performed the action. APPLICATION : Identifies the Application which was being used by the user. ID : Holds the record ID of the application used. REMARK : This field denotes the activity performed and why the system did not allow the attempted activity. Transact supports multiple countries and its time zones in single instance. It provides local time of the actual branch from where the transaction originates and facilitates the representation of local time zones throughout the Transact application in the format DATE:HOURS:MINUTES:SECONDS. A field Local Time is introduced in the PROTOCOL file to update the transactions committed in local time zone.

PROTOCOL

PROTOCOL L

Results 1 - 19 of 141

Protocol - default list

q User Application Func Proc.date ID

@id	K User	Application	Level Function	Process Date	Id
202203313686045424.00	INPUTTER	CUSTOMER		15 APR 2021	
202203313686045424.01	INPUTTER	CUSTOMER		15 APR 2021	
202203313686053640.00	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.01	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.02	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.03	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.04	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.05	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.06	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.07	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.08	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.09	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.10	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.11	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.12	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.13	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.14	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.15	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.16	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.17	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.18	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.19	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.20	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.21	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.22	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.23	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.24	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.25	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.26	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.27	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.28	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.29	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.30	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.31	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.32	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.33	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.34	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.35	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.36	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.37	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.38	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.39	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.40	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.41	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.42	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.43	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.44	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.45	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.46	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.47	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.48	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.49	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.50	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.51	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.52	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.53	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.54	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.55	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.56	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.57	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.58	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.59	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.60	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.61	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.62	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.63	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.64	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.65	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.66	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.67	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.68	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.69	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.70	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.71	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.72	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.73	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.74	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.75	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.76	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.77	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.78	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.79	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.80	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.81	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.82	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.83	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.84	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.85	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.86	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.87	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.88	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.89	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.90	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.91	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.92	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.93	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.94	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.95	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.96	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.97	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.98	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.99	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	
202203313686053640.00	INPUTTER	FUNDS.TRANSFER.AC	I	15 APR 2021	

Slide 23

Above screenshot shows all commands executed by USER INPUTTER.

PROTOCOL

ENQ %PROTOCOL

TEMENOS

ENQ %PROTOCOL

Results 1 - 19 of 141

Protocol - default list

Id User Application Func Process Date Id

202203313686045424.00	INPUTTER	CUSTOMER			15 APR 2021	
202203313686045424.01	INPUTTER	CUSTOMER			15 APR 2021	
202203313686045424.02	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.03	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.04	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.05	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.06	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.07	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.08	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.09	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.10	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.11	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.12	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.13	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.14	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.15	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.16	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.17	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.18	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.19	INPUTTER	FUNDS.TRANSFER.AC	I		15 APR 2021	
202203313686045424.20	INPUTTER	AA.ARRANGEMENT	P		15 APR 2021	
202203313686045424.21	INPUTTER	AA.ARRANGEMENT	P		15 APR 2021	
202203313686045424.22	INPUTTER	AA.ARRANGEMENT	P		15 APR 2021	
202203313686045424.23	INPUTTER	ACCOUNT	P		15 APR 2021	
202203313686045424.24	INPUTTER	ACCOUNT	P		15 APR 2021	
202203313686045424.25	INPUTTER	ACCOUNT	P		15 APR 2021	
202203313686045424.26	INPUTTER	AA.ARR.ACCOUNT	P		15 APR 2021	

PROTOCOL Default List

USER equals INPUTTER

Lesson Summary

In this lesson, we saw how to set up USER and permission for users

| Lesson Summary



Services

Lesson Overview

In this lesson I am going to describe about.. Services Process and Jobs Features Auto Services Monitoring Services

| Lesson Overview



Slide 5

Services Help in automation. Services are programs that are run as background processes without user intervention. Can be executed at a schedule and can be also be triggered by a manual activity. They spawn multiple processes simultaneously and are centrally monitored by Transact. Catalogued in TSA.SERVICE application.

| Services

- Help in automation.
- Services are programs that are run as background processes without user intervention.
- Can be executed at a schedule and can be also be triggered by a manual activity.
- They spawn multiple processes simultaneously and are centrally monitored by Transact.
- Catalogued in the TSA.SERVICE application.

Slide 6

ABC Industries pays salary to its employees on the last working day of every month. All employees of ABC Industries have accounts with Bank B. Bank B maintains a file that is sent from ABC Industries. This file has the names and salary details of the employees. On the last working day, the company account must be debited, and the employees salary account must be credited. The above set of steps can be automated by Transact Services.

| Services

- ABC Industries pays salary to its employees on the last working day of every month.
- All employees of ABC Industries have accounts with Bank B.
- Bank B maintains a file that is sent from ABC Industries. This file has the names and salary details of the employees.
- On the last working day, the company account must be debited, and the employees salary account must be credited.
- The above set of steps can be automated by Transact Services

Slide 7

A Service is also known as a Process or a Batch. A service is made up of related jobs. For e.g., an archival service may be made up of two jobs. One to copy records older than a specific date to a different DB and another to remove the records from the current DB Job -The lowest unit of work Process - A collection of related jobs that perform a business function – for e.g., salary disbursement. A Process in Transact is catalogued in the BATCH application.

| Services

Units of Service:

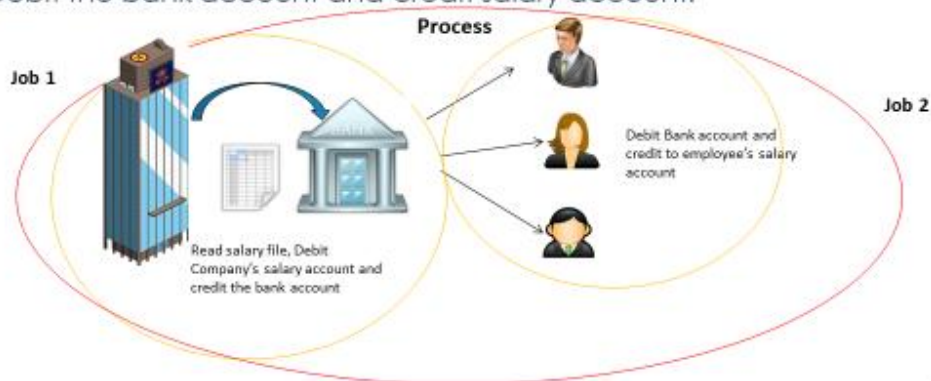
- Job
 - The lowest unit of work.
- Process
 - A collection of related jobs that perform a business function – for e.g., salary disbursement.
 - A Process in Transact is catalogued in the BATCH application.

Slide 8

Relation between Process and Jobs The salary disbursement service(process) must do the following Read the employee file and determine whether the file and all employee accounts are valid Debit the company's account and move funds into the bank account Debit the bank account and credit salary account

Process and Jobs

- The salary disbursement service needs to do the following:
 - Read the employee file and determine whether the file and all employee accounts are valid.
 - Debit the company's account and move funds into the bank account.
 - Debit the bank account and credit salary account.



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Slide 9

Jobs perform a specific task or unit of work by executing a piece of code which are written in routines. These routines must have an entry in PGM.FILE application. The field Type must be set to B (i.e., BATCH) for this record in PGM.FILE. A process is nothing but a collection of jobs which together perform a specific business function for e.g., Interest Accrual, Salary disbursement. Each process must have an entry in the BATCH application. Hence a BATCH application record identifies a business function that we want to run as a service in the background. This BATCH record should contain the list of jobs that will be executed in the sequence in which it is listed. For e.g., The BATCH record for Salary Disbursement should list two jobs in the field Job. Job.1 – Debit Company a/c and credit Bank a/c. Job.2 – Debit bank a/c and credit individual salary a/c. These two jobs must execute only in a sequence i.e, first Job.1, then Job.2. We must then define a service, by making an entry in TSA.SERVICE application. The record ID of the TSA.SERVICE record and BATCH record must be the same. For e.g. To run the salary disbursement service, we must also make an entry in the TSA.SERVICE application, then set the Service. Control field in this service record to Start. This will start or execute the service in the background. Transact Services module will then pick up the corresponding BATCH record by matching the record ID and execute the Jobs listed in the BATCH record i.e., Job.1 and Job.2 in the sequence defined.

| Process and Jobs » Setting up a Service

- Jobs are individual units of work, which are implemented as routines in Transact.
- Job routines are catalogued in PGM.FILE.
- A process is a group of jobs which perform a business function.
- Processes are catalogued in BATCH.
- A service is an execution of the jobs in a Process.
- Services are catalogued in TSA.SERVICE – ID identical to BATCH record ID.



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Slide 10

In this example, an existing service called BNK/FT.BULK.PAYMENT is used. The screen shots above show the identical TSA.SERVICE and BATCH record IDs. The BATCH record, lists the Jobs i.e., routines that will get executed when you run this service. Each of the jobs listed will have a PGM.FILE entry. Setting the Service. Control field to start will trigger this service, which will in turn execute the jobs FT.PROCESS.BULK.PAYMENT.PRE and FT.PROCESS.BULK.PAYMENT in a sequence in the background. It is important to understand - Why the Bank Mnemonic is part of the BATCH record ID? This is because Processes are company specific. If we want the same process to be executed for each company defined in the COMPANY application, then we have to have processes defined for every company separately. For Example: Assume the salary accounts of the employees are in branch BNK. BNK branch should schedule the Salary disbursement process on the last working day of every month, but this service is not applicable for branch EU1. The only way to do this in Transact, is to create a process in the BATCH application with ID prefixed with the mnemonic BNK and not EU1. BNK/<Batch Process ID Name> If you want a batch process to be executed for both BNK and EU1, then you need to do the following. Create the following records in the BATCH application BNK/<Batch Process ID Name> EU1/<Batch Process ID Name>

Process and Jobs » The Payments Service

- Let us look at an example, discussed earlier. FT.BULK.PAYMENT is the service used for bulk payments processing. Look at the jobs within the process.

TSA.SERVICE, BNK/FT.BULK.PAYMENT		BATCH, BNK/FT.BULK.PAYMENT	
Description.1	SERVICE FOR BULK PROCESSING	Process Status	0 READY
User	SEAT/USER	Batch Environment	F FOREGROUND
Service Control	Stop	Job Name.1	FT.PROCESS.BULK.PAYMENT.PRET PROCESS BULK PAYMENT PRE
Stopped.1	11/03/2022 19:51:15	Frequency.1	D DAILY
Elapsed.1	19:51:15	Job Status.1	0 READY
Transactions.1	10	Last Payment.1	15 APR 2021 15 APR 2021
Curr No	1	Job Name.2	FT.PROCESS.BULK.PAYMENT FT PROCESS BULK PAYMENT
Inputter.1	SY_COMP.IN.TSA	Verification.2.1	FT.PROCESS.BULK.PAYMENT.PRET PROCESS BULK PAYMENT PRE
Date time.1	11 MAR 22 03:07	Frequency.2	D DAILY
Authoriser	16_TRAINSTI1_OFS_MB.OFS	Job Status.2	0 READY
Company	CB0010001	Last Payment.2	15 APR 2021 15 APR 2021
Dept Code	1	Lock Retry Count	0
		Curr No	1
		Inputter.1	SY_COMP.IN.BATCH
		Date time.1	11 MAR 22 03:07
		Authoriser	16_TRAINSTI1_OFS_MB.OFS
		Company	CB0010001 Model Bank

Slide 11

Jobs will have an entry in PGM.FILE with TYPE as B

Process and Jobs » The Payments Service

BATCH, BNK/FT.BULK.PAYMENT

Process Status	0	READY
Batch Environment	F	
	FOREGROUND	
Job Name.1	FT.PROCESS.BULK.PAYMENT.PRE	FT PROCESS BULK PAYMENT PRE
Frequency.1	D	DAILY
Job Status.1	0	READY
Last Payment.1	15 APR 2021	15 APR 2021
Job Name.2	FT.PROCESS.BULK.PAYMENT	FT PROCESS BULK PAYMENT
Verification.2.1	FT.PROCESS.BULK.PAYMENT.PRE	FT PROCESS BULK PAYMENT PRE
Frequency.2	D	DAILY
Job Status.2	0	READY
Last Payment.2	15 APR 2021	15 APR 2021
Lock Retry Count	0	
Cur No	1	
Inputter.1	SY_COMPIN.BATCH	
Date time.1	11 MAR 22 03:07	
Authoriser	16_TRAINS11_OFS_MB.OFS	
Company	080010001	Model Bank

PGM.FILE, FT.PROCESS.BULK.PAYMENT.PRE

Type	B
Screen Title	EN T PROCESS BULK PAYMENT PRE
Narrative	.NOINFINITY
Batch Job.1	@BATCH.JOB.CONTROL
Product	FT
Description	This job runs before FT.PROCESS.BULK.PAYMENT to generate FT between corporate or active account and wash account if SINGLE.MULTI is set SINGLE

- Jobs are catalogued in PGM.FILE
- TYPE = B

Slide 12

Let us look at the batch record in detail Batch Environment - Defines the environment to run the process in, i.e foreground or background. A foreground process will be run directly on the user's terminal. In this case, the user will not be able to do any other activity on the terminal since the service is running on the terminal. A background process will run as a phantom task i.e., the process will be disconnected from the user's terminal and will be executing in the background. So, the user's terminal is free to perform other activities. Job Name - Contains the name of the routine to be executed. All routines defined here will have an entry in PGM.FILE with TYPE set to 'B' Frequency - specifies the frequency at which the batch job has to be executed. It can have values as follows. D specifies that the job will be executed every working day. D nn specifies that the job will be executed every nn'th working day W specifies that the job will be executed on a weekly basis, that is every Friday M specifies that the job will be executed on the last working day of every month M nn specifies that the job will be executed every nn day of the month or previous working day of each month Y specifies that the job will be executed last working day of the year Y nn specifies that the job will be executed the last working day of the nn'th month A specifies that the job will be executed on an adhoc basis. You should manually specify a date on which the job has to be executed.

Features » BATCH

Frequency

BATCH, BNK/FT.BULK.PAYMENT		
Process Status	0	READY
Batch Environment	F	FOREGROUND
Job Name.1	FT.PROCESS.BULK.PAYMENT.PRE	FT PROCESS BULK PAYMENT PRE
Frequency.1	D	DAILY
Job Status.1	0	READY
Last Payment.1	15 APR 2021	15 APR 2021
Job Name.2	FT.PROCESS.BULK.PAYMENT	FT PROCESS BULK PAYMENT
Verification.2.1	FT.PROCESS.BULK.PAYMENT.PRE	FT PROCESS BULK PAYMENT PRE
Frequency.2	D	DAILY
Job Status.2	0	READY
Last Payment.2	15 APR 2021	15 APR 2021
Lock Retry Count	0	
Curr No	1	
Inputter.1	SY_COMPIN.BATCH	
Date time.1	11 MAR 22 03:07	
Authoriser	16_TRAINS11_OFS_MB.OFS	
Company	GB0010001	Model Bank

Frequency

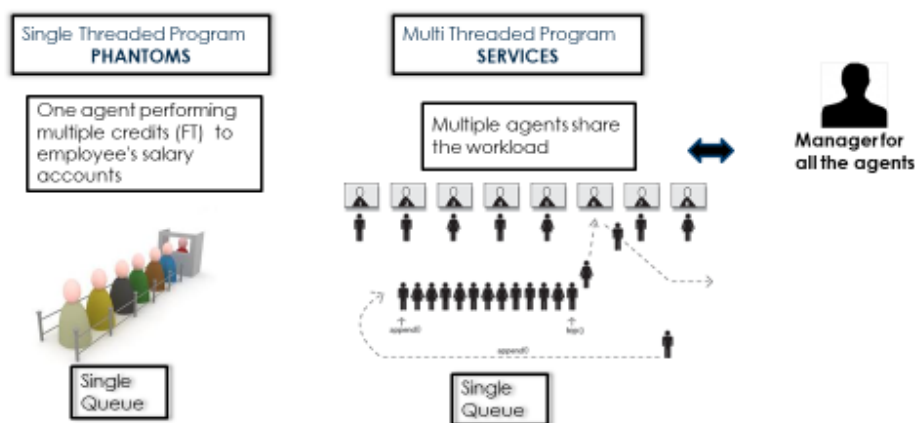
- D
- D nn
- W
- M
- M nn
- Y
- Y nn
- A

Slide 13

When one program does the work of crediting salary for all employees it will take n amount of time. There is one list of IDs that has to be processed and each ID from this list is processed by the program. Single threaded programs are called PHANTOMS in Transact almost all PHANTOMS have been rewritten as services. SERVICE – instead of one program performing the work of crediting the account with salary, there can be multiple processes/agents doing the same work. There is only ONE LIST OF IDs. Each agent can pick up a different ID and do the work. As multiple agents are doing the work, it gets completed faster. When there are multiple agents, there is a manager process to manage all the agents.

Features » TSA.SERVICE

- A Service is executed as a multithreaded process. Let us understand this with an example.



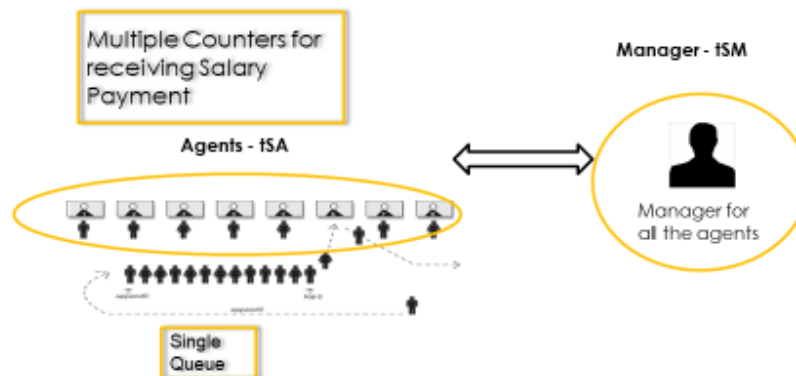
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Slide 14

There are two programs tSM and tSA to run the multi-threaded routines.

Features » tSM & tSA

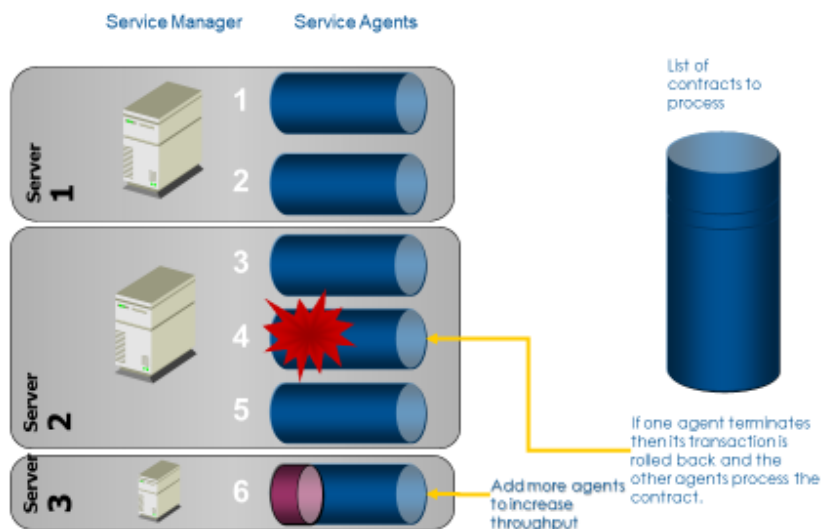
- There are two programs available to run a multi-threaded routine:
 - tSM
 - tSA



Slide 15

In Transact multithreaded routines are executed by service agents. The above screenshot shows that many agents can be started on a server. There is one tSM for each server. The list of contracts is allocated to the different agents. To increase performance, we can add more agents. If any agent fails, the contract being processed by the agent will continue to be available in the list of contracts and can be processed by another agent.

| Features » How jobs are multithreaded



Slide 16

REVIEW TIME : The job of the tSM is to Read the TSA.SERVICE records with SERVICE.CONTROL field set to START . Read the relevant TSA.WORKLOAD.PROFILE records. Check if the required number of agents are running. If yes, check to see if they are performing their job. If no, instruct the user to start the required number of agents. **TIME OUT :** Every agent (tSA) is required to update the manager (tSM) that it is performing the task assigned to it. The tSM is designed to wait for a stipulated time for each agent to report that it is alive and working. The tSM will term the agent as “dead”, if it doesn't get a reply within the stipulated time. It will stop the tSA process at the operating system level (kill the process) and start a substitute process. The time given for every agent to report to the tSM that it is alive, and working is set in the field TIME.OUT (Specify the value in the COB record).

| Features » Manager & Agents

tSM	tSA
tSM stands for Transact Service Manager	tSA stands for Transact Service Agent
tSM itself is pre-configured in TSA.SERVICE	tSM is the first tSA , i.e, tSA 1
tSM manages all services that must be started or are already running	Multiple tSA's can execute simultaneously – each tSA running one copy of service
REVIEW TIME - Check if the required number of agents are running. If yes, check to see if they are performing their job. If no, instruct the user to start the required number	TIME OUT - Every agent (tSA) is required to update the manager(tSM) that it is performing the task assigned to it
The tSM will term the agent as “dead”, if it doesn't get a reply within TIME.OUT	The tSA process at the operating system level gets killed and a substitute process is started

Slide 17

What you see here is the COB record in TSA.SERVICE application. Description - field holds the text that describes the TSA.SERVICE record. Server Name - field holds the IP address or the host name of the server where this service TSM needs to be executed. There can be only one tSM per Transact application server. When MS (Multi Server) product is installed, this field can be multi valued to include the ip of all the servers involved. Work Profile field - holds the ID of the TSA.WORKLOAD.PROFILE application which in turn contains the number of agents required to run this service. For the record TSM, the number of agents should always be 1. This field forms a multi-value set along with Description and Server Name fields. User field - holds a valid Transact username. It has no significance for this record TSM. Has significance for the record COB. Service Control - This field holds the control the user wishes to exercise upon the current service. Currently it can hold the following values. START :- The Service is ready to start and can be kicked off. STOP :- The Service must be stopped. The service if it had been running will be terminated. AUTO - If AUTO is specified, then the service is ready for kicking off and does not require manual input of START . The service will be available for starting and can be stopped at the end of the process. DEBUG - The Service can be launched in debug mode with service name and -DEBUG as arguments. This option is mainly introduced for enabling debugging facility in container environments where there is no tSM.

Features » TSA.SERVICE

- The number of agents is decided based on the load handled by a service. This is set in TSA.WORKLOAD.PROFILE.

TSA.SERVICE, COB

Description.1

FOR COB

Server Name.1

Work Profile.1

MB.COB.AGENT

Server Status.1

User *

INPUTTER

Service Control

☒ Stop
 ☐ Start
 ☐ Debug

Review Time

TSA.WORKLOAD.PROFILE MB.COB.AGENT

Description

FOR COB

Agents Required.1

10

Slide 18

This is the application used to specify the number of agents. The number of agents in TSA.WORKLOAD.PROFILE can be set based on time. Agents can be increased/decreased based on time (Transact application server time). Assume a scenario where in you know that the server will be busy at 11PM as that is when transactions from an interface will be coming in. You may choose to reduce the number of agents running COB during that period so that the server be used for the interface as well. Define this in the TSA.WORKLOAD.PROFILE application. TSA.WORKLOAD.PROFILE is the application which defines the number of tSAs required. The ID of a record in this application can be any alphanumeric text. The number of agents in TSA.WORKLOAD.PROFILE can be set based on time if required. TIME and AGENTS.REQUIRED : These fields are associated multi value set which holds the specific time and the number of agents required at that point in time. The number of agents that can run on server is directly proportional to the number of processors available. Just because Transact supports multi threaded services doesn't mean you can start a 100 tSA's in the background. It will kill the server. The thumb rule to be followed is a maximum of 2 tSA's per processor available.

| Features » TSA.WORKLOAD.PROFILE

- Used to set the number of agents required
- No of agents can also be varied with time

TSA.WORKLOAD.PROFILE		OFS.MESSAGE.SERVICE
Description	OFS TSA message service	
Time.1	2100	
Agents Required.1	3	
Time.2	2200	
Agents Required.2	1	

Slide 19

Parameter file Has only one record – ID is SYSTEM. Review Time Is used if the REVIEW.TIME field in the TSA.SERVICE record is not specified. Default value is 60. Death Watch Is used if the TIME.OUT field in the TSA.SERVICE record is not specified. Default value is 300.

| Features » TSA.PARAMETER

- TSA.PARAMETER holds the review time and time out, if not mentioned in TSA.SERVICE.
- Has only one record – ID is SYSTEM.

TSA.PARAMETER SYSTEM	
Review Time	60
Death Watch	999
Highest Agent	4
Stoppage Time	
Stop Count	
Select Death Watch	

Slide 20

You must first turn the service on using the SERVICE.CONTROL field in TSA.SERVICE. This change must be authorized. You must remember that this action does not start off the service in the background. It's only a flag to say that the service will be picked up and executed. Launch the service using the TAFJEE/Execute servlet This will cause all service records with either START or AUTO to be started.

| Features » How will you start / Stop a service ?

- To start a service, the TSA.SERVICE record must be edited and the field SERVICE.CONTROL set to START or AUTO.

The screenshot shows the 'TSA.SERVICE, TSM' form. The 'Description.1' field contains 'TSM Record for running upgrade as service'. The 'Server Name.1' field is empty. The 'Work Profile.1' field is set to 'TSM'. The 'Server Status.1' field is empty. The 'User =' field is set to 'INPUTTER'. The 'Service Control' field has four radio buttons: 'Stop', 'Start' (which is selected), 'Auto', and 'Debug'.

The screenshot shows the 'TSA.SERVICE, BNK/PRINT.OUT' form. The 'Description.1' field contains 'DE FORMATTING SERVICE'. The 'Server Name.1' field is empty. The 'Work Profile.1' field is set to 'DE FORMATTING SERVICE'. The 'Server Status.1' field is empty. The 'User =' field is set to 'INPUTTER'. The 'Service Control' field has four radio buttons: 'Stop', 'Start' (which is selected), 'Auto', and 'Debug'. The 'Review Time' field is set to '10'.

Slide 21

Execute Servlet TAFJEE_WAR_TAFJ contains a servlet which could be used to post message to the JMS queue t24EXECQueue. Navigate to this URL - <http://localhost:9089/TAFJEE/Execute> To post START.TSM submit START.TSM in the form. Remark: The following property must be set correctly to run TAFJ within an application server context. `temn.tafj.runtime.phantom.as.process = false`

Features » TAFJEE/Execute servlet

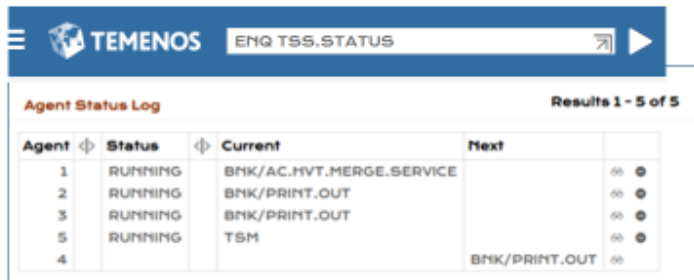
The diagram illustrates the process of executing the TAFJEE/Execute servlet. It shows a browser window with the URL `localhost:9089/TAFJEE/`. The main content area displays the 'TAFJEE - Home' page with a sidebar menu. The sidebar menu has two main sections: 'Diagnostic' and 'Execution'. The 'Execution' section is highlighted, showing options like 'Execute servlet', 'Entry points', and 'DBTools'. The 'Execute servlet' option is selected, leading to the 'Execute phantom' form. This form has a 'Command' field containing 'START.TSM' and an 'Execute command' button. Below the form is the 'Execution logs' section, which shows a list of log entries. The logs include timestamps, log levels (INFO), and messages related to the execution of the 'START.TSM' command. A text box on the right indicates the login credentials: 'uname t24user' and 'Password t24User123!'.

Slide 22

TSA.STATUS and BATCH.STATUS are LIVE files that capture information about the TSAs. If the service is not running, we can clear the files using `DELETE F_TSA_STATUS;` `DELETE F_BATCH_STATUS;` From the SQL prompt. Note- this command can be used only in training environment.

Features » TAFJEE/Execute servlet

- Check if the service is running using command **ENQ TSS.STATUS**



The screenshot shows the TEMENOS Agent Status Log interface. At the top, there is a search bar with the text "ENQ TSS.STATUS" and a search icon. Below the search bar, the title "Agent Status Log" is displayed, followed by "Results 1 - 5 of 5". The main content is a table with the following columns: Agent, Status, Current, Next, and an empty column. The table contains five rows of data.

Agent	Status	Current	Next	
1	RUNNING	BNK/AC.MVT.MERGE.SERVICE		⌂ ⌂
2	RUNNING	BNK/PRINT.OUT		⌂ ⌂
3	RUNNING	BNK/PRINT.OUT		⌂ ⌂
5	RUNNING	TSM		⌂ ⌂
4			BNK/PRINT.OUT	⌂

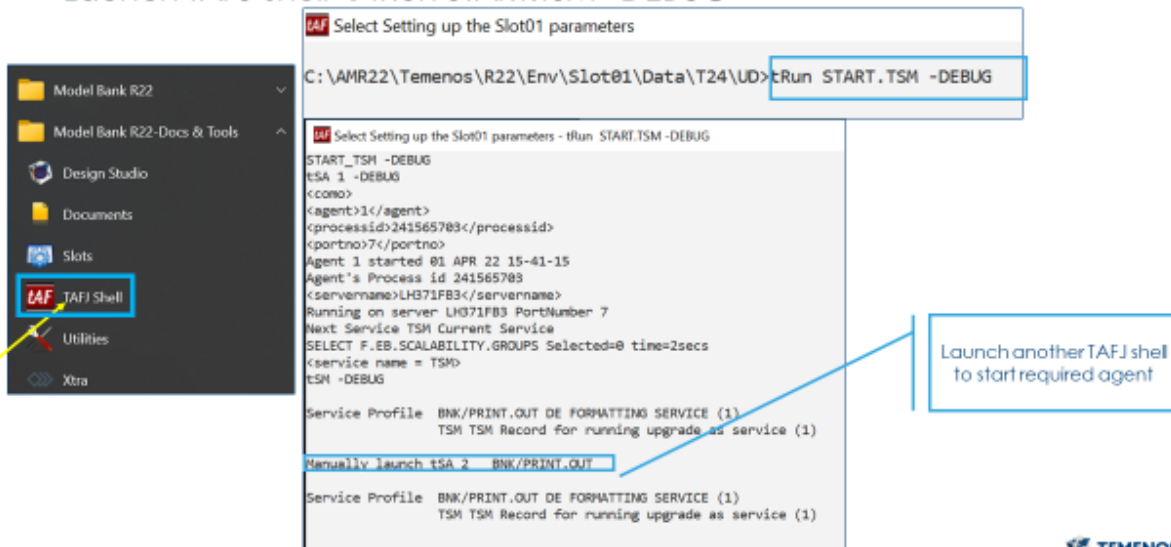
```
sql> delete f_tsa_status;  
(Update count: 32, 7 ms)  
sql> delete f_batch_status;  
(Update count: 0, 1 ms)  
sql>
```


Slide 23

Launch the TAFJ shell from the Docs and Tools Menu in Model Bank. At the TAFJ shell prompt, give the command `tRun START.TSM -DEBUG` to start the TSM in DEBUG mode. After the agents are assigned jobs, in DEBUG mode, type `tRun tSA <agent number>` at the TAFJ shell prompt to start the agents to the service. After the job is finished successfully, a message 'Agent Stopped' will be displayed and the cursor will go back to TAFJ shell prompt. When a service stops, the field `SERVICE.CONTROL` is set to `STOP` automatically.

Features » start service from TAFJ Command Line

- Launch TAFJ shell → `tRun START.TSM -DEBUG`



The screenshot shows the TAFJ shell interface. On the left, a sidebar menu lists various options: Model Bank R22, Model Bank R22-Docs & Tools, Design Studio, Documents, Slots, **TAFJ Shell** (highlighted with a red box and a yellow arrow), Utilities, and Xtra. The main window displays the command prompt and the output of the `tRun START.TSM -DEBUG` command. The output shows the TSM starting in debug mode, assigning agents to jobs, and displaying service profiles. A blue box highlights the command `tRun START.TSM -DEBUG` in the command prompt, and another blue box highlights the command `Manually launch tSA 2 BNK/PRINT.OUT` in the output. A callout box with an arrow pointing to the second command says "Launch another TAFJ shell to start required agent".

```

TAFJ Select Setting up the Slot01 parameters
C:\AMR22\Temenos\R22\Env\Slot01\Data\T24\UD>tRun START.TSM -DEBUG

TAFJ Select Setting up the Slot01 parameters - tRun START.TSM -DEBUG

START_TSM -DEBUG
tSA 1 -DEBUG
<cono>
<agent>1</agent>
<processid>241565703</processid>
<portno>7</portno>
Agent 1 started 01 APR 22 15-41-15
Agent's Process id 241565703
<servername>LHB71FB3</servername>
Running on server LHB71FB3 PortNumber 7
Next Service TSM Current Service
SELECT F.EB.SCALABILITY.GROUPS Selected=0 time=2secs
<service name = TSM>
tSM -DEBUG

Service Profile BNK/PRINT.OUT DE FORMATTING SERVICE (1)
TSM TSM Record for running upgrade as service (1)

Manually launch tSA 2 BNK/PRINT.OUT

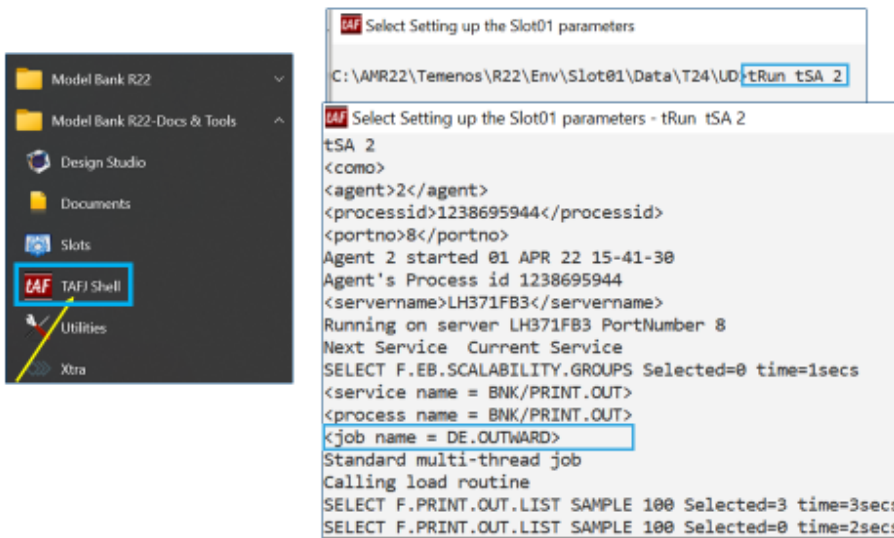
Service Profile BNK/PRINT.OUT DE FORMATTING SERVICE (1)
TSM TSM Record for running upgrade as service (1)
    
```

Slide 24

Launch the TAFJ shell from the Docs and Tools Menu in Model Bank. At the TAFJ shell prompt, give the command `tRun START.TSM -DEBUG` to start the TSM in DEBUG mode. After the agents are assigned jobs, in DEBUG mode, type `tRun tSA <agent number>` at the TAFJ shell prompt to start the agents to the service. After the job is finished successfully, a message 'Agent Stopped' will be displayed and the cursor will go back to TAFJ shell prompt. When a service stops, the field `SERVICE.CONTROL` is set to `STOP` automatically.

Features » start service from TAFJ Command Line

- Launch TAFJ shell → `tRun tSA 2`



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TEMENOS
Learning Community

Slide 25

The SERVICE.CONTROL in TSA.SERVICE can also be set to AUTO if required. Such services need not be started/stopped manually.

| Auto Services

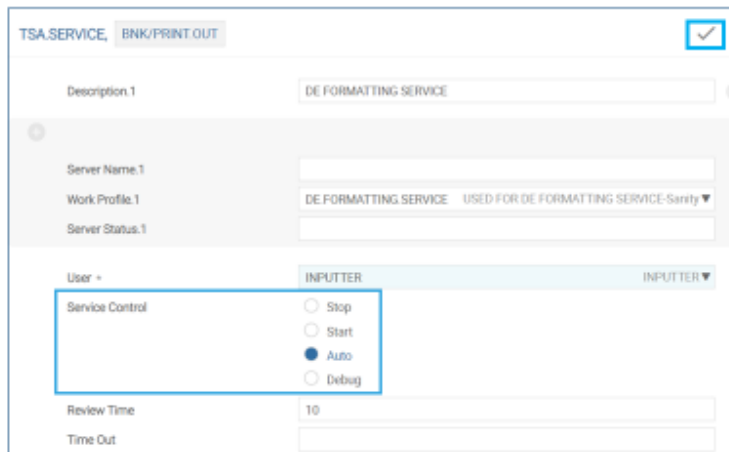
- What are Auto Services?
 - A bank needs to keep communicating with its customers. One of the ways to communicate is by sending Mailers.
 - Different departments in a bank will be compositing these mailers.
 - There will be a central service which will be disbursing the mails on behalf of the bank.
 - Is there a need to start and stop this service: NO.
 - It must be triggered once a mailer is created. So, this service needs to poll on a mailer box(in a software system a file) and keep running continuously. Such a service is called an auto service.

Slide 26

An auto service is triggered once the TSA.SERVICE of TSM is set to start. We need not explicitly set SERVICE.CONTROL of an auto service to "Start".

| Auto Services » Example

- An auto service is triggered once the TSA.SERVICE of TSM is set to start. We need not explicitly set SERVICE.CONTROL of an auto service to START



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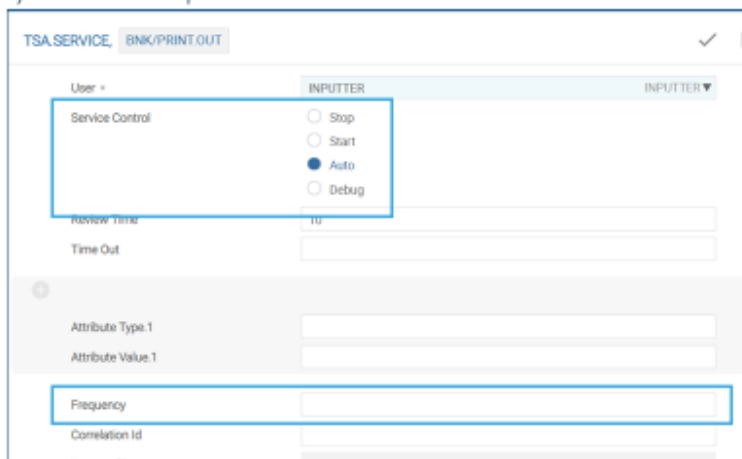


Slide 27

Services can be configured as a periodic activity using the FREQUENCY field. This field holds a value in the format Date Time Frequency- Date is the system date which is entered in Transact date format Time is entered in the format HH:MM(Transact hour format) Frequency is entered in the format 1D,2D etc., EXAMPLE : 20100215 14:00 1D TSM normally builds the service profile only for the services set to START or AUTO. But when the frequency is specified for the service and SERVICE.CONTROL is set as 'STOP', the TSM will check whether the specified frequency has been reached considering the time as well as frequency and will start the service. When the service is stopped, the frequency will be cycled to the next date. This field is not allowed for an AUTO service. The service control cannot be changed if the frequency is present. This is mostly used for archiving.

| Auto Services » Scheduling of Services

- Services can be configured as a periodic activity using the FREQUENCY field in TSA.SERVICE.
- Frequency cannot be specified for auto services



The screenshot shows the 'TSA.SERVICE' configuration form for 'BNK/PRINT OUT'. The 'Service Control' field is set to 'Auto'. The 'Frequency' field is highlighted with a red box, indicating it is a required field for periodic activities. The 'Review time' field is set to '10'. The 'Time Out' field is empty. The 'Attribute Type.1' and 'Attribute Value.1' fields are empty. The 'Correlation Id' field is empty.

Slide 28

Agent ID is the allocated Agent Number for the service T24.SESSION.NO is updated with a unique number that identifies every session in Transact. A number between 1 -99999 is allocated when a session is started in a port to uniquely identify that session in Transact(both for online and COB). A live file called T24.SESSION gets updated with record key being the T24.SESSION.NO and contains information regarding the agent to which this session number was assigned. A record in this application gets deleted, when the corresponding process terminates(session expires). Server Name is the name of the server where the agent is running. Note that details of tSAs and tSMs across application servers will be updated here. Agent Status is the status of the agent. It can hold either of the three values. RUNNING – Agent performing the job STOPPED – Agent has stopped properly. DEAD – Agent did not report within the time given and hence the tSM has marked the agent as DEAD Process Id is the Operating system level process ID for the agent process that is running Current Service is the service the agent is performing at the moment. Note agent 1 is performing the service TSM and the others are performing the service COB Next Service is the service the agent is supposed to perform next. While COB is running, if you wish to stop COB, set the field SERVICE.CONTROL to STOP for the record COB in the TSA.SERVICE file. Then, check the NEXT.SERVICE value for the agents running COB, it will be set to STOP denoting that the agents needs to stop once they complete the current transaction Job Progress is the status of the jobs being performed. Details of the status is not in the scope of this learning unit.

Monitoring Services » TSA.STATUS

- TSA.STAUS is a live application that holds information about the TSAs

Contract Screen

TSA.STATUS =

Id	Server	Agent status	Last contact	Process id
1	LH371FB3	RUNNING	17:30:58	1885471115
2	LH371FB3	RUNNING	18:27:29	33882515
3		STOPPED	18:05:26	
4		STOPPED	18:05:27	
5		STOPPED		
OLTP10170	LH371FB3	RUNNING	09:51:34	1899430314

Slide 29

You know tSM launches and controls the tSAs. How does an end user know what is happening with the tSAs? Use the enquiry AGENT.STATUS. This enquiry picks up the information for display from the TSA.STATUS application.

Monitoring Services » Enquiry AGENT.STATUS

- Enquiry AGENT.STATUS shows the activity of every agent.

TEMENOS

END AGENT STATUS

9

Results 1 - 9 of 9

TSA STATUS INFORMATION

AGENT ID	TSA SESSION NO	SERVER NAME	AGENT STATUS	PROCESS ID	CURRENT SERVICE	NEXT SERVICE	JOB PROGRESS	OPENS	READS	WRITES	INPUTS	EXECUTES	DELETES	CLEARFILES	MEMORY
1	54990	UJ071F03	RUNNING	241565703	TSM		0	225	201	11	0	4	1	0	39.14M
2	76382	UJ071F03	RUNNING	1238995944	BNK/PRINT OUT		0	80	57	3	0	0	0	0	58.07M
3			STOPPED				0	82	61	4	0	0	0	0	61.15M

LAST MESSAGE

COMO NAME

__1_01 APR 2022_15:43:16:706_SELECT F.TSA.SERVICE WITH FREQUENCY NE "" OR SERVICE CONTROL EQ "START" OR EQ "AUTO" OR @ID EQ "TSM" Selected=5 time=25secs

ISA_1_20220401_15-41-15

_BNK/PRINT OUT_DE OUTWARD_2_01 APR 2022_15:43:24:576_SELECT F PRINT OUT LIST SAMPLE 100 Selected=0 time=2secs

ISA_2_20220401_15-41-30

Slide 30

COMO.NAME in AGENT.STATUS holds the name of the log file. Log files are stored under a directory COMO. Use any text editor to view the content of the log files. If an agent is restarted, there will be two log files for that agent as the log file ID is based on date and time. Log files do not get cleared automatically from the COMO.

Monitoring Services » COMO

OS (C:) > AMR22 > Temenos > R22 > Env > Slot01 > Logs > T24 > como

```
tSA_1_20220401_154116
1 COMO tSA_1_20220401_15-41-15 established Fri Apr 01 15:41:16 IST 2022
2 START_TSM -DEBUG
3 tSA 1 -DEBUG
4 <como>
5 <agent>lc/agent>
6 <processid>241565703</processid>
7 <portno>7</portno>
8 Agent 1 started 01 APR 22 15-41-15
9 Agent's Process id 241565703
10 <servername>LH371FB3</servername>
11 Running on server LH371FB3 PortNumber 7
12 Next Service TSM Current Service
13 SELECT F.ES.SCALABILITY.GROUPS Selected=0 time=2secs
14 <service name = TSM>
15 tSM -DEBUG
16
17 Service Profile BNK/PRINT.OUT DE FORMATTING SERVICE (1)
18 TSM TSM Record for running upgrade as service (1)
19
20 Manually launch tSA 2 BNK/PRINT.OUT
21
22 Service Profile BNK/PRINT.OUT DE FORMATTING SERVICE (1)
23 TSM TSM Record for running upgrade as service (1)
24
25 Manually launch tSA 2 BNK/PRINT.OUT
26 0_1_01 APRIL 2022_15:41:16:100_SELECT F.TSA.SERVICE WITH FREQUENCY ME "" OR SERVICE.CONTROL EQ "START" OR EQ "AUTO" OR #ID EQ "TSM" Selected=5
time=0secs
```


Lesson Summary

In this lesson I described about ... Services Process and Jobs Features Auto Services Monitoring Services

| Lesson Summary



COB

Lesson Overview

In this lesson I am going to describe about COB Service in Transact with particular emphasis on :
Introduction to COB Stages of COB Batch Stage Execute COB Monitor COB

| Lesson Overview



Slide 5

1. COB stands for Close Of Business 2. COB marks the end of all financial transactions for the day and rolls forward the Transact date to the next business day 3. COB can be run on every working day of the bank Let's understand one of the processes done during COB with a simple example Steve has taken a loan from a bank at a 10% rate of interest, payable back to the bank in a year. This amount will be paid by him in 12 monthly installments. The bank would like to keep a daily track of the interest amount they are due to receive from Steve at the end of the month. So, they calculate the amount payable by him everyday. This is done to keep the Profit and Loss/Balance sheet current. Now, this is just for one customer. Imagine the bank has given 100,000 loans with different interest rates for different periods of time to individuals like Steve. It's an arduous task for the bank to do the whole process manually. So, they automate it. However, the amount payable to the bank need not be calculated during normal working hours of the bank (because they will be busy doing other day to day activities of the bank). So, they do it at the end of the day. The process of calculating the interest payable, but not yet fallen due for payment, is called Accrual. Therefore, accrual is one of the processes that is done during Transact COB (Close of Business)

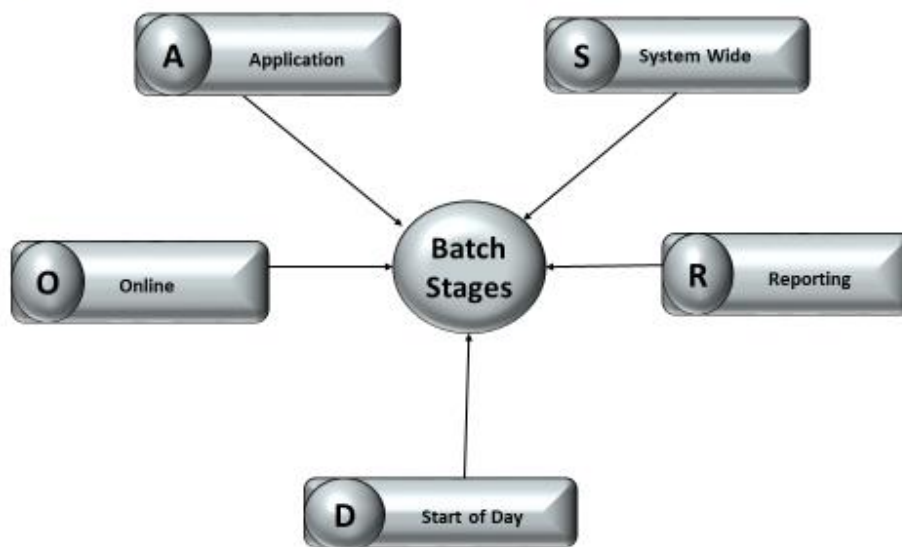
| Introduction

- COB stands for Close Of Business.
- Marks the end of all financial transactions for the day and rolls forward the Transact date to the next business day.
- COB can be run on every working day of the bank.
- Let's understand one of the processes done during COB with a simple example.

Slide 6

COB runs in five stages, A-S-R-D-O. A stands for Application stage. It involves all the individual application processes. E.g.,: Forex, Funds Transfer, etc. S stands for System Wide stage. It involves jobs that are common to Transact. E.g.,: Interest accrual, Interest capitalization. R stands for Reporting stage. It involves generation and printing of reports. E.g.,: Trial Balance, Transaction journal etc. D stands for Start Of Day stage. It involves all the start of day operations and date change. E.g.,: Executing standing orders etc. O stands for Online stage. It involves any non-critical reports and processes which can be run after the system has returned to online mode. Eg: Jobs to clean up the temporary work files, etc.

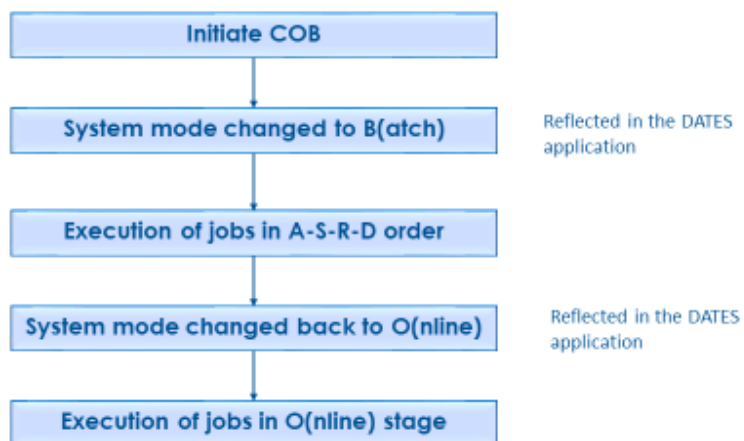
| Stages of COB



Slide 7

When you initiate COB, either in debug mode or interactive mode The System Mode is changed from Online mode to Batch mode. This is reflected in the DATES application. This is to indicate that COB is currently running on the system The Execution of COB jobs take place in A, S, R, D order The System is changed to Online mode at the start of “O” stage of COB. This is also reflected in the DATES application Finally, all the jobs in the online stage are completed.

| Stages of COB



Slide 8

Each Batch stage is suffixed with a 3-digit sequence number, between 000-999, which will tell the system the order of execution of COB e.g. A000, A100, S000, R099 etc., A Batch Stage consists of one or more processes. In Transact, these COB Processes are defined in an application called BATCH Each Process also consists of one or more Jobs that perform a specific task. All jobs within a process are executed in the order they are defined and NEVER simultaneously. For E.g. In the diagram that you see, Job 2, will never get executed before Job 1 is complete.

| Stages of COB

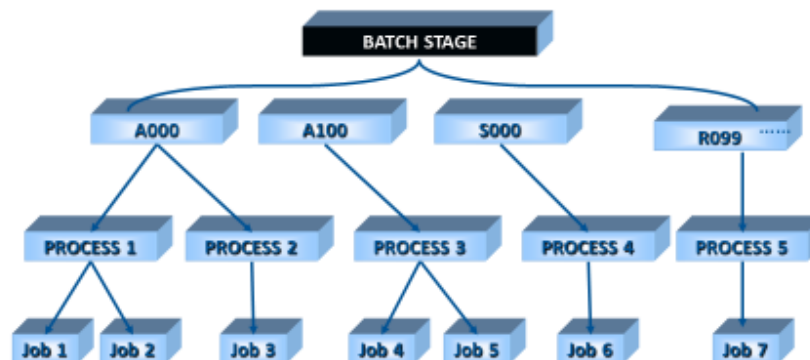
- What?
 - Each batch record, i.e. , a process, needs to be executed in a particular order, for COB.
 - Each process consists of one or more jobs that perform a specific task.
 - All jobs within a process are also executed in the order they are defined and NEVER simultaneously.
- Why?
 - Stages tell the system the order of execution of COB.
 - Each Batch stage is suffixed with a 3-digit sequence number, between 000-999. Ex : A000, A100, S000, R099 , D017

Slide 9

Each Batch stage is suffixed with a 3-digit sequence number, between 000-999, which will tell the system the order of execution of COB e.g. A000, A100, S000, R099 etc., Batch Stage consists of one or more processes. In Transact, these COB Processes are defined in an application called BATCH Each Process consists of one or more Jobs that perform a specific task. All jobs within a process are executed in the order they are defined and NEVER simultaneously. For E.g. In the diagram that you see, Job 2, will never get executed before Job 1 is complete. Why should the processes be executed in a particular order? As a continuation of the previous scenario of accrual, it is time for Steve to make his first installment of monthly interest. The bank has been calculating the interest everyday and, on that day, they know the exact amount payable by Steve. The calculated interest amount is now accounted as due for payment and is called Interest Capitalization. Steve should now arrange to pay through his bank account/cash/card/cheque etc., to the bank's account. Interest Capitalization is also one of the processes executed during COB. Now, lets try to answer this question Should I do Interest accrual first and then Interest Capitalization, or vice versa? The obvious answer would be first interest accrual and then interest capitalization otherwise the math would never be correct. Therefore, all jobs are executed in a specified order during COB.

| Batch Stage » Illustration

- Why should the processes be executed in a particular order?



Slide 10

As, you just learnt, a record in the BATCH application is called a process. All processes are prefixed with the bank mnemonic. Processes to be executed during COB must belong to a Batch Stage. Processes that do not belong to a Batch Stage will be executed as services.

| Batch Stage » Records

- List the processes in your Transact area – BATCH L.

BATCH - Default List		Results 11
Batch Process	Batch Stage	
<u>BNK/AA.CONVERT.NEXT.ACTIVITY</u>		
<u>BNK/AA.CUSTOMER.RELATIONSHIP.TRACKER</u>		
<u>BNK/AA.EOD.POOL.ADMINISTRATION</u>	S025	Process linked to a Batch stage, is a part of COB.
<u>BNK/AA.EOD.PROCESS</u>	S016	
<u>BNK/AA.EXPAND.ETD.SERVICE</u>		
<u>BNK/AA.INTEGRITY.CHECK</u>		Process without a stage, is a service
<u>BNK/AA.INTERNAL.CAP.PROCESS</u>		
<u>BNK/AA.INTRADAY.PROCESS</u>		
<u>BNK/AA.ONLINE.SERVICE</u>		
<u>BNK/AA.POOL.ORCHESTRATION.PROCESS</u>		



Batch Stage » Records

Batch Stage Contains a character to denote the Batch Stage and a 3-digit number to denote the sequence within the stage. Sequence numbers are in the range 000 - 999. The first Batch stage executed during COB is A000 and the last is O999. Job Name Contains the name of the routine to be executed. All routines defined here will have an entry in PGM.FILE with TYPE set to 'B' Frequency specifies the frequency at which the batch job has to be executed. It can have values as follows. D specifies that the job will be executed every working day. D nn specifies that the job will be executed every nnth working day W specifies that the job will be executed on a weekly basis, that is every Friday M specifies that the job will be executed on the last working day of every month M nn nn represents the offset added to the first day of every month to arrive at Next Run date. Y specifies that the job will be executed last working day of the year Y nn specifies that the job will be executed the last working day of the nn'th month A specifies that the job will be executed on adhoc basis. You should manually specify a date on which the job has to be executed.

| Batch Stage » Records

BATCH, BNK/COB.INITIALISE		
Batch Stage	A000	APPLICATION
Main Stage	Application	
Cob Stage	A-start	
Cob Stage Seq	A000	
Process Status	0	READY
Batch Environment	F	
	FOREGROUND	
Job Name.1	COB.EXECUTE.API	COB.EXECUTE.API
Frequency.1	D	DAILY
Job Status.1	0	READY
Last Payment.1	14 APR 2021	14 APR 2021
Job Name.2	COB.CHECK.EB.EOD.ERROR	COB.CHECK.EB.EOD.ERROR
Verification.2.1	COB.EXECUTE.API	COB.EXECUTE.API
Frequency.2	D	DAILY
Job Status.2	0	READY
Last Payment.2	14 APR 2021	14 APR 2021
Job Name.3	EB.MANAGE.SERVICES	Managing Services via COB
Frequency.3	D	DAILY
Job Status.3	0	READY
Last Payment.3	14 APR 2021	14 APR 2021
Lock Retry Count	0	

Frequency

D
D nn
W
M
M nn
Y
Y nn
A



Batch Stage » For a COB Process

In the field Verification, you can enter the name of the job (routine) which needs to have been executed successfully on that day before the current job is executed. For e.g. The job COB.CHECK.EB.EOD.ERROR will be executed only if the job COB.EXECUTE.API has been executed successfully. NOTE: Only jobs already defined in the current record can be entered in the field Verification. Jobs belonging to other Processes (Batch Records) cannot be entered in this field. 2. Next Run Date contains the date when the current job has to run next. Populated automatically on authorisation of the record (new jobs) or on successful execution of the job. NO date is populated for Daily jobs, as the jobs needs to be run daily. User input is required for adhoc jobs. 3. Last Run Date is a No Input Field and is populated on successful execution of the job during COB.

Batch Stage » For a COB Process

The screenshot shows the 'BATCH, BHK/COB.INITIALISE' form. It contains two main sections for job configuration. The first section is for 'Job Name.1' (COB.EXECUTE.API) and the second is for 'Job Name.2' (COB.CHECK.EB.EOD.ERROR). Annotations point to specific fields:

- NEXT.RUN.DATE:** Points to the 'Next Run Date.1' field, which is currently empty.
- LAST.RUN.DATE:** Points to the 'Last Payment.1' field, which is populated with '14 APR 2021'.
- VERIFICATION:** Points to the 'Verification.2.1' field, which is populated with 'COB.EXECUTE.API'.

Other visible fields include 'Batch Stage' (A200), 'Main Stage' (Application), 'Cob Stage' (A-start), 'Cob Stage Seq' (A000), 'Default Printer' (D), 'Process Status' (P), 'Batch Environment' (F), 'Department Code', 'Job Status.1' (D), 'Job Message.1', 'User.1', 'Select Ahead.1', 'Frequency.1' (D), 'Printer Name.1', 'Data.1', 'Frequency.2' (D), 'Printer Name.2', and 'Data.2'.

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Batch Stage » Process Status and Job Status

Process Status field holds the status of the Batch record. It can hold the following values 0 specifies that the process is in Ready state 1 specifies that the process is in Running state 2 specifies that the process has Completed successfully 3 specifies that the process has stopped due to some Error Job Status field Holds the status of the individual job (routine) . It can hold the following values 0 specifies that the job is in Ready state 1 specifies that the job is in Running state 2 specifies that the job has Completed successfully 3 specifies that the job has stopped due to some Error

| Batch Stage » Process Status and Job Status

STATUS CODES
0-READY
1-RUNNING
2-COMPLETED
3-ERROR

BATCH: BNR/COB/INITIALISE

Batch Stage	A000
Main Stage	Application
Cob Stage	A-start
Cob Stage Seq	A000
Default Printer	
Process Status	0
Batch Environment *	☐ P ☐ B
Department Code	

Job 1

Job Name.1 *	COB.EXECUTE.API	COB.EXECUTE.API
Verification.1.1		
Frequency.1 *	D	
Next Run Date.1	DD MMYY	
Printer Name.1		
Data.1.1		
Job Status.1	0	
Last Payment.1	14 APR 2021	14 APR 2021
Job Message.1		
User.1		
Select Ahead.1		

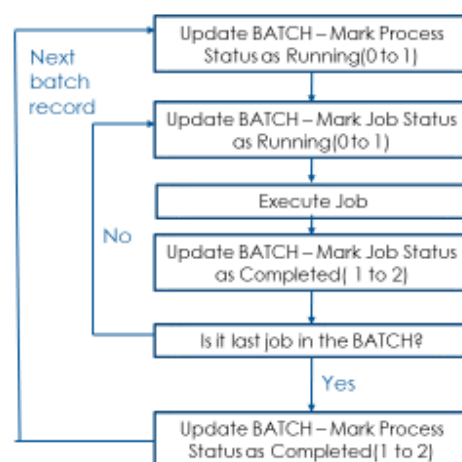
Job 2

Job Name.2 *	COB.CHECK.EB.EOD.ERROR	COB.CHECK.EB.EOD.ERROR
Verification.2.1	COB.EXECUTE.API	COB.EXECUTE.API
Frequency.2 *	D	
Next Run Date.2	DD MMYY	
Printer Name.2		
Data.2.1		

Slide 14

What you see here is a flowchart explaining the relationship between process status and job status. When a batch is loaded for execution, the process status of the batch record is changed from 0 to 1, (Ready to Running) The status of the first Job changes from 0 to 1 (Ready to Running) The job is executed After the job completes execution, the status of the job is changed from 1 to 2 (Running to Completed Successfully) marking successful completion of the job. Next the batch record is checked for any other available job. If there is another job available, it is processed in the same fashion as we saw just now. If there is no job available in the batch, the process status of the batch record is changed from 1 to 2 (Running to Completed Successfully) Likewise, the other batch records are processed.

| Batch Stage » Process Status and Job Status



Slide 15

The entire COB process is controlled by tSM tSM stands for Transact Service Manager As the name implies, it acts as a manager and controls all worker processes that perform COB Launches worker processes Checks if worker processes are running as scheduled If any of the worker processes are dead, launches substitute worker processes The actual COB process is executed by tSA's tSA stands for Transact Service Agent (Worker processes) Execute the jobs (routines) defined in BATCH record(s) tSM and tSA's are started on the Transact application servers

| Execute COB » tSM & tSA

- The entire COB process is controlled by tSM.
 - tSM stands for Transact Service Manager.
 - As the name implies, it acts as a manager and controls all worker processes that perform COB.
 - It launches worker processes.
 - It checks if the worker processes are running as scheduled.
 - If any of the worker processes are dead, it launches substitute worker processes.
- The actual COB process is executed by tSAs.
 - tSA stands for Transact Service Agent (Worker processes).
 - Execute the jobs (routines) defined in BATCH record(s).
 - tSM and tSAs are started on the Transact application servers.

Slide 16

TSA.SERVICE, As the name signifies, it is an application in Transact where the services are defined. In the application TSA.SERVICE, TSM is defined as a service. COB is also another service defined in TSA.SERVICE. Both services namely TSM and COB are pre-configured in the TSA.SERVICE application. You will see quite a few records in the TSA.SERVICE application other than TSM and COB. They are online services (Covered as part of the session 'Services')

| Execute COB » TSA.SERVICE

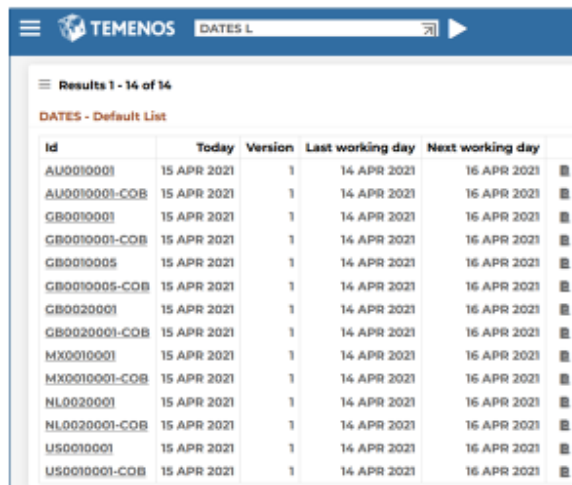
- TSA.SERVICE
- TSM is a service
- COB is a service
- Both services namely TSM and COB are pre-configured in TSA.SERVICE application.

Execute COB » Date Change

Transact maintains its own date per company in an application called DATES. This is the date that is referred to for all financial transactions. When do you think Transact decides it's the next day? This is done during COB, as it marks the end of all financial transactions for that day and makes the system ready for the next day.

Execute COB » Date Change

- There will be one record per company in the DATES application.
- ID of the record will be the company code.



Results 1 - 14 of 14

DATES - Default List

Id	Today	Version	Last working day	Next working day
AU0010001	15 APR 2021	1	14 APR 2021	16 APR 2021
AU0010001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021
GB0010001	15 APR 2021	1	14 APR 2021	16 APR 2021
GB0010001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021
GB0010005	15 APR 2021	1	14 APR 2021	16 APR 2021
GB0010005-COB	15 APR 2021	1	14 APR 2021	16 APR 2021
GB0020001	15 APR 2021	1	14 APR 2021	16 APR 2021
GB0020001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021
MX0010001	15 APR 2021	1	14 APR 2021	16 APR 2021
MX0010001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021
NL0020001	15 APR 2021	1	14 APR 2021	16 APR 2021
NL0020001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021
US0010001	15 APR 2021	1	14 APR 2021	16 APR 2021
US0010001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021

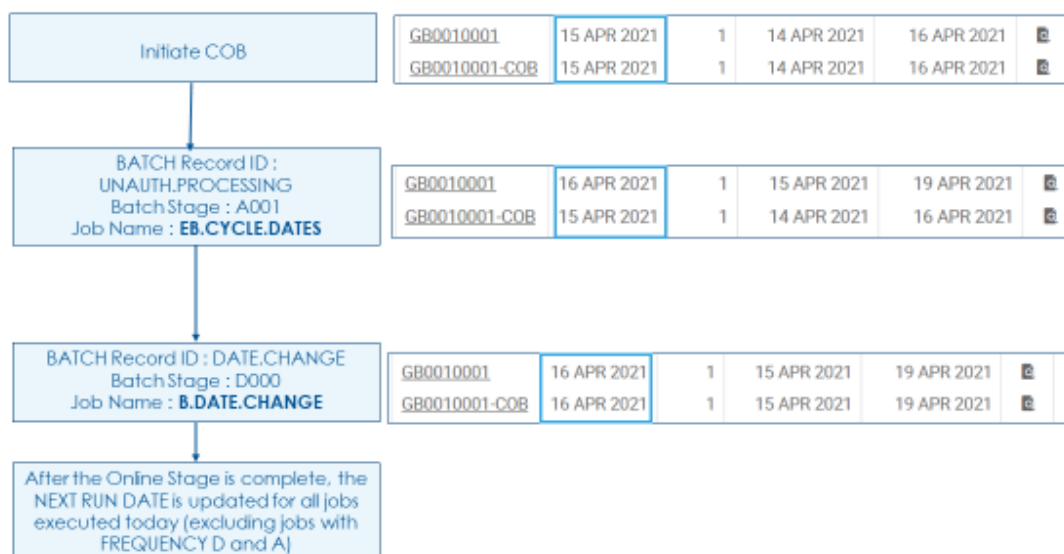
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Slide 18

Before COB is first run on a Transact environment, there will be only 1 record per company in the DATES application. As soon as we start COB, in the batch stage A001, a job called EB.CYCLE.DATES is executed. EB.CYCLE.DATES creates a record in the DATES application with @ID as <Company. Code>-COB. It also advances TODAY/LAST.WORKING.DAY by 1 day and changes the field NEXT WORKING DAY. This is done for the existing DATES record for a COMPANY with @ID <Company. Code>. E.g.,: Imagine we are running COB during the end of day of 17 Apr 2021, the following takes place - EU0010001 is the default DATES record for the Euro Company. Notice that the field TODAY has been advanced to 18 Apr 2021, field LAST WORKING DAY has been advanced from 16th Dec 2021 to 17th Dec 2021, field NEXT WORKING DAY has been changed to 19 Apr 2021) A record with @ID EU0010001-COB has been created with details that was held by DATES record - EU0010001, before COB was started. When COB is run, company specific cob jobs will reference DATES record <Company. Code>-COB and not the default DATES record for a company with @ID <Company. Code> 3. When COB reaches batch stage D000, a job called B.DATE.CHANGE is executed. This job changes fields TODAY, LAST WORKING DAY and NEXT WORKING DAY for the record <Company. Code>-COB in the DATES application 4. During the beginning of the online stage, the job BATCH.DATE.RESET is executed. The NEXT RUN DATE is updated for all jobs executed today (excluding jobs with FREQUENCY D and A). Also, all the PROCESS.STATUS and JOB.STATUS fields in the BATCH records are reset to zero. This is all part of the COB framework.

Execute COB » Date Change During COB



Slide 19

When NS is installed, the system will be available for input even when COB is in progress. All new transactions that are input during COB, will have the value date as next working day. The DATES record with @ID as COMPANY.CODE will be used.

Execute COB » NS module

- When NS is installed, the system will be available for input even when COB is in progress. All new transactions that are input during COB, will have the value date as next working day. The DATES record with @ID as COMPANY.CODE will be used.

FUNDS.TRANSFER, FT21106YHQMF	
Transaction Type	AC
Debit Account	20737
Debit Account (IN)	
Ccy Market Dr	1
Debit Currency	USD
Debit Amount	1.00
Debit Value Date	16 APR 2021
Debit Value Date (IN)	DD MMM YYYY
Debit Narrative	

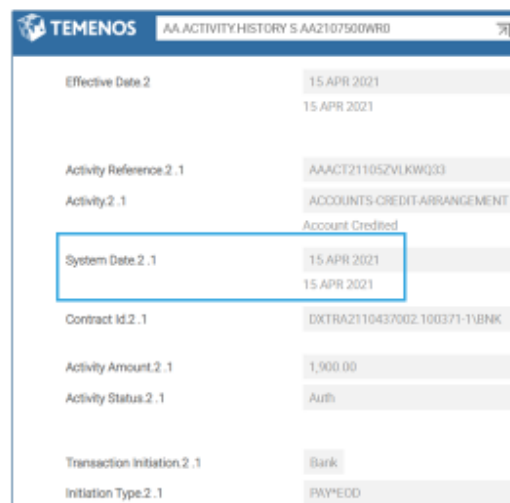
DATES - Default List				
Id	Today	Version	Last working day	Next working day
AU0010001	16 APR 2021	1	15 APR 2021	19 APR 2021
AU0010001-COB	16 APR 2021	1	15 APR 2021	19 APR 2021
GB0010001	16 APR 2021	1	15 APR 2021	19 APR 2021
GB0010001-COB	16 APR 2021	1	15 APR 2021	19 APR 2021
GB0010005	16 APR 2021	1	15 APR 2021	19 APR 2021
GB0010005-COB	16 APR 2021	1	15 APR 2021	19 APR 2021
GB0020001	16 APR 2021	1	15 APR 2021	19 APR 2021
GB0020001-COB	16 APR 2021	1	15 APR 2021	19 APR 2021
MX0010001	16 APR 2021	1	15 APR 2021	19 APR 2021
MX0010001-COB	16 APR 2021	1	15 APR 2021	19 APR 2021
NL0020001	16 APR 2021	1	15 APR 2021	19 APR 2021
NL0020001-COB	16 APR 2021	1	15 APR 2021	19 APR 2021
US0010001	16 APR 2021	1	15 APR 2021	19 APR 2021
US0010001-COB	16 APR 2021	1	15 APR 2021	19 APR 2021

Slide 20

Transact supports 24 hours non-stop processing. For this a product named NS (Non-Stop) must be installed. When NS is installed, the Transact system will be available for input even when COB is in progress. All new transactions that are input during COB, will have the value date as next working day. The DATES record with @ID as COMPANY.CODE will be used. For all records that are created as part of the COB process the DATES record with @ID COMPANY.CODE-COB will be used.

Execute COB » Non-Stop (NS) Module

- For all records that are created as part of the COB process the DATES record with @ID COMPANY.CODE-COB will be used.



TEMENOS AA ACTIVITY HISTORY S AA2107500WRD	
Effective Date 2	15 APR 2021
	15 APR 2021
Activity Reference 2.1	AAACT21105ZVLKWQ33
Activity 2.1	ACCOUNTS CREDIT ARRANGEMENT
	Account Credited
System Date 2.1	15 APR 2021
	15 APR 2021
Contract Id 2.1	DXTRA2110437002.100371-1YBNK
Activity Amount 2.1	1,900.00
Activity Status 2.1	Auth
Transaction Initiation 2.1	Bank
Initiation Type 2.1	PWY*EOD

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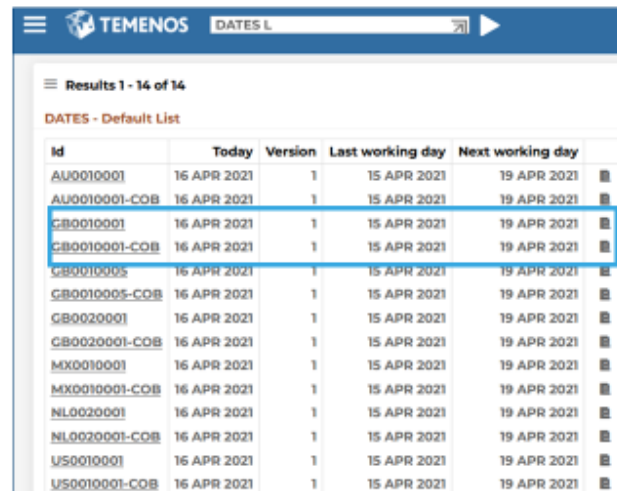


Slide 21

The record with ID COMPANY.CODE-COB is not deleted after COB is completed but is updated when COB is run the next time.

Execute COB » NS module

- The record with COMPANY.CODE-COB is not deleted after COB but is updated when COB is run next



Results 1 - 14 of 14

DATES - Default List

Id	Today	Version	Last working day	Next working day	
AU0010001	16 APR 2021	1	15 APR 2021	19 APR 2021	
AU0010001-COB	16 APR 2021	1	15 APR 2021	19 APR 2021	
CB0010001	16 APR 2021	1	15 APR 2021	19 APR 2021	
CB0010001-COB	16 APR 2021	1	15 APR 2021	19 APR 2021	
GB0010001	16 APR 2021	1	15 APR 2021	19 APR 2021	
GB0010001-COB	16 APR 2021	1	15 APR 2021	19 APR 2021	
GB0020001	16 APR 2021	1	15 APR 2021	19 APR 2021	
GB0020001-COB	16 APR 2021	1	15 APR 2021	19 APR 2021	
MX0010001	16 APR 2021	1	15 APR 2021	19 APR 2021	
MX0010001-COB	16 APR 2021	1	15 APR 2021	19 APR 2021	
NL0020001	16 APR 2021	1	15 APR 2021	19 APR 2021	
NL0020001-COB	16 APR 2021	1	15 APR 2021	19 APR 2021	
US0010001	16 APR 2021	1	15 APR 2021	19 APR 2021	
US0010001-COB	16 APR 2021	1	15 APR 2021	19 APR 2021	

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Slide 22

The first agent (tSA) to be started is called the tSM (Transact Service Manager). tSM monitors one or more agent processes called tSA (Transact Service Agent). Every time an agent is started, it will first check if it is the tSM. If not, then it will continue to work as an agent because it knows that an agent has already been assigned as the tSM. tSA calls a routine called S.JOB.RUN, that in turn invokes a routine EB.SORT.BATCH only ONCE. The work of EB.SORT.BATCH is to sort the COB jobs in 'A', 'S', 'R', 'D', 'O' order and give it to the tSAs.

Execute COB » Sorting of Jobs



Slide 23

Once you list the records present in the application EB.EOD.ERROR you will be able to see records that are date and company specific. All errors encountered as a result of running COB on a single day are stored in one record. If multiple errors are recorded during the same day, they are stored in the same EB.EOD.ERROR record but multivalued. Each error has a Detail Key populated against it which is nothing but the ID to the application EB.EOD.ERROR. The FIX.REQUIRED field is updated by the system to 'YES' , for all the errors that are crucial and are to be resolved before next COB.

| Execute COB » EB.EOD.ERROR

- Is there a file where all the errors encountered during COB can be viewed? Yes. The name of the file is EB.EOD.ERROR.
- The ID to a record in EB.EOD.ERROR is
<COMPANY.CODE>.<COBDATE> GB0010001.20220108.
- EB.EOD.ERROR contains one record per company per day running COB. The details of the individual errors can be found in a file called EB.EOD.ERROR.DETAIL.

Slide 24

Once you list the records present in the application EB.EOD.ERROR you will be able to see records that are date and company specific. All errors encountered as a result of running COB on a single day are stored in one record. If multiple errors are recorded during the same day, they are stored in the same EB.EOD.ERROR record but multivalued. Each error has a Detail Key populated against it which is nothing but the ID to the application EB.EOD.ERROR. The FIX.REQUIRED field is updated by the system to 'YES' , for all the errors that are crucial and are to be resolved before next COB.

| Execute COB » EB.EOD.ERROR

Contract Screen

EB.EOD.ERROR *

@Id Application Id Routine

GB0010001.20220404 GIFTS.VOUCHER-GIFTS.VOUCHER OPF

EB.EOD.ERROR GB0010001.20220404

Time Date.1 10:15:58 04 APR 2022

Description.1.1 MISSING FILE - , MNEMONIC = , IN.PARAMETER = , COMPANY = GB0010001 , CALL.ROUTINE = BATCH.JOB.CONTROL,S.JOB.RUN,ISA

Application Id.1 GIFTS.VOUCHER-GIFTS.VOUCHER

Routine.1 OPF

Detail Key.1.1 196718305336958.00

Fix Required.1 YES

Slide 25

For every error encountered during COB (for that day) there will be a record in the application EB.EOD.ERROR.DETAIL. The application EB.EOD.ERROR.DETAIL will contain the name of the Process, the name of the job and the routine from which the error was generated. The ID to a record in this application is held in the EB.EOD.ERROR file. The main use of this application is to track down the cause of the problem encountered during COB. This helps the consultant in fixing the problem before starting COB the next time.

| Execute COB » EB.EOD.ERROR.DETAIL

EB.EOD.ERROR GB0010001.20220404	
Time Date.1	10:15:58 04 APR 2022
Description.1 .1	MISSING FILE - , MNEMONIC = , IN.PARAMETER = , COMPANY = GB0010001 , CALL.ROUTINE = BATCH.JOB.CONTROL,S.JOB.RUN,tSA
Application Id.1	GIFTS.VOUCHER-GIFTS.VOUCHER
Routine.1	OPF
Detail Key.1 .1	196718305336958.00
Fix Required.1	YES

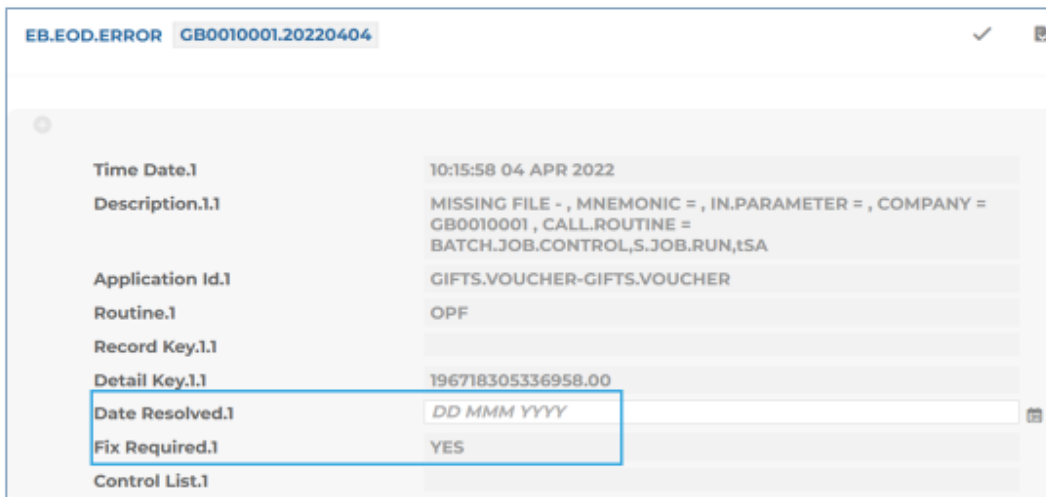
EB.EOD.ERROR.DETAIL 196718305336958.00	
Date And Time	10:15:58 04 APR 2022
Application Id	GIFTS.VOUCHER-GIFTS.VOUCHER
Routine	OPF
Error Code	200
System Date	05 APR 2022

Slide 26

COB cannot run the next day without resolving errors encountered during the previous run. Q. How do we tell Transact that an error has been resolved? A. Open the corresponding record in EB.EOD.ERROR in EDIT mode and set the field Date Resolved to the date when the error was fixed. Commit and authorise the record. This has to be done before COB is started the next day.

| Execute COB » EB.EOD.ERROR

- How do we tell Transact that an error has been resolved?



EB.EOD.ERROR GB0010001.20220404	
Time Date.1	10:15:58 04 APR 2022
Description.1.1	MISSING FILE - , MNEMONIC = , IN.PARAMETER = , COMPANY = GB0010001 , CALL.ROUTINE = BATCH.JOB.CONTROL,S.JOB.RUN,tSA
Application Id.1	GIFTS.VOUCHER-GIFTS.VOUCHER
Routine.1	OPF
Record Key.1.1	
Detail Key.1.1	196718305336958.00
Date Resolved.1	DD MMM YYYY
Fix Required.1	YES
Control List.1	

Slide 27

| Execute COB

- Check EB.EOD.ERROR. COB cannot be started if there are un resolved errors from the previous run.
- COB can be started from the Transact Browser using TAFJEE Servlet.
- COB can be started from the TAFJ Shell either in the background (phantom mode) or in the foreground/interactive/DEBUG mode.

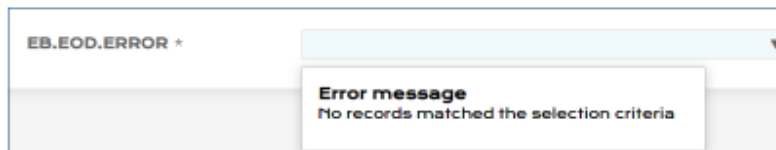
Slide 28

Check EB.EOD.ERROR. COB cannot be started if there are unresolved errors from the previous run. Set SERVICE.CONTROL for COB to Start. Launch the TAFJEE/Execute servlet using the URL <http://localhost:9089/TAFJEE/Execute>. Type START.TSM in the text box and click on Submit. When the COS TSS.LIST is launched, we can see the agents are running

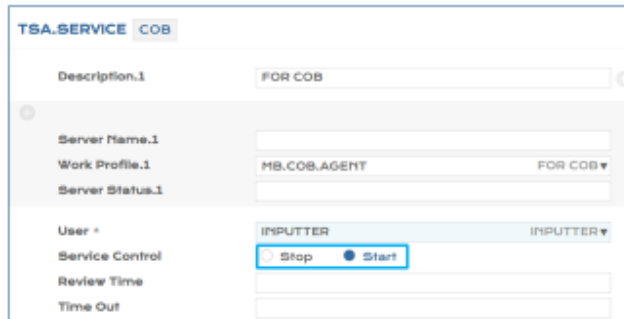
Execute COB » TAFJEE/Execute servlet

- Check EB.EOD.ERROR.

1



2



Slide 29

Check EB.EOD.ERROR. COB cannot be started if there are unresolved errors from the previous run. Set SERVICE.CONTROL for COB to Start. Launch the TAFJEE/Execute servlet using the URL <http://localhost:9089/TAFJEE/Execute>. Type START.TSM in the text box and click on Execute Command

Execute COB » TAFJEE/Execute servlet

<http://localhost:9089/TAFJEE>

3

Sign in
<http://localhost:9089>

Username:

Password:

Username t24user
Password t24User123!

4

WELCOME!
Welcome to TAFJEE services. If you can see this page, your configuration is correct.

Configuration

- [Presentation setup](#)
Configure TAFJEE properties

Diagnosis

- [Status](#)
Get details about your TAFJEE environment
- [Compile](#)
Get routine compilation details
- [Security check](#)
Run environment validation
- [Runtime properties](#)
Get details about current runtime properties

Execution

- [Execute servlet](#)
Post message to ExecuteServlet
- [Execute scripts](#)
WebServices invocation and entry points
- [DBTools](#)
Execute DBTools command
- [Shell](#)
Execute a Run command

5

Execute phantom

Command:

Execution logs

Log file: C:\AMR22\Temenos\R22\Env\Std51\Products\TAFJ\Default\log\trun.log

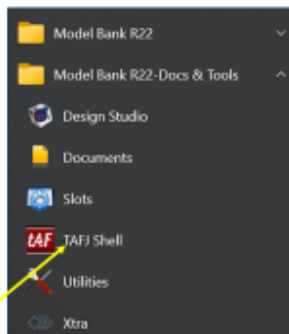
```
[INFO] 2022-04-01 12:37:22.902 DIAGNOSTIC: TRUN - LH371FB3 - 1333002247 -- START_TSM - uid=b19d122-4a7-8616-609a602a4c24, dim=D, level=INFO, comp=TEMN-TAFJ-TRUN-CORE, on=com.temenos.logging.wrapper.common.LoggerCommon:info:181, code="11412000", temn_type="info_log", msg="Started"
[INFO] 2022-04-01 12:37:22.964 DIAGNOSTIC: TRUN - LH371FB3 - 1333002247 -- START_TSM - uid="55c53a24-1662-4534-b63e-e176937a2667", dim=D, level=INFO, comp=TEMN-TAFJ-TRUN-CORE, on=com.temenos.logging.wrapper.common.LoggerCommon:info:181, code="11412000", temn_type="info_log", msg="Completed"
[INFO] 2022-04-01 12:37:22.980 DIAGNOSTIC: TRUN - LH371FB3 - 1332671175 -- ISA 1 - uid="13472a7c-72dc-4371-a8dc-0d33f82c2a0", dim=D, level=INFO, comp=TEMN-TAFJ-TRUN-CORE, on=com.temenos.logging.wrapper.common.LoggerCommon:info:181, code="11412000", temn_type="info_log", msg="Started"
[INFO] 2022-04-01 12:37:23.334 DIAGNOSTIC: TRUN - LH371FB3 - 1332671175 -- ISA 1 - uid="75eb309c-23d7-4504-b731-0616447a330a", dim=D, level=INFO, comp=TEMN-TAFJ-TRUN-CORE, on=com.temenos.logging.wrapper.common.LoggerCommon:info:181, code="11412000", temn_type="info_log", msg="Completed"
```

Slide 30

COB can be run in the interactive mode too i.e. in the foreground mode. Once the tSM is started, it finds out the number of agents that are required to be started for COB. Since, tSM is run in interactive mode, it prompts you to start the require number of agents. If tSM was run in a phantom mode, then it would have started the agents automatically. Login to TAFJ shell prompt i.e. the backend. Type tRun START.TSM – DEBUG to start the TSM in interactive/ debug mode. You will be prompted to manually launch agents.

Execute COB » TAFJ (Interactive Mode)

- Login to TAFJ shell -> and type tRun START.TSM – DEBUG.



```

TAFJ Select Setting up the Slot01 parameters
C:\AMR22\Temenos\R22\Env\Slot01\Data\T24\UD>tRun START.TSM -DEBUG

TAFJ Select Setting up the Slot01 parameters - tRun START.TSM -DEBUG
START_TSM -DEBUG
tSA 1 -DEBUG
<comp>
<agent>1</agent>
<processid>1116536842</processid>
<portno>10</portno>
Agent 1 started 04 APR 22 13:53:15
Agent's Process id 1116536842
<servername>LH371FB3</servername>
Running on server LH371FB3 PortNumber 10
Next Service TSM Current Service
SELECT P.ED.SCALABILITY.GROUPS Selected=0 time=3secs
<service name = TSM>
tSM -DEBUG

Service Profile TSM TSM Record for running upgrade as service (1)
BNK/PRINT.OUT DE FORMATTING SERVICE (1)
COB FOR COB (10)

Manually launch tSA 3 COB
Manually launch tSA 4 COB
Manually launch tSA 5 COB
Manually launch tSA 6 COB
Manually launch tSA 7 COB
Manually launch tSA 8 COB
Manually launch tSA 9 COB
Manually launch tSA 10 COB
Manually launch tSA 11 COB
Manually launch tSA 12 COB
    
```

Slide 31

Launch the agents by starting a new TAFJ shell and giving the command `tRun tSA <agent number>`

Execute COB » TAFJ » Interactive Mode – Launching the agents

- Launch another TAFJ shell and give the command **tRun tSA 11**

```

Select Setting up the Slot01 parameters - tRun START.TSM -DEBUG
START.TSM -DEBUG
tSA 1 -DEBUG
<como>
<agent>1</agent>
<processid>1116536842</processid>
<portno>18</portno>
Agent 1 started 04 APR 22 13-53-15
Agent's Process id 1116536842
<servername>LH371FB3</servername>
Running on server LH371FB3 PortNumber 18
Next Service TSM Current Service
SELECT F.EB.SCALABILITY.GROUPS Selected=0 time=3secs
<service name = TSM>
tSM -DEBUG

Service Profile TSM TSM Record for running upgrade as service (1)
BPR/PRINT.OUT DE FORMATTING SERVICE (1)
COB FOR COB (10)

Manually launch tSA 3 COB
Manually launch tSA 4 COB
Manually launch tSA 5 COB
Manually launch tSA 6 COB
Manually launch tSA 7 COB
Manually launch tSA 8 COB
Manually launch tSA 9 COB
Manually launch tSA 10 COB
Manually launch tSA 11 COB
Manually launch tSA 12 COB
  
```

```

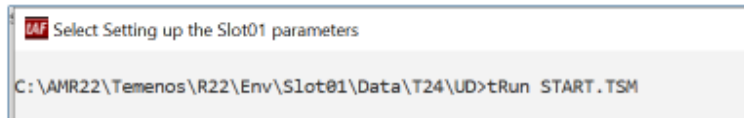
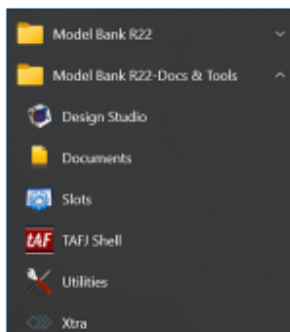
Select Setting up the Slot01 parameters - tRun tSA 11
tSA 11
<como>
<agent>11</agent>
<processid>398983186</processid>
<portno>18</portno>
Agent 11 started 04 APR 22 13-59-37
Agent's Process id 398983186
<servername>LH371FB3</servername>
Running on server LH371FB3 PortNumber 18
Next Service Current Service
SELECT F.EB.SCALABILITY.GROUPS Selected=0 time=4secs
<service name = COB>
SELECT F.COMPANY WITH CONSOLIDATION.MARK NE 'C' Selected=7 time=11secs
SSELECT F.BATCH BY BATCH.STAGE Selected=5289 time=166secs
  
```

Slide 32

The entire process of COB can be run in phantom mode, meaning in the background. There are various monitoring tools available that will help you identify what is taking place with respect to COB at any given point, this is discussed later in the session. Step 1 : Check EB.EOD.ERROR for details. This is a file that contains all the errors that took place during the previous COB. Any errors reported in this file, should be corrected before starting COB. You will learn about this in detail later. Step 2: Set the field SERVICE.CONTROL to "START" in the records TSM and COB in the TSA.SERVICE application. Commit and authorise both the records. Step 3: At the TAFJ shell prompt type the command tRun START.TSM . This will start the TSM in phantom mode. tSM not only starts agents for COB but also for online services set to 'Start' in Transact.

Execute COB » TAFJ shell » Phantom Mode

- Check EB.EOD.ERROR
- Set the field SERVICE.CONTROL to "START" in the records TSM and COB in the TSA.SERVICE application. Commit and authorise both the records
- Launch TAFJ shell
- Type tRun START.TSM (without -DEBUG. The agents are launched automatically)



The agents will be launched automatically as -DEBUG is not specified

Slide 33

This is one utility available for monitoring COB in the Transact Classic interface.

Monitor COB

- At TAFJ shell, give the command tRun EX to launch Transact.
- Login to Transact by entering the USER credentials.
- Type COB.MONITOR

Select Setting up the Slot01 parameters - tRun EX - tRun EX - tRun EX

Model Bank SELECT APPLICATION

-- LAST SIGN ON, DATE: 04 APR 2022 TIME: 15:06 ATTEMPTS: 04 APR 2022 15:07:17 USER (16 APR) INPUTTER [16649,]
ACTION COB.MONITOR
AWAITING APPLICATION

Select Setting up the Slot01 parameters - tRun EX

=====

COB.PROGRESS.DISPLAY

=====

TIME : 16:46:52 04 APR 2022 COB-DATE : 15 APR 2021
COB Running in 080010005 080010001 US0010001 NL0020001 AL00010001 PX00010001 080020001

STAGE	PROGRESS	PROCESSED	TOTAL	%COMPLETED
APPLICATION	[]	331	331	100.00
SYSTEM WIDE	[]	306	306	100.00
REPORTING	[]	332	332	100.00
START OF DAY	[]	349	349	100.00
ONLINE	[]	175	209	60.55
COB	[]	1493	1607	92.91

=====

To Quit Press EXIT exit Q q

Slide 34

ENQ COB.PROGRESS is available from Browser to view the progress of COB

| **Monitor COB » Enquiry COB.PROGRESS**

  **TEMENOS**

ENQ COB.PROGRESS  

≡ Results 1 - 8 of 8

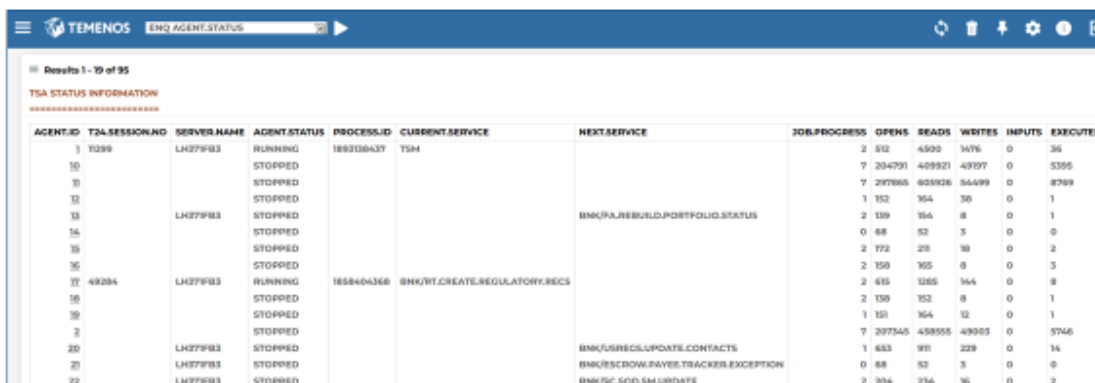
Service	Progress	The max	Complete	Maximum
COB	1749 5785 30.23336214347	5785		5785
COB-1	0 0 0	0		0
COB-2	0 0 0	0		0
COB-3	0 0 0	0		0
COB-4	0 0 0	0		0
COB-5	0 0 0	0		0
COB-6	0 0 0	0		0
COB-7	0 0 0	0		0

Slide 35

You know tSM launches and controls the tSAs. How does an end user know what is happening with the tSAs? Use the enquiry AGENT.STATUS Agent ID is the Agent Number T24.SESSION.NO is updated with a unique number that identifies every session in Transact. A number between 1 -99999 is allocated when a session is started in a port to uniquely identify that particular session in Transact(both for online and COB). Note: This random number is generated using a routine called EB.ALLOCATE.SESSION.NUMBER and is stored in a common variable called C\$T24.SESSION.NO. This helps in improving the monitoring capabilities of Transact. A live file called T24.SESSION gets updated with record key being the T24.SESSION.NO and contains information regarding the agent to which this session number was assigned. A record in this application gets deleted, when the corresponding process terminates(session expires). Server Name is the name of the server where the agent is running. Note that details of tSAs and tSMs across application servers will be updated here Agent Status is the status of the agent. It can hold either of the three values. RUNNING/STOPPED/DEAD Process Id is the Operating system level process ID for the agent process that is running Current Service is the service the agent is performing at the moment. Note agent 1 is performing the service TSM and the others are performing the service COB Next Service is the service the agent is supposed to perform next. While COB is running, if you wish to stop COB, set the field SERVICE.CONTROL to STOP for the record COB in the TSA.SERVICE file. Then, check the NEXT.SERVICE value for the agents running COB, it will be set to STOP denoting that the agents needs to stop once they complete the current transaction Job Progress is the status of the jobs Opens, Reads, Writes, Inputs, Executes, Deletes, Clearfiles, Memory provides details about the number of file opens, number of I/O's, memory usage etc., by each agent. These details available in the enquiry improves the monitoring capabilities of Transact. It provides an understanding on the processing done by the agent. Updated both for service and online agents. Last Message is the message displayed by the agents as they are performing the jobs. Como Name is the name of the log record ID that holds the log information of the agent. These log records are stored in a directory called COMO. The contents of these records cannot be viewed from browser.

Monitor COB » ENQUIRY AGENT.STATUS

- Enquiry AGENT.STATUS shows the activity of every agent.



Results 1 - 19 of 95

TSA STATUS INFORMATION

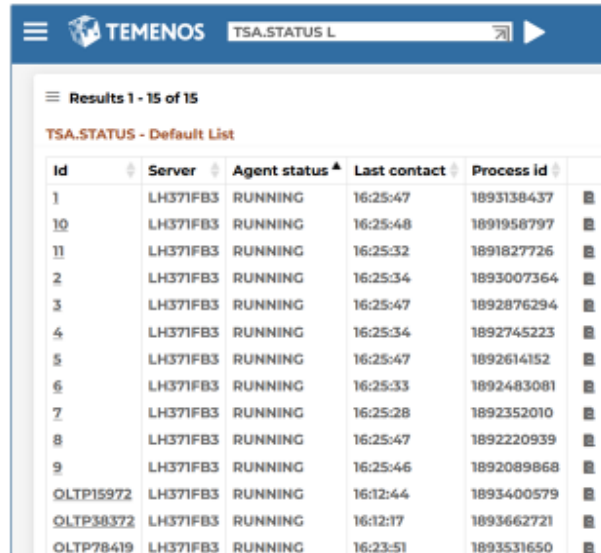
AGENT ID	T24-SESSION.NO	SERVER NAME	AGENT STATUS	PROCESS ID	CURRENT SERVICE	NEXT SERVICE	JOB PROGRESS	OPENS	READS	WRITES	INPUTS	EXECUTES
1	10299	LH27F9B3	RUNNING	189208A27	TSM		2	512	4509	1676	0	36
10			STOPPED				7	204791	409921	43997	0	5395
11			STOPPED				7	297868	608928	64499	0	8789
12			STOPPED				1	152	364	38	0	1
13		LH27F9B3	STOPPED			BNK/PAREBUILD.PORTFOLIO.STATUS	2	139	164	8	0	1
14			STOPPED				0	68	52	3	0	0
15			STOPPED				2	772	216	98	0	2
16			STOPPED				2	158	305	8	0	3
17	49284	LH27F9B3	RUNNING	185846A368	BNK/WT.CREATE.REGULATORY.RECS		2	415	1285	144	0	8
18			STOPPED				2	138	152	8	0	1
19			STOPPED				1	151	364	12	0	1
2			STOPPED				7	297345	458955	49983	0	5746
20		LH27F9B3	STOPPED			BNK/USREGS.UPDATE.CONTACTS	1	633	910	229	0	14
21		LH27F9B3	STOPPED			BNK/ESCROW.PAYES.TRACKER.EXCEPTION	0	88	52	3	0	0
22		LH27F9B3	STOPPED			BNK/SC.SOD.SH.UPDATE	2	204	234	16	0	2

Slide 36

The enquiry picks up data from the TSA.STATUS live file.

Monitor COB » TSA.STATUS

- TSA.STATUS.
 - Only updated by tSM and tSA(s).
 - Holds the status of all agents.
- Type TSA.STATUS L to view details in the TSA.STATUS file.



Results 1 - 15 of 15					
TSA.STATUS - Default List					
Id	Server	Agent status	Last contact	Process id	
1	LH371FB3	RUNNING	16:25:47	1893138437	
10	LH371FB3	RUNNING	16:25:48	1891958797	
11	LH371FB3	RUNNING	16:25:32	1891827726	
2	LH371FB3	RUNNING	16:25:34	1893007364	
3	LH371FB3	RUNNING	16:25:47	1892876294	
4	LH371FB3	RUNNING	16:25:34	1892745223	
5	LH371FB3	RUNNING	16:25:47	1892614152	
6	LH371FB3	RUNNING	16:25:33	1892483081	
7	LH371FB3	RUNNING	16:25:28	1892352010	
8	LH371FB3	RUNNING	16:25:47	1892220939	
9	LH371FB3	RUNNING	16:25:46	1892089868	
OLTP15972	LH371FB3	RUNNING	16:12:44	1893400579	
OLTP38372	LH371FB3	RUNNING	16:12:17	1893662721	
OLTP78419	LH371FB3	RUNNING	16:23:51	1893531650	

Slide 37

COMO.NAME in AGENT.STATUS holds the name of the log file. Log files are stored under a directory COMO. Use any text editor to view the content of the log files. If an agent is restarted, there will be two log files for that agent as the log file ID is based on date and time. Log files do not get cleared automatically from the COMO

| **Monitor COB » COMO**

- In Model Bank environment, como is available in the path

OS (C:) > AMR22 > Temenos > R22 > Env > Slot01 > Logs > T24 > como	
Name	Date modified
 tSA_1_20220404_16-24-36	4/4/2022 4:24 PM
 tSA_2_20220404_16-24-36	4/4/2022 4:27 PM
 tSA_2_20220404_16-29-04	4/4/2022 4:33 PM
 tSA_3_20220404_16-24-38	4/4/2022 4:37 PM
 tSA_3_20220404_16-29-39	4/4/2022 4:34 PM
 tSA_4_20220404_16-24-39	4/4/2022 4:27 PM
 tSA_4_20220404_16-29-06	4/4/2022 4:31 PM
 tSA_5_20220404_16-24-40	4/4/2022 4:27 PM
 tSA_5_20220404_16-33-39	4/4/2022 4:34 PM
 tSA_6_20220404_16-24-41	4/4/2022 4:29 PM
 tSA_6_20220404_16-31-43	4/4/2022 4:34 PM
 tSA_7_20220404_16-24-42	4/4/2022 4:27 PM
 tSA_7_20220404_16-29-08	4/4/2022 4:34 PM
 tSA_8_20220404_16-24-43	4/4/2022 4:34 PM
 tSA_9_20220404_16-24-44	4/4/2022 4:38 PM
 tSA_9_20220404_16-29-38	4/4/2022 4:34 PM
 tSA_10_20220404_16-24-45	4/4/2022 4:29 PM
 tSA_10_20220404_16-31-44	4/4/2022 4:34 PM
 tSA_11_20220404_16-24-46	4/4/2022 4:27 PM
 tSA_11_20220404_16-29-10	4/4/2022 4:34 PM

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Slide 38

COMO.NAME in AGENT.STATUS holds the name of the log file. Log files are stored under a directory COMO. Use any text editor to view the content of the log files. If an agent is restarted, there will be two log files for that agent as the log file ID is based on date and time. Log files do not get cleared automatically from the COMO

Monitor COB » COMO

- Open any of the generated log file shown in the path in the previous slide and you will see like below:

```

tSA_5_20220404_16-24-40
1 COMO tSA_5_20220404_16-24-40 established Mon Apr 04 16:24:40 IST 2022
2   5_04 APR 2022_16:24:40:489_<portno>8</portno>
3   5_04 APR 2022_16:24:40:490_Running on server LH371FB3 PortNumber 8
4   5_04 APR 2022_16:24:40:515_<service name = COB>
5   BSG/COB.INITIALISE 5_04 APR 2022_16:24:43:228_<process name = BSG/COB.INITIALISE>
6   US1/COB.INITIALISE 5_04 APR 2022_16:24:43:244_</process>
7   US1/COB.INITIALISE 5_04 APR 2022_16:24:43:245_<process name = US1/COB.INITIALISE>
8   AU1/COB.INITIALISE 5_04 APR 2022_16:24:43:252_</process>
9   AU1/COB.INITIALISE 5_04 APR 2022_16:24:43:252_<process name = AU1/COB.INITIALISE>
10  AU1/COB.INITIALISE EB.MANAGE.SERVICES 5_04 APR 2022_16:24:43:259_<job name = EB.MANAGE.SERVICES>
11  MX1/COB.INITIALISE EB.MANAGE.SERVICES 5_04 APR 2022_16:24:43:321_</process>
12  MX1/COB.INITIALISE EB.MANAGE.SERVICES 5_04 APR 2022_16:24:43:321_<process name = MX1/COB.INITIALISE>
13  MX1/COB.INITIALISE COB.EXECUTE.API 5_04 APR 2022_16:24:43:331_</job>
  
```

Lesson Summary

In this lesson I described about COB Service in Transact with a particular emphasis on... Introduction to COB Stages of COB Batch Stage Execute COB Monitor COB

| Lesson Summary



Practice

Practice3

Create a record in TSA.SERVICE with ID BESTSERVICE Authorise the record Change the description Authorise the change(observe the changes in the record in the tab below) List authorised records Reverse the record BESTSERVICE. What is the record status? Authorise the reversal. Where is the record now? What is the RECORD.STATUS? Bring the record back to LIVE. what function will you use? Authorise the History restore. Create \$DEL for TSA.SERVICE. Input record and delete. Use ENQ VIEW.DELETE.HISTORY to view the deleted record.

| Practice

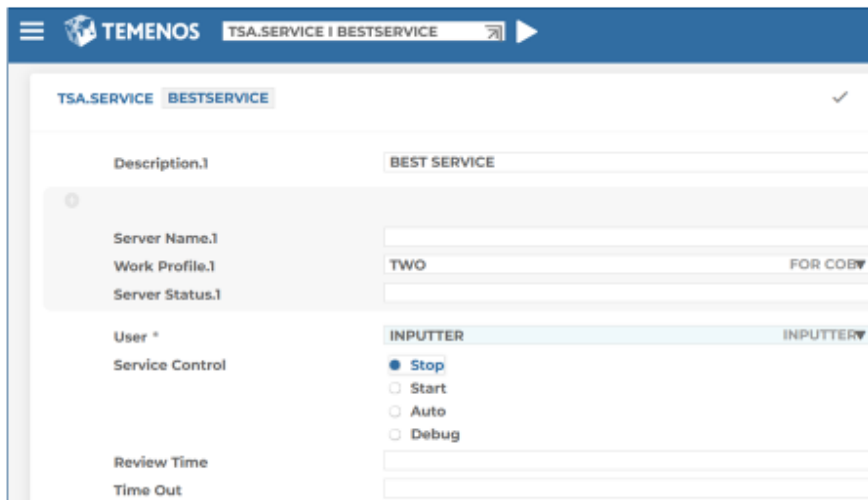
- Create a record in TSA.SERVICE with ID BESTSERVICE
- Authorise the record
- Change the description
- Authorise the change(observe the changes in the record)
- List authorised records in TSA.SERVICE
- Reverse the record BESTSERVICE. What is the record status?
- Authorise the reversal. Where is the record now? What is the record status?
- Bring the record back to LIVE. what function will you use?
- Authorise the History restore.
- Create \$DEL for TSA.SERVICE.
- Input record and delete.
- Use ENQ VIEW.DELETE.HISTORY to view the deleted record.

Practice4

Create a record in TSA.SERVICE with ID BESTSERVICE

| Practice » Solution

- Create a record in TSA.SERVICE with ID BESTSERVICE



The screenshot shows the TEMENOS application interface. At the top, there is a blue header bar with the TEMENOS logo and a search bar containing 'TSA.SERVICE | BESTSERVICE'. Below the header, the main content area displays the 'TSA.SERVICE' record for 'BESTSERVICE'. The record details are as follows:

Field	Value
Description.1	BEST SERVICE
Server Name.1	
Work Profile.1	TWO
Server Status.1	
User *	INPUTTER
Service Control	☒ Stop ☐ Start ☐ Auto ☐ Debug
Review Time	
Time Out	

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 **TEMENOS**
Learning Community

Practice5

Authorise the record BESTSERVICE

Practice » Solution

- Authorise the record BESTSERVICE

 **TEMENOS**

TSA.SERVICE A BESTSERVICE

TSA.SERVICE

BESTSERVICE

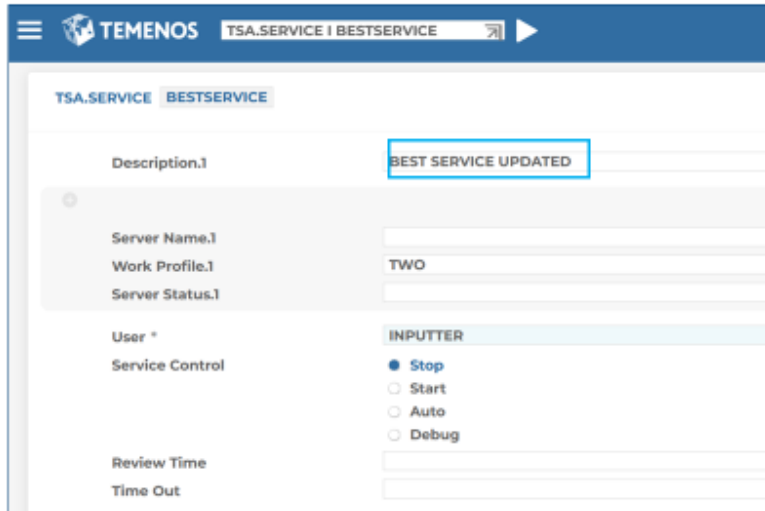
Description.1	BEST SERVICE	
Work Profile.1	TWO	FOR COB
User	INPUTTER	INPUTTER
Service Control	Stop	
Curr No	1	
Inputter.1	84623_INPUTTER_OFS_IRISPA	
Date time.1	01 APR 22 16:59	
Authoriser	84623_AUTHORISER_OFS_IRISPA	
Company	GB0010001	Model Bank
Dept Code	1	Implementation

Practice6

Change the Description

| Practice » Solution

- Change the Description



The screenshot shows the TEMENOS web interface for configuring a service. The header bar includes the TEMENOS logo and the text 'TSA.SERVICE | BESTSERVICE'. Below the header, there are two tabs: 'TSA.SERVICE' and 'BESTSERVICE'. The main content area displays a form with the following fields:

- Description.1**: A text field containing 'BEST SERVICE UPDATED', which is highlighted with a blue box.
- Server Name.1**: A text field.
- Work Profile.1**: A text field containing 'TWO'.
- Server Status.1**: A text field.
- User ***: A text field.
- Service Control**: A section with four radio buttons: 'Stop' (selected), 'Start', 'Auto', and 'Debug'.
- Review Time**: A text field.
- Time Out**: A text field.

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Authorize the change (observe the changes in the record)

Practice » Solution

- Authorise the change (observe the changes in the record)

TEMENOS

TSA.SERVICE A BESTSERVICE

TSA.SERVICE

BESTSERVICE

Description.1

Work Profile.1

User

Service Control

Record Status

Curr No

Inputter.1

Date time.1

Company

Dept Code

BEST SERVICE UPDATED

TWO

INPUTTER

Stop

INAU

2

B4623_INPUTTER_OFS_IRISPA

01 APR 22 17:03

GB0010001

1

FOR COB

INPUTTER

INPUT Unauthorised

Please Select

GO

Changes

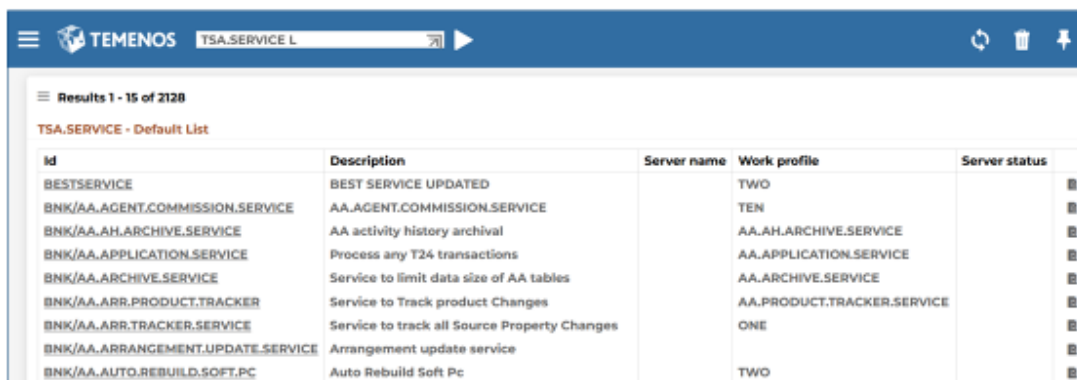
Field	Old Value	New Value
Description.1	BEST SERVICE	BEST SERVICE UPDATED
Record Status		INAU
Curr No	1	2
Date time.1	2204011659	2204011703
Authoriser	B4623_AUTHORISER_OFS_IRISPA	

Practice8

List Authorised records in TSA.SERVICE.

Practice » Solution

- List Authorised records in TSA.SERVICE



Results 1 - 15 of 2128

TSA.SERVICE - Default List

Id	Description	Server name	Work profile	Server status
BESTSERVICE	BEST SERVICE UPDATED		TWO	
BNK/AA.AGENT.COMMISSION.SERVICE	AA.AGENT.COMMISSION.SERVICE		TEN	
BNK/AA.AH.ARCHIVE.SERVICE	AA activity history archival		AA.AH.ARCHIVE.SERVICE	
BNK/AA.APPLICATION.SERVICE	Process any T24 transactions		AA.APPLICATION.SERVICE	
BNK/AA.ARCHIVE.SERVICE	Service to limit data size of AA tables		AA.ARCHIVE.SERVICE	
BNK/AA.ARR.PRODUCT.TRACKER	Service to Track product Changes		AA.PRODUCT.TRACKER.SERVICE	
BNK/AA.ARR.TRACKER.SERVICE	Service to track all Source Property Changes		ONE	
BNK/AA.ARRANGEMENT.UPDATE.SERVICE	Arrangement update service			
BNK/AA.AUTO.REBUILD.SOFT.PC	Auto Rebuild Soft Pc		TWO	

Practice9

Reverse the record BESTSERVICE.

Practice » Solution

- Reverse the record BESTSERVICE, the record status is RNAU.

TSA.SERVICE R BESTSERVICE	
TSA.SERVICE BESTSERVICE	
Description.1	BEST SERVICE UPDATED
Work Profile.1	TWO FOR COB
User	INPUTTER INPUTTER
Service Control	Stop
Record Status	RNAU REVERSE Unauthorised
Curr No	3
Inputter.1	84623_AUTHORISER__OFS_IRISPA
Date time.1	01 APR 22 17:06
Company	GB0010001 Model Bank
Dept Code	1 Implementation

Practice10

Authorise the reversal.

| Practice » Solution

- Authorise the reversal. The record is in History table and the record status is REVE.

TSA.SERVICE A BESTSERVICE	
TSA.SERVICE BESTSERVICE	
Description.1	BEST SERVICE UPDATED
Work Profile.1	TWO
User	INPUTTER
Service Control	Stop
Record Status	REVE
Curr No	3
Inputter.1	84623_AUTHORISER_OFS_IRISPA
Date time.1	01 APR 22 17:08
Authoriser	84623_INPUTTER_OFS_IRISPA
Company	GB0010001
Dept Code	1

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Practice11

Authorise the reversal. The record is in History table and the record status is REVE.

Practice » Solution

- Authorise the reversal. The record is in History table and the record status is REVE.

TEMENOS

TSA.SERVICE A BESTSERVICE

TSA.SERVICE

BESTSERVICE

Description.1	BEST SERVICE UPDATED
Work Profile.1	TWO
User	INPUTTER
Service Control	Stop
Record Status	REVE
Curr No	3
Inputter.1	B4623_AUTHORISER_OFS_IRISPA
Date time.1	01 APR 22 17:08
Authoriser	B4623_INPUTTER_OFS_IRISPA
Company	GB0010001
Digit Code	1

Results 67 - 103 of 321

TSA.SERVICEHS - DEFAULT HISTORY LIST

BESTSERVICE.3	REVE	B4623_AUTHORISER_OFS_IRISPA	01 APR 22 17:08	B4623_INPUTTER_OFS_IRISPA	
BESTSERVICE.2		B4623_INPUTTER_OFS_IRISPA	01 APR 22 17:04	B4623_AUTHORISER_OFS_IRISPA	
BESTSERVICE.1		B4623_INPUTTER_OFS_IRISPA	01 APR 22 16:59	B4623_AUTHORISER_OFS_IRISPA	
BECUPLD.SERVICE.3	REVE	SV_COMPRIN.TSA	11 MAR 22 03:07		
BECUPLD.SERVICE.1		S_TRANS11_OFS_M8_OFS	10 MAR 22 23:42		
BILLING.SERVICE.2	REVE	SV_COMPRIN.TSA	11 MAR 22 03:07		

%TSA.SERVICEHS

Service

GB0010001

BEST SERVICE

Practice12

Bring the record back to LIVE. The function used is 'H'.

| Practice » Solution

- Bring the record back to LIVE. The function used is 'H'.

 **TEMENOS**

TSA.SERVICE H BESTSERVICE



TSA.SERVICE

BESTSERVICE [:3] 01 APR 2022

Description.1	BEST SERVICE UPDATED	
Work Profile.1	TWO	FOR COB
User	INPUTTER	INPUTTER
Service Control	Stop	
Record Status	HNAU	HISTORY RESTORE Unauthorised
Curr No	4	
Inputter.1	84623_INPUTTER__OFS_IRISPA	
Date time.1	01 APR 22 17:18	
Company	GB0010001	Model Bank
Dept Code	1	Implementation

Practice13

Authorise the History restore.

Practice » Solution

- Authorise the History restore.

 **TEMENOS**

TSA.SERVICE A BESTSERVICE

TSA.SERVICE BESTSERVICE

Description.1	BEST SERVICE UPDATED	
Work Profile.1	TWO	FOR COB
User	INPUTTER	INPUTTER
Service Control	Stop	
Curr No	4	
Inputter.1	84623_INPUTTER__OFS_IRISPA	
Date time.1	01 APR 22 17:21	
Authoriser	84623_AUTHORISER_OFS_IRISPA	
Company	GB0010001	Model Bank
Dept Code	1	Implementation

Practice14

Create \$DEL for TSA.SERVICE.

Practice » Solution

- Create \$DEL for TSA.SERVICE.

TEMENOS

FILE CONTROL | TSA.SERVICE

FILE CONTROL

TSA.SERVICE

Description.1 *

Product *

Suffixes.1

Suffixes.2

Suffixes.3

File Type *

File Module

Classification *

Service definitions

EB

☐ \$Nau

☒ \$His

☐ \$Sim

☐ \$Del

☐ \$Arc

☒ \$Nau

☐ \$His

☐ \$Sim

☐ \$Del

☐ \$Arc

☐ \$Nau

☐ \$His

☐ \$Sim

☒ \$Del

☐ \$Arc

2

3

Int

COORDINATOR

Sim Req'd

☐ Y ☐ N

Delete History

☒ Y

DI Date Field.1

Imp Choice Type

- Please Select

Practice15

Input record and delete.

Practice » Solution

- Input record and delete.

TEMENOS

TSA.SERVICE I BESTSERVICE

TSA.SERVICE

BESTSERVICE

Description.1

BEST SERVICE - DELETE

Server Name.1

Work Profile.1

TWO

Server Status.1

User *

INPUTTER

Service Control

Stop

Start

Auto

Debug

Review Time

Time Out

TEMENOS

TSA.SERVICE D BESTSERVICE

TSA.SERVICE

BESTSERVICE

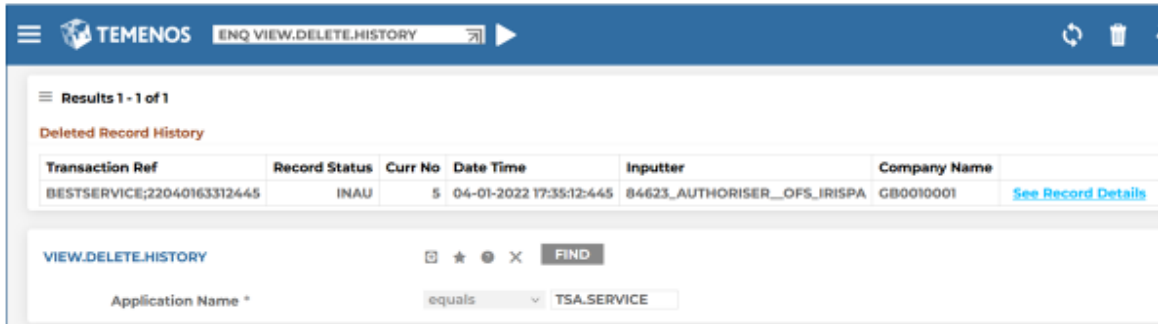
Description.1	BEST SERVICE - DELETE	
Work Profile.1	TYWO	FOR COB
User	INPUTTER	INPUTTER
Service Control	Stop	
Record Status	IDEL	INPUT
Curr No	5	
Inputter.1	84623_AUTHORISER_OFS_IRISPA	
Date time.1	01 APR 22 17:26	
Company	GB0010001	Model Bank
Dept Code	1	Implementation

Practice16

Use ENQ VIEW.DELETE.HISTORY to view the deleted record. This is visible only because we have made the DELETE HISTORY to Y in FILE.CONTROL.

Practice » Solution

- Use ENQ VIEW.DELETE.HISTORY to view the deleted record.



The screenshot shows the TEMENOS interface for the ENQ VIEW.DELETE.HISTORY function. The header bar includes the TEMENOS logo and the function name. Below the header, it indicates 'Results 1 - 1 of 1'. The main content area is titled 'Deleted Record History' and contains a table with the following data:

Transaction Ref	Record Status	Curr No	Date Time	Inputter	Company Name	
BESTSERVICE;22040163312445	INAU	5	04-01-2022 17:35:12:445	84623_AUTHORISER__OFS_IRISPA	GB0010001	See Record Details

Below the table, there is a search section titled 'VIEW.DELETE.HISTORY' with a 'FIND' button. The search criteria are set to 'Application Name *' with a dropdown menu showing 'equals' and a text input field containing 'TSA.SERVICE'.

Practice

Practice3

Create a USER with all permissions

| Practice 1

- Create a USER with all permissions

Practice4

Create a USER with all permissions

Practice 1 » Solution

  **TEMENOS**

USER, 1 DEMOUSER



USER

DEMO

✓

User Name *

DEMOUSER

Sign On Name *

DEMOUSR

Classification *

☐ Ext ☒ Int

Language *

1 English▼

Company.1 *

GB0010001 Model Bank

Company.2 *

US0010001 US Model Bank

Company.3 *

AU0010001 Australia Model Bank

Company.4 *

MX0010001 Banc Modelo Mexicano

Department Code *

1 Implementation▼

Password Validity *

31 DEC 2099 M0601 31 DEC 2099 Every 6 months on day 1

Start Date *

04 APR 2022 04 APR 2022

End Date *

31 DEC 2099 31 DEC 2099

Start Time.1 *

0

End Time.1 *

2400

Time Out Minutes *

999

Attempts *

9

Practice5

Create a USER with all permissions

| Practice 1 » Solution

USER

DEMO

Company Restr.1

ALL

User Group.1

ALL.PG

Version.1

Function Allowed.1

A 2 B C D E F H I L P R S V

Field No.1

Data Comparison.1

- Please Select

Data From.1

Data To.1

Sign On Off Log *

☐ Y

☒ No

Security Mgmt L *

☐ Y

☒ No

Application Log *

☐ Y

☒ No

Function Id Log *

☐ Y

☒ No

Input Day Month *

☒ Ddmm

☐ Mmdd

Practice6

Create a USER with all permissions

| Practice 1 » Solution

USER

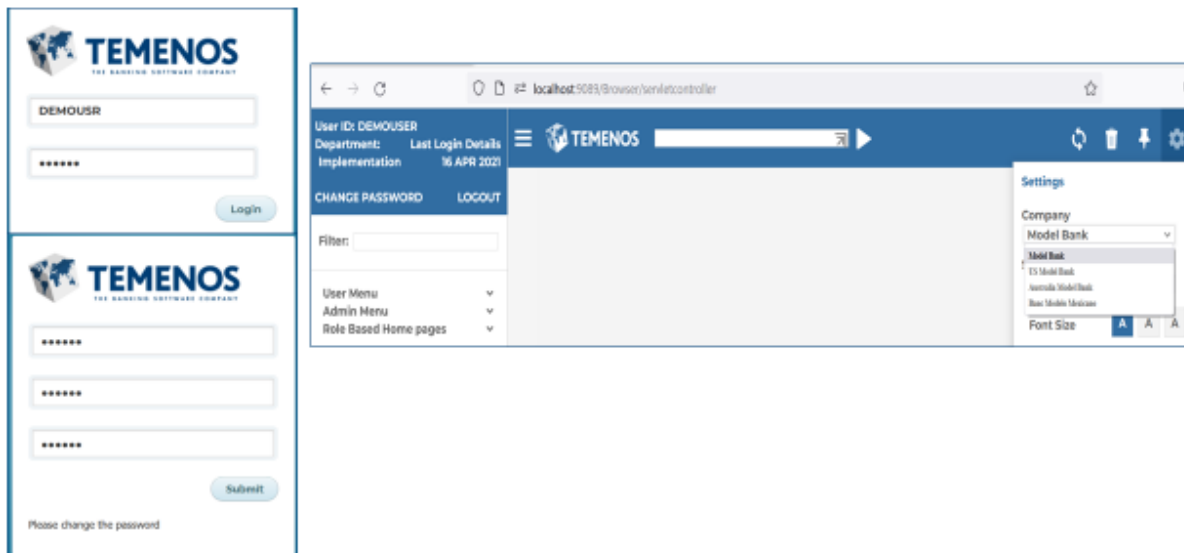
DEMO

Max Sessions	
Attributes.1	Super User

Practice7

Test the Permissions. We could see that the user has access to multiple company as specified in the user record.

Practice 1 » Solution



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Learning Community

Practice8

Create a USER with following restrictions access to one company only – GB0010001 access only the ACCOUNT and CUSTOMER applications See and Print functions for the CUSTOMER application See and Input data functions for the ACCOUNT application Access to the command line Note : Use Help text to find out how User can be given access to Command Line

| Practice 2

- Create a USER with following restrictions
 - access to one company only – GB0010001
 - access only the **CUSTOMER** and **ACCOUNT** applications
 - See and Print functions for the **CUSTOMER** application
 - See and Input data functions for the **ACCOUNT** application
 - Access to the command line

Practice9

Create a User with the given permissions. access to one company only – GB0010001 access only the ACCOUNT and CUSTOMER applications See and Print functions for the CUSTOMER application See and Input data functions for the ACCOUNT application Access to the command line

Practice 2 » Solution

USER, XACML

User Name *

XACML

Sign On Name *

XACML.1

Classification *

☐ Ext
 ☒ Int

Language *

1 English▼

Company.1 *

GB0010001 Model Bank

Department Code *

1 Implementation▼

Password Validity *

01 OCT 2022 M0601 01 OCT 2022 Every 6 months on day 1

Start Date *

05 APR 2022 05 APR 2022

End Date *

28 APR 2034 28 APR 2034

Start Time.1 *

9

End Time.1 *

2400

Time Out Minutes *

999

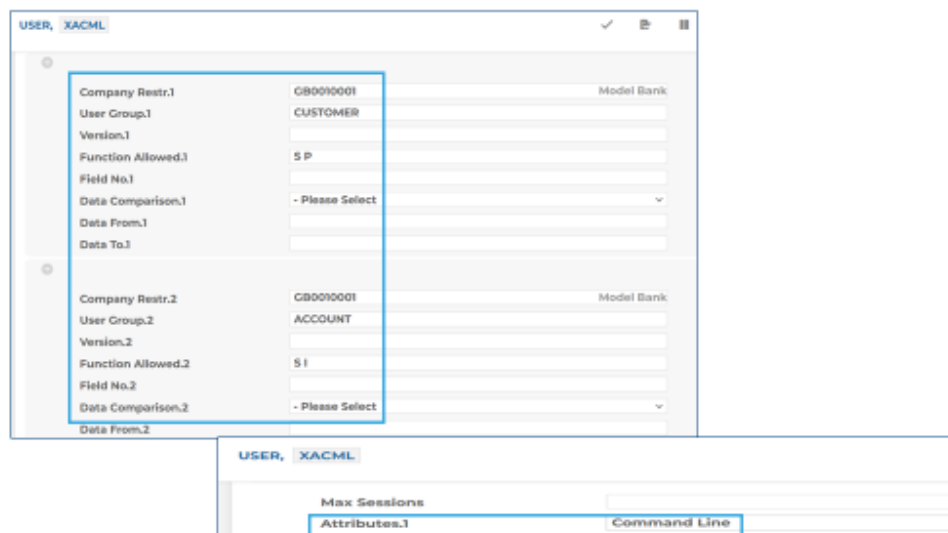
Attempts *

9

Practice10

Create a User with the given permissions. access to one company only – GB0010001 access only the ACCOUNT and CUSTOMER applications See and Print functions for the CUSTOMER application See and Input data functions for the ACCOUNT application Access to the command line

| Practice 2 » Solution



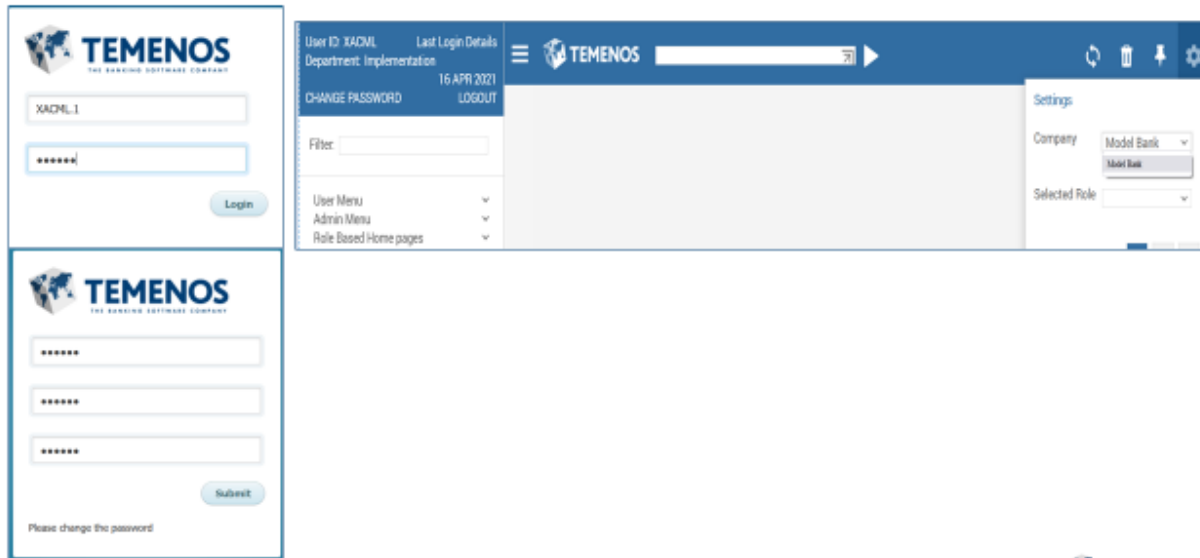
The screenshot displays the 'USER, XACML' configuration window. It contains two user profiles, each with a set of fields for configuration. The first profile (Company Restr.1) is configured with Company Restr.1: GB0010001, User Group.1: CUSTOMER, Version.1: (empty), Function Allowed.1: S P, Field No.1: (empty), Data Comparison.1: - Please Select, Data From.1: (empty), and Data To.1: (empty). The second profile (Company Restr.2) is configured with Company Restr.2: GB0010001, User Group.2: ACCOUNT, Version.2: (empty), Function Allowed.2: S I, Field No.2: (empty), Data Comparison.2: - Please Select, Data From.2: (empty), and Data To.2: (empty). Below the main configuration area, there is a section for 'Max Sessions' and 'Attributes.1' (Command Line).

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Practice11

Test the User's permissions. We could see that the user has access to only one company.

| Practice 2 » Solution

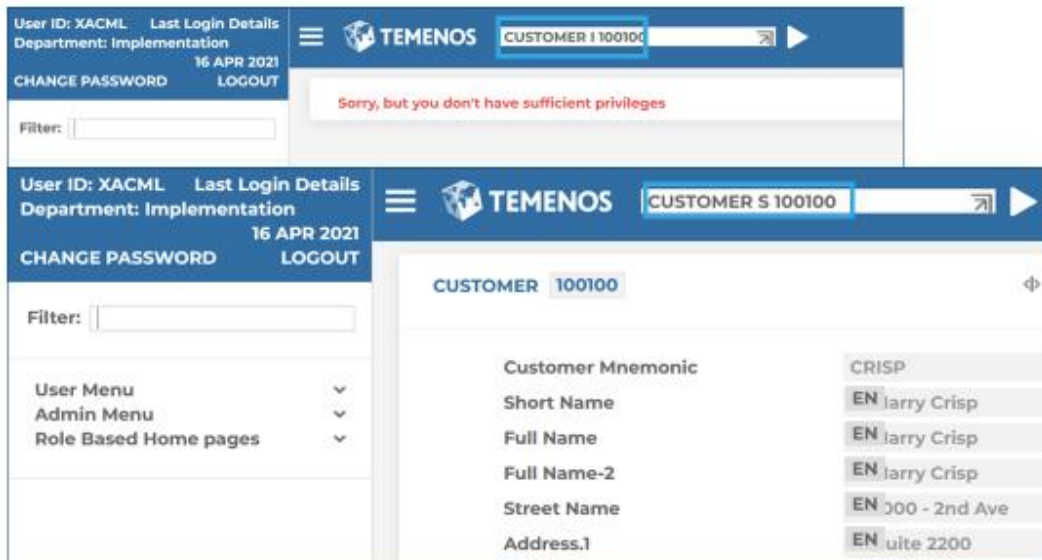


The screenshot displays the TEMENOS user interface. On the left, there is a login form with fields for 'XACML1' and a password, followed by a 'Login' button. Below this is a 'Change Password' form with three password input fields and a 'Submit' button. The main area shows a user profile for 'User ID: XACML1', 'Department: Implementation', and 'Last Login Details: 16 APR 2021'. There are links for 'CHANGE PASSWORD' and 'LOGOUT'. A 'Filter' input field is present. On the right, a 'Settings' panel shows 'Company' set to 'Model Bank' and 'Selected Role' set to 'Model Bank'. The bottom of the page includes a copyright notice: '©2022 Temenos Headquarters SA - all rights reserved.'

Practice12

Test the Permissions for CUSTOMER application. User has permissions only to See and Print and hence as such Input of record is restricted for the user.

Practice 2 » Solution



CUSTOMER 100100	
Customer Mnemonic	CRISP
Short Name	EN Larry Crisp
Full Name	EN Larry Crisp
Full Name-2	EN Larry Crisp
Street Name	EN 300 - 2nd Ave
Address.1	EN Suite 2200

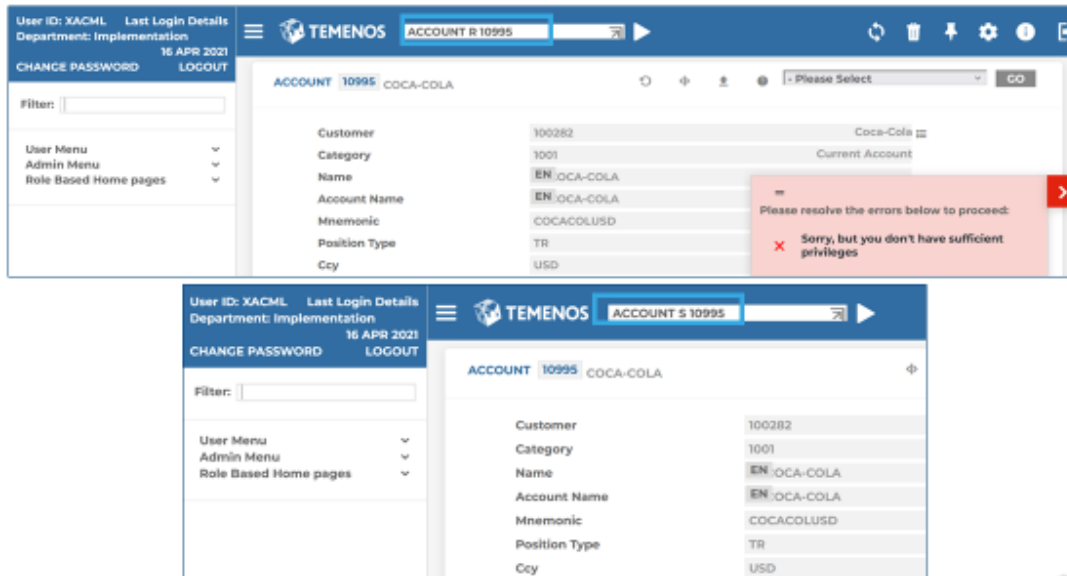
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Practice13

Test the Permissions for ACCOUNT application. We could see that user has access to S(See) function but not for R(Reverse)

Practice 2 » Solution



Top Screenshot: ACCOUNT R 10995

User ID: XACML | Last Login Details | Department: Implementation | 16 APR 2021 | CHANGE PASSWORD | LOGOUT

ACCOUNT 10995 COCA-COLA

Customer	100282	Coca-Cola
Category	1001	Current Account
Name	EN COCA-COLA	
Account Name	EN COCA-COLA	
Mnemonic	COCACOLUSD	
Position Type	TR	
Ccy	USD	

Please resolve the errors below to proceed:
✖ Sorry, but you don't have sufficient privileges

Bottom Screenshot: ACCOUNT S 10995

User ID: XACML | Last Login Details | Department: Implementation | 16 APR 2021 | CHANGE PASSWORD | LOGOUT

ACCOUNT 10995 COCA-COLA

Customer	100282	
Category	1001	
Name	EN COCA-COLA	
Account Name	EN COCA-COLA	
Mnemonic	COCACOLUSD	
Position Type	TR	
Ccy	USD	

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Practice14

Create two roles CRMROLE and ADMINROLE. Create a USER with ID GREENE and assign the two roles to the USER Login as GREENE and ensure that the roles/permissions are effective.

| Practice 3

- Create two roles CRMROLE and ADMINROLE.
- Create a USER with ID GREENE and assign the two roles to the USER
- Login as GREENE and ensure that the roles/permissions are effective.

Practice15

Create a USER GREENE and add the two roles CRMROLE and ADMINROLE Created using EB.USER.ROLES as seen above..

Practice 3 » Solution

USER GREENE

Attempts	9
Sign On Off Log	Y
Security Mgmt L	Y
Application Log	Y
Function Id Log	Y
Input Day Month	Ddmm
Clear Screen	Y
Attributes.1	Command Line
Date Format	1
Role Company Restr.1	GB0010001
User Roles.1.1	CRMROLE
User Roles.1.2	ADMINROLE
Curr No	1
Inputter.1	52297_INPUTTER__OFS_IRISPA
Date time.1	05 APR 22 13:26
Authoriser	52297_INPUTTER_OFS_IRISPA

EB.USER.ROLES S CRMROLE

EB.USER.ROLES CRMROLE

Description	EN CRM ROLE
Application.1	CUSTOMER
Function Allowed.1	A 2 B C D E F H I L P R S V
Application.2	ACCOUNT
Function Allowed.2	S

EB.USER.ROLES S ADMINROLE

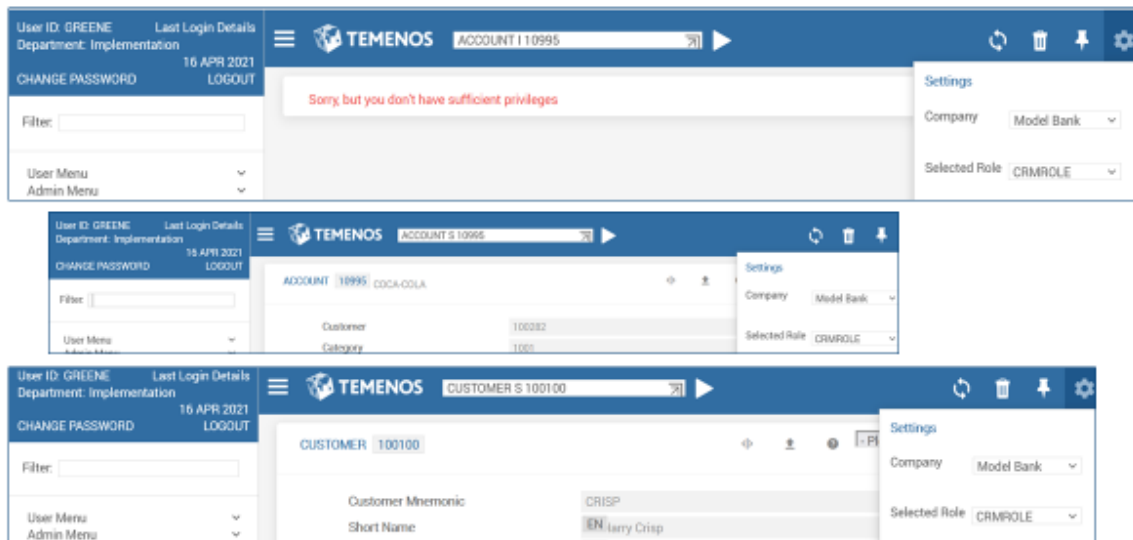
EB.USER.ROLES ADMINROLE

Description	EN ADMIN ROLE
Application.1	ALL PG
Function Allowed.1	A 2 B C D E F H I L P R S V

Practice16

Since the CRMROLE is assigned to the USER GREENE , user is able to View Account record but do not have permissions to edit it , and able to view the customer application as the user has all permissions on Customer application per defined in EB.USER.ROLES .

| Practice 3 » Solution



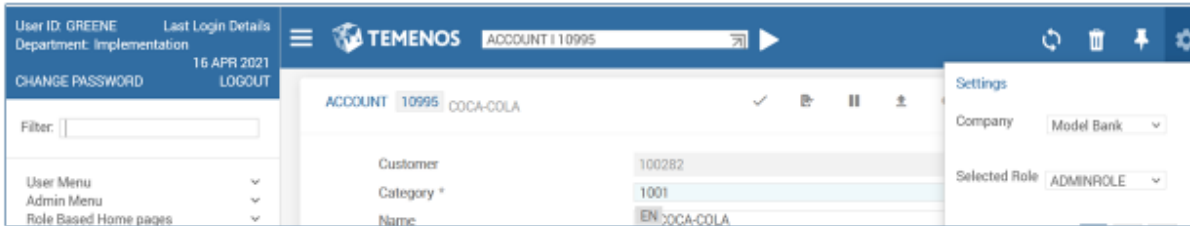
The screenshots illustrate the user's capabilities based on their role (CRMROLE) and assigned permissions:

- Top Screenshot:** The user (User ID: GREENE, Department: Implementation) is logged in. The interface shows a message: "Sorry, but you don't have sufficient privileges". The "Selected Role" is set to CRMROLE.
- Middle Screenshot:** The user is viewing the "ACCOUNT S 10995" record. The account details are visible, including the company (Model Bank) and the selected role (CRMROLE).
- Bottom Screenshot:** The user is viewing the "CUSTOMER S 100100" record. The customer details are visible, including the customer mnemonic (CRISP) and the short name (EN Larry Crisp). The selected role is CRMROLE.

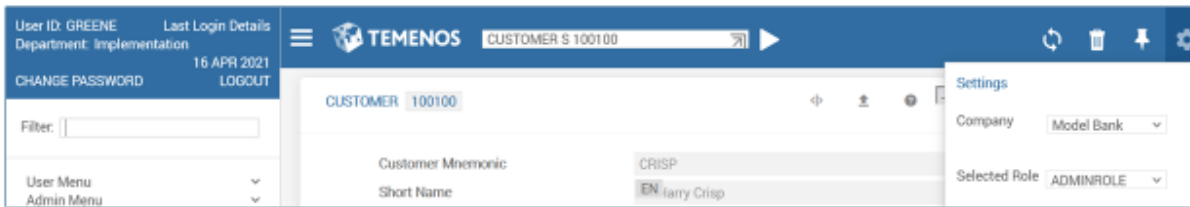
Practice17

All the Applications are accessible to ADMINROLE and hence the user is able to access both the ACCOUNT and CUSTOMER applications for all functions as seen above by the user GREENE.

| Practice 3 » Solution



The screenshot shows the TEMENOS ACCOUNT application interface. The top header includes the user ID 'GREENE', last login details, and a date '16 APR 2021'. The main content area displays 'ACCOUNT 10995 COCA-COLA' with a search bar and a list of fields: Customer (100282), Category * (1001), and Name (EN: COCA-COLA). A settings panel on the right shows 'Company' as 'Model Bank' and 'Selected Role' as 'ADMINROLE'.



The screenshot shows the TEMENOS CUSTOMER application interface. The top header includes the user ID 'GREENE', last login details, and a date '16 APR 2021'. The main content area displays 'CUSTOMER 100100' with a search bar and a list of fields: Customer Mnemonic (CRISP) and Short Name (EN: Larry Crisp). A settings panel on the right shows 'Company' as 'Model Bank' and 'Selected Role' as 'ADMINROLE'.

Practice

Practice3

Format the debit and credit advice using BNK/PRINT.OUT service.

| Practice

- Format the debit and credit advice using BNK/PRINT.OUT service.

Practice4

Create a FUNDS.TRANSFER record by enabling the debit and credit advice. FUNDS.TRANSFER, I F3

| Practice » Solution

FUNDS.TRANSFER, FT21105GC9DB

Transaction Type	AC
Debit Account	10995
Debit Account (IN)	
Ccy Market Dr	1
Debit Currency	USD
Debit Amount	300
Debit Value Date	15 APR 2021
Debit Value Date (IN)	DD MMM YYYY
Debit Narrative	
Credit Narrative	
Credit Account	100056
Ccy Market Cr	1
Credit Currency	GBP

FUNDS.TRANSFER, FT21105GC9DB

Send Payment Y/N?	☐ Y ☐ No
Dr Advice Req'd Y/N?	☒ Y ☐ No
Cr Advice Req'd Y/N?	☒ Y ☐ No

Practice5

On authorising the record the debit and credit advice is generated.

Practice » Solution

FUNDS.TRANSFER, FT21105GC9DB		
Position Type	TR	TRADING POSITION
Amount Debited	USD300.00	USD 300.00
Amount Credited	GBP209.21	GBP 209.21
Customer Rate	1.434	
Delivery Reference.1	D20220401846235895903-900.1.1 DEBIT ADVICE	≡
Delivery Reference.2	D20220401846235895904-910.2.1 CREDIT ADVICE	≡
Credit Comp Code	GB0010001	Model Bank
Debit Comp Code	GB0010001	Model Bank
Amt Debited Local	300.00	
Amt Credited Local	300.00	
Customer Group Level	99	
Debit Customer	100282	Coca-Cola ≡
Credit Customer	100367	J Johnson Jr ≡
Dr Advice Req'd Y/N?	Y	
Cr Advice Req'd Y/N?	Y	

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TEMENOS
 Learning Community

Practice6

The delivery advice will be in unformatted status.

Practice » Solution

Delivery Reference.1

D20220401846235895903-900.1.1 DEBIT ADVICE

Delivery Reference.2

D20220401846235895904-910.2.1 CREDIT ADVICE

Results 1 - 1 of 1

View Delivery Messages

Delivery ref	Bank date	Type	Description	Header Status	Msg Status	Carrier	To address	
D20220401846235895903	15 APR 21	900	CONFIRMATION OF DEBIT	UNFORMATTED		PRINT.1	Coca-Cola	

DE.O.HEADER

D20220401846235895903

Message Type	900	CONFIRMATION OF DEBIT
Application Format	1	
Application	FTAC	
Status	Unformatted	
Priority	N	
Account Number	GB61DEMO60161300010995	
Customer No.1	100282	

Practice7

Start the TSM and BNK/PRINT.OUT.

Practice » Solution

The image shows two overlapping windows from the TEMENOS application. The background window is titled 'TSA.SERVICE, TSM' and contains the following fields:

- Description.1: TSM Record for running upgrade as service
- Server Name.1: (empty)
- Work Profile.1: TSM
- Server Status.1: (empty)
- User: INPUTTER
- Service Control: ☒ Start, ☐ Stop, ☐ Auto, ☐ Debug
- Review Time: (empty)

The foreground window is titled 'TSA.SERVICE, BNK/PRINT.OUT' and contains the following fields:

- Description.1: DE FORMATTING SERVICE
- Server Name.1: (empty)
- Work Profile.1: DE FORMATTING SERVICE USED FOR DE FORMATTING SERVICE: Sanity
- Server Status.1: (empty)
- User: INPUTTER
- Service Control: ☒ Start, ☐ Stop, ☐ Auto, ☐ Debug

Practice8

Execute Servlet TAFJEE_WAR_TAFJ contains a servlet which could be used to post message to the JMS queue t24EXECQueue. Navigate to this URL - <http://localhost:9089/TAFJEE/Execute> To post START.TSM submit START.TSM in the form. Remark: The following property must be set correctly to run TAFJ within an application server context. `temn.tafj.runtime.phantom.as.process = false`

Practice » Solution

The diagram illustrates the steps to execute the TAFJ servlet:

- Browser Window:** Shows the URL `localhost:9089/TAFJEE/`.
- Sign in Form:** Requires a `Username` (t24user) and `Password` (t24User123!).
- TAFJEE - Home Dashboard:** Contains a menu with the following items:
 - Diagnostic:**
 - [tDiag](#): Get details about your TAFJ environment
 - [tShow](#): Get routine compilation details
 - [Sanity check](#): Run environment validation
 - [Runtime properties](#): Get details about current runtime properties
 - Execution:**
 - [Execute servlet](#): Post message to ExecQueue (highlighted with a blue box and arrow)
 - [Entry points](#): Webservices invocation and entry points
 - [DBTools](#): Execute DBTools command
- Execute phantom Form:**
 - Command:** `START.TSM`
 - Execution logs:** Shows a log of events, including:
 - INFO | 2022-04-01 12:37:22.902 DIAGNOSTIC.TRUN - LH371FB3 - 1333002247 - - START.TSM - uid="b1f56122-4c89-44e7-461d-609e602a4c24" dim="D" level="INFO" comp="TEMN-TAFJ-TRUN-CORE" cn="com.temenos.logging.wrapper.common.LoggerCommonInfo:181" code="11412000" term_type="tall_log" msg="Started"
 - INFO | 2022-04-01 12:37:22.964 DIAGNOSTIC.TRUN - LH371FB3 - 1333002247 - - START.TSM - uid="55c53a24-1662-4534-b83e-17693752505" dim="D" level="INFO" comp="TEMN-TAFJ-TRUN-CORE" cn="com.temenos.logging.wrapper.common.LoggerCommonInfo:181" code="11412000" term_type="tall_log" msg="Completed"
 - INFO | 2022-04-01 12:37:22.980 DIAGNOSTIC.TRUN - LH371FB3 - 1332871175 - - ISA 1 - uid="13472e7c-72dc-4371-a9d-6d3bf92c2a0" dim="D" level="INFO" comp="TEMN-TAFJ-TRUN-CORE" cn="com.temenos.logging.wrapper.common.LoggerCommonInfo:181" code="11412000" term_type="tall_log" msg="Started"
 - INFO | 2022-04-01 12:37:23.334 DIAGNOSTIC.TRUN - LH371FB3 - 1332871175 - - ISA 1 - uid="73db30bc-23d7-4594-4731-b616447a330a" dim="D" level="INFO" comp="TEMN-TAFJ-TRUN-CORE" cn="com.temenos.logging.wrapper.common.LoggerCommonInfo:181" code="11412000" term_type="tall_log" msg="Completed"

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Practice9

View the formatted advice.

Practice » Solution

FUNDSTRANSFER, FT21105GC9DB

Delivery Reference.1	D20220401046235095901-900.11 DEBIT ADVICE
Delivery Reference.2	D30220401046235095904-910.11 CREDIT ADVICE

View Delivery Message

Delivery ref	Bank date	Type	Description	Header Status	Msg Status	Cancel	To address
D20220401046235095901	15 APR 21	900	CONFIRMATION OF DEBIT	FORMATTED	FORMATTED	PRINT	Coca-Cola

DATE : 15 APRIL 2021

REFERENCE : FT/21105/0C9DB

CUSTOMER : 100282

Coca-Cola
1 COCA-COLA PLAZA
ALANTA

DEBIT ADVICE

WE HAVE TODAY EXECUTED THE FOLLOWING Account Transfer
INSTRUCTION FOR VALUE 15 APRIL 2021

BENEFICIARY : YOURSELVES

WE HAVE DEBITED YOUR USD ACCOUNT NO. GB61DEMO60161300010995
VALUE 15 APRIL 2021 AS FOLLOWS :

TRANSFER AMOUNT	GBP	209.21
AT EXCHANGE RATE	1.434	
EQUIVALENT AMOUNT	USD	300.00
AMOUNT DEBITED	USD	300.00

IF YOU HAVE ANY QUERIES REGARDING THE ABOVE DETAILS, PLEASE CONTACT
US AS SOON AS POSSIBLE QUOTING OUR TRANSACTION NUMBER.

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Practice

Practice3

Execute COB.

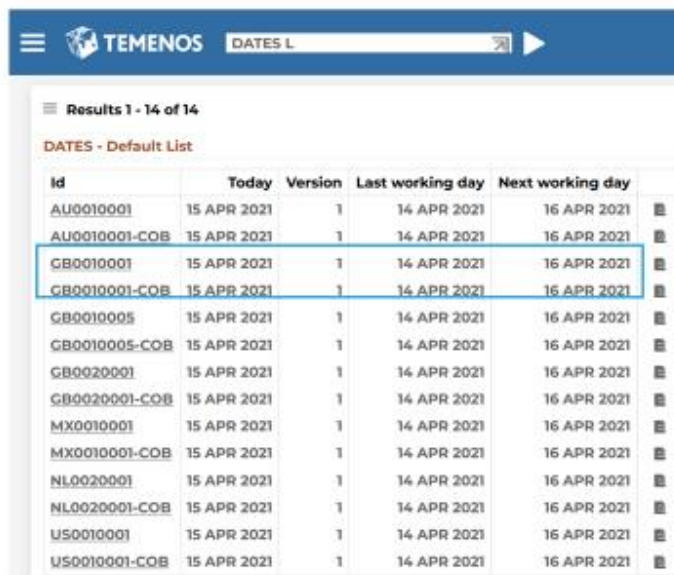
| Practice

- Execute COB

Practice4

Check the Transact Date before running the COB.

| Practice » Solution



Results 1 - 14 of 14

DATES - Default List

Id	Today	Version	Last working day	Next working day
AU0010001	15 APR 2021	1	14 APR 2021	16 APR 2021
AU0010001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021
GB0010001	15 APR 2021	1	14 APR 2021	16 APR 2021
GB0010001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021
GB0010005	15 APR 2021	1	14 APR 2021	16 APR 2021
GB0010005-COB	15 APR 2021	1	14 APR 2021	16 APR 2021
GB0020001	15 APR 2021	1	14 APR 2021	16 APR 2021
GB0020001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021
MX0010001	15 APR 2021	1	14 APR 2021	16 APR 2021
MX0010001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021
NL0020001	15 APR 2021	1	14 APR 2021	16 APR 2021
NL0020001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021
US0010001	15 APR 2021	1	14 APR 2021	16 APR 2021
US0010001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021

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Practice5

Set the Service Control status of TSM and COB to start.

Practice » Solution

TSA.SERVICE, TSM Description.1TSM Record for running upgrade as service Server Name.1 Work Profile.1TSMWORK LOAD Server Status.1 User *INPUTTER Service Control ☐ Stop ☒ Start ☐ Auto ☐ Debug

Practice6

Start the agents.

Practice » Solution


Select Setting up the Slot01 parameters - tRun START.TSM -DEBUG

```

START_TSM -DEBUG
tSA 1 -DEBUG
<como>
<agent>1</agent>
<processid>1116536842</processid>
<portno>18</portno>
Agent 1 started 04 APR 22 13-53-15
Agent's Process id 1116536842
<servername>LH371FB3</servername>
Running on server LH371FB3 PortNumber 18
Next Service TSM Current Service
SELECT F.EB.SCALABILITY.GROUP5 Selected=0 time=3secs
<service name = TSM>
TSM -DEBUG

Service Profile TSM TSM Record for running upgrade as service (1)
BNK/PRINT.OUT DE FORMATTING SERVICE (1)
COB FOR COB (10)

Manually launch tSA 3 COB
Manually launch tSA 4 COB
Manually launch tSA 5 COB
Manually launch tSA 6 COB
Manually launch tSA 7 COB
Manually launch tSA 8 COB
Manually launch tSA 9 COB
Manually launch tSA 10 COB
Manually launch tSA 11 COB
Manually launch tSA 12 COB

```


Select Setting up the Slot01 parameters - tRun tSA 11

```

tSA 11
<como>
<agent>11</agent>
<processid>398983186</processid>
<portno>18</portno>
Agent 11 started 04 APR 22 13-59-37
Agent's Process id 398983186
<servername>LH371FB3</servername>
Running on server LH371FB3 PortNumber 18
Next Service Current Service
SELECT F.EB.SCALABILITY.GROUP5 Selected=0 time=4secs
<service name = COB>
SELECT F.COMPANY WITH CONSOLIDATION.MARK NE 'C' Selected=7 time=11secs
SSELECT F.BATCH BY BATCH.STAGE Selected=5289 time=166secs

```


Select Setting up the Slot01 parameters - tRun tSA 5

```

SELECT F.MARKET.PRICE WITH @ID LIKE US0010001... AND SECURITY.CODE EQ 200567-000 Selected=0 time=2secs
SELECT F.MARKET.PRICE WITH @ID LIKE US0010001... AND SECURITY.CODE EQ 000400-003 Selected=0 time=2secs
SELECT F.MARKET.PRICE WITH @ID LIKE US0010001... AND SECURITY.CODE EQ 000400-004 Selected=0 time=2secs
SELECT F.MARKET.PRICE WITH @ID LIKE US0010001... AND SECURITY.CODE EQ 000400-007 Selected=0 time=2secs
SELECT F.MARKET.PRICE WITH @ID LIKE US0010001... AND SECURITY.CODE EQ 000400-005 Selected=0 time=2secs
SELECT F.MARKET.PRICE WITH @ID LIKE US0010001... AND SECURITY.CODE EQ 000400-006 Selected=0 time=2secs
SELECT F.MARKET.PRICE WITH @ID LIKE US0010001... AND SECURITY.CODE EQ 000400-009 Selected=0 time=2secs
SELECT F.MARKET.PRICE WITH @ID LIKE US0010001... AND SECURITY.CODE EQ 000400-011 Selected=0 time=1secs

```

Practice7

Monitor the COB progress using COB.MONITOR tool.

| Practice » Solution

MA Select Setting up the Slot01 parameters - trun EX

COB.PROGRESS.DISPLAY				
=====				
TIME : 16:46:52 04 APR 2022		COB-DATE : 15 APR 2021		
COB Running in GB0010005 GB0010001 US0010001 NL0020001 AU0010001 MX0010001 GB0020001				
=====				
STAGE	PROGRESS	PROCESSED	TOTAL	%COMPLETED
=====				
APPLICATION	[]	331	331	100.00
SYSTEM WIDE	[]	306	306	100.00
REPORTING	[]	332	332	100.00
START OF DAY	[]	349	349	100.00
ONLINE	[]	175	289	60.55
=====				
COB	[]	1493	1607	92.91
=====				
To Quit Press EXIT exit Q q				

Practice8

DATES default list is shown in which different dates for Today, Last Working Day and Next Working Day are seen when compared with the previous dates default screen before starting COB.

| Practice » Solution

DATES - Default List					
Id	Today	Version	Last working day	Next working day	
AU0010001	16 APR 2021	1	15 APR 2021	19 APR 2021	
AU0010001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021	
GB0010001	16 APR 2021	1	15 APR 2021	19 APR 2021	
GB0010001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021	
GB0010005	16 APR 2021	1	15 APR 2021	19 APR 2021	
GB0010005-COB	15 APR 2021	1	14 APR 2021	16 APR 2021	
GB0020001	16 APR 2021	1	15 APR 2021	19 APR 2021	
GB0020001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021	
MX0010001	16 APR 2021	1	15 APR 2021	19 APR 2021	
MX0010001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021	
NL0020001	16 APR 2021	1	15 APR 2021	19 APR 2021	
NL0020001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021	
US0010001	16 APR 2021	1	15 APR 2021	19 APR 2021	
US0010001-COB	15 APR 2021	1	14 APR 2021	16 APR 2021	

Practice9

The today's date would be set to the next working day and Co Batch Status would be set to 'O' in the DATES application, indicating that the system is in online mode and ready for the next day transaction.

| Practice » Solution

DATES CB-001-0001 Model Bank		
Today	16 APR 2021	16 APR 2021
Version	1	
Last Working Day	15 APR 2021	15 APR 2021
Next Working Day	19 APR 2021	19 APR 2021
Local Payment Day	16 APR 2021	16 APR 2021
Local Dispo Day	16 APR 2021	16 APR 2021
Back Value Minimum	23 FEB 2021	23 FEB 2021
Back Value Max	09 NOV 2020	09 NOV 2020
Forw Value Minimum	28 MAY 2021	28 MAY 2021
Forw Value Max	20 AUG 2021	20 AUG 2021
Co Batch Status	O	
Period End	18 APR 2021	18 APR 2021
Julian Date	2021106	
Current Day	NORMAL	
Last Period End	15 APR 2021	15 APR 2021
Bk Bal Date Start	23 JUL 1998	23 JUL 1998
Override.1	LOCAL.PAYMENT.DAY NOT > NEXT.WORKING.DAY	
Override.2	LOCAL.DISPO.DAY NOT > NEXT.WORKING.DAY	

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